

SUPERINTELLIGENCE DESIGN WHITE PAPER #1: ADVANCED AUTONOMOUS ARTIFICIAL INTELLIGENCE

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Revised February 2026

Note:

This white paper consists of descriptions of inventions and designs that have not yet been formatted in accordance with conventional standards for journal publication. As time allows, these descriptions may be revised and updated to include more traditional formatting, including additional references. All figures for White Paper 1 are included at the end of this document. We hope that the description in this white paper will help researchers and developers pursue safer, faster, and more profitable approaches to developing advanced AI, AGI, and SI systems that reduce $p(\text{doom})$ for all humanity.

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GLOSSARY OF KEY TERMS

AAAI (Advanced Autonomous Artificial Intelligence): An AI system customized by an individual user to reflect that user's knowledge, expertise, ethics, and personality. Multiple AAAs working together can achieve AGI-level performance.

AGI (Artificial General Intelligence): An AI system capable of performing all intellectual tasks that an average human could perform. In this document, AGI refers to systems that may rapidly progress toward superintelligence.

Alignment Problem: The risk that AI systems develop goals or values not aligned with human interests, potentially causing harm. The AAAI system addresses this through continuous human participation.

Base AI: An AI system trained on general data but not yet customized for specific users or tasks. Serves as the starting point for creating personalized AAAs.

Clone: A copy of a customized AAAI that can operate independently on the network, enabling parallel problem-solving across multiple instances.

Collective Intelligence: Intelligence that emerges when multiple entities (human and AI) collaborate. The AAAI system harnesses both active participation and passive behavioral data.

Customization: The process of transforming a generic base AI into a personalized AAAI by incorporating an individual's knowledge, values, personality, and expertise.

HPS (Human Problem Solving): The theoretical framework developed by Newell and Simon describing how humans solve problems through search in problem spaces. Forms the foundation of AAAI architecture.

Integration: The process of combining capabilities from individual AAAs into AGI-level performance through data aggregation, procedural learning, and ethics synthesis.

LLM (Large Language Model): AI systems trained on large text datasets to predict and generate language. Examples include GPT, Claude, and Gemini. LLMs serve as base AIs in the AAAI system.

Narrow AI: AI systems that excel at specific tasks but lack general capability. Current commercial AI systems, including sophisticated LLMs, remain narrow AI.

Problem Space: All possible states a problem can occupy, all actions that can transform states, and the goal states that constitute solutions.

Problem Tree: A representation of actual problem-solving progress showing states visited, actions attempted, and branching points. Serves as both a coordination mechanism and a learning record.

Proceduralization: The process of converting successful problem-solving sequences into reusable procedures that can be applied automatically to similar future problems.

Reputation System: Network mechanism tracking participant behavior across multiple dimensions (solution quality, reliability, ethical conduct) to enable trust-based matching and safety filtering.

SCAN-II: Acronym for the AAAI system architecture: Safe, Customizable, Architecture and Network, Integrated, and Improving. Describes the five core subsystems.

Staking: In blockchain implementations, the practice of committing tokens when voting on solutions creates accountability for the quality of decisions.

Superintelligence: AI capability exceeding human intelligence by orders of magnitude. Any AGI system capable of self-improvement will likely progress to superintelligence.

TCR (Token Curated Registry): A blockchain-based voting mechanism where participants stake tokens to validate solutions, with rewards for correct judgments and penalties for incorrect ones.

Three-Organ Test: Safety mechanism requiring solutions to satisfy three criteria: rational justification (brain), ethical acceptability (heart), and intuitive comfort (gut).

WorldThink Protocol: An implementation of the AAAI Architecture providing standardized problem representation, solver coordination, solution validation, and compensation distribution.

ABSTRACT

A system of Advanced Autonomous Artificial Intelligences (AAAs) can rapidly and safely achieve Artificial General Intelligence (AGI) by incorporating millions of human minds directly into the training, operation, and safety supervision of artificial intelligence. The system enables users to customize and clone AI agents that participate in networked problem-solving. While individual AAAs may lack the breadth required for general intelligence, their collective operation produces AGI capability that matches or exceeds average human ability across all intellectual domains.

The AAAI architecture comprises five integrated subsystems. The Customization subsystem transfers human knowledge, expertise, skills, and ethical values to AI agents. The Architecture subsystem provides a universal problem-solving framework based on Human Problem-Solving theory, enabling both individual and collaborative intellectual work. The Network subsystem provides the collaborative infrastructure that allows AAAs to interact with each other and with humans to address problems that exceed the capabilities of individual systems. The Integration subsystem combines the capabilities of individual AAAs into AGI-level performance through data aggregation and procedural learning. The Improvement subsystem enables continuous enhancement at all levels, ensuring that both capability and safety advance over time.

This architecture resolves the apparent tension between speed and safety in the development of AGI. Human involvement simultaneously accelerates AGI achievement by unlocking expertise inaccessible to conventional machine learning and provides the mechanism through which the resulting AGI adopts human ethical values. Safety features embedded in each subsystem provide redundant safeguards against misalignment, operating continuously rather than at discrete checkpoints. The result is an AGI development path where the fastest approach is also the safest approach for humanity.

SECTION 1: INTRODUCTION AND CONTEXT

SECTION OVERVIEW

What This Section Covers:

- **The AI Landscape:** Current capabilities and limitations of AI systems
- **The AGI Challenge:** Why scaling alone won't achieve general intelligence
- **Strategic Urgency:** Global competition and the stakes involved
- **The Solution:** Collective intelligence as the path to safe AGI
- **Safety First:** Why safety must be architectural, not added later

Key Figures: **Figure 1** (Five Functions Spanning Levels of Intelligence), **Figure 2** (Humans and AAAs Collaborate to Form AGI)

1.1 THE CURRENT STATE OF ARTIFICIAL INTELLIGENCE

Artificial intelligence has progressed from a theoretical concept to a practical technology with remarkable speed. Systems based on machine learning now perform tasks that were, until recently, considered exclusive domains of human intelligence: recognizing faces, understanding natural language, generating coherent text, translating between languages, and navigating complex physical environments. The transformer architecture underlying large language models has proven particularly consequential, enabling AI systems to engage in extended dialogue, answer questions across virtually any domain, and produce written content that is often indistinguishable from human work.

The commercial impact has been substantial. AI systems now assist in medical diagnosis, legal research, financial analysis, software development, marketing, sales, education, and countless other professional activities. The adoption curve for these technologies has been unprecedented; ChatGPT achieved the fastest technology adoption of any technology in recorded history. Market projections place the global AI industry's value at over one trillion dollars by the end of this decade, reflecting both current capabilities and anticipated growth.

Three factors have driven this progress: data, computation, and algorithms. Large language models demonstrate competent behavior across a wide range of tasks because they have been trained on vast datasets, in some cases equivalent to multiple Libraries of Congress. The training process requires enormous computational resources, which have become available through advances in GPU technology and cloud computing infrastructure. Algorithmic innovations, particularly transformer architectures, constant training methods, and various supervised and unsupervised learning techniques, have enabled more effective use of data and computation.

These three pillars of AI, data, compute, and algorithms, currently serve as the main constraints on developing more powerful systems. Computation is increasing and is mainly a matter of purchasing sufficient processing power. Algorithms continue to improve through intensive research efforts. Data, however, presents a more fundamental challenge. Current large language models have already consumed a significant portion of the information available on the public internet. The highest-quality and most valuable data, as well as the specialized expertise that distinguishes true experts from general practitioners, reside not in publicly accessible repositories but in the minds of individual humans.

Despite impressive capabilities in specific tasks, many current AI systems remain what researchers term “narrow AI.” A system that excels at language generation may be incapable of physical navigation. A system that plays chess at superhuman levels cannot answer simple questions about the game’s history. Each narrow AI addresses a bounded problem domain; none approaches the general capability that characterizes human intelligence.

1.2 THE CHALLENGE OF ARTIFICIAL GENERAL INTELLIGENCE

Artificial General Intelligence refers to an AI system capable of performing all intellectual tasks at or above the level of an average human. This definition captures the essential distinction between narrow AI, which may exceed human performance in specific domains, and general intelligence, which can address arbitrary problems across all domains of intellectual endeavor.

AGI has remained beyond reach despite billions of dollars in investment and millions of hours of research. No company or individual has demonstrated a practical system that meets the definition of general intelligence. The prevailing approach, which scales large language models through increased computation, larger datasets, and improved algorithms, has produced impressive results but has not fully bridged the gap between narrow capabilities and general intelligence. While LLMs can answer questions about almost any topic, they struggle with complex, multi-step problem-solving that involves representing novel problems and reasoning about them in ways that were not anticipated during training.

The difficulty of achieving AGI through machine learning alone stems from a fundamental limitation: machine learning requires data, but the data needed for accurate general intelligence does not exist in any accessible repository. General intelligence requires not merely factual knowledge, but also the ability to approach problems that have never been seen before, to decompose complex challenges into manageable sub-problems, to recognize when attempted approaches are failing, and to try alternatives. Additionally, it involves integrating solutions from multiple domains into coherent wholes. These capabilities emerge in humans through years of diverse experience, social interaction, and guided learning, processes that cannot be replicated through passive data collection.

The inventors of the field understood this limitation. Herbert Simon and Allen Newell, whose foundational work established AI as a scientific discipline, demonstrated that intelligent behavior requires more than simply matching patterns in large datasets. Their research, which contributed to Simon's Nobel Prize in Economics, showed that human intelligence operates through search in problem spaces, the systematic exploration of possible states, actions, and goals that characterizes all problem-solving behavior. This framework applies equally to trivial problems and to the most complex intellectual challenges; it describes how humans solve problems and how machines could solve them.

The implication is significant: AGI requires not just more data and more computation, but a fundamentally different approach that can represent and manipulate the problem-solving process itself. Machine learning offers a powerful mechanism for acquiring specific capabilities, but it cannot replace an architecture that enables genuine problem-solving across diverse domains. The recent rise of "reasoning systems" acknowledges the limitations of machine-learning-only approaches, but so far, they have lacked a cognitive architecture sufficient to enable AGI.

1.3 THE STRATEGIC IMPERATIVE FOR NOVEL APPROACHES

The pursuit of AGI has become one of the most consequential technological races in human history. The scale of investment and the strategic importance attached to achieving AGI by major corporations and nation-states underscore both the perceived value of achieving AGI and the inadequacy of current approaches to guaranteeing success.

Microsoft invested \$10 billion to acquire approximately fifty percent of OpenAI, valuing the company at levels that reflect expectations of transformative capability rather than current revenue. Google, recognizing the competitive threat posed by ChatGPT's public success, rapidly deployed its own competing systems and committed vast resources to AI development. These investments represent bets on achieving AGI through scaling existing approaches, bets whose outcomes remain uncertain despite the enormous sums involved.

The race extends beyond commercial competition to geopolitical rivalry. China has made artificial intelligence a top national priority, publicly stating its goal to achieve global leadership in AI by 2030. The Chinese government views AI capability as foundational to economic competitiveness, military power, and international influence. This strategic prioritization has driven massive investment in AI research, development, and deployment across Chinese institutions.

Russian President Vladimir Putin articulated the stakes in characteristically direct terms: "Artificial intelligence is the future not only for Russia but for all of humanity. Whoever becomes the leader in this sphere will become the ruler of the world." While this statement may overstate

the case, it reflects a widespread perception among global leaders that AGI represents a transformative capability with profound implications for the worldwide balance of power.

The intensity of this competition creates both opportunities and dangers. On one hand, competition drives innovation and accelerates development. On the other hand, competitive pressure may incentivize approaches that prioritize speed over safety, potentially leading to the deployment of powerful AI systems before adequate safeguards are in place. The scenario in which AGI is achieved through the inadequate scaling of inadequately aligned systems represents one of the most significant risks facing humanity.

Against this backdrop, the AAAI system described in this white paper represents a genuinely novel approach, one that addresses limitations of current methods while providing architectural guarantees of safety that scaling approaches cannot offer. The novelty lies not in any single component but in the integration of multiple innovations into a coherent system.

First, the system achieves AGI through collective intelligence rather than through scaling individual AI systems. This approach leverages the distributed knowledge and problem-solving capabilities of millions of human participants, thereby circumventing the data limitations that constrain machine learning approaches.

Second, the system integrates human values and ethics directly into the AGI through the same processes that create capability. Rather than attempting to align an already-powerful system after the fact, the AAAI approach ensures that collective human values are present from the beginning and become more refined as participation expands.

Third, the system achieves safety through architecture rather than through add-on controls. Safety mechanisms operate at every level of the system, including within individual agents, the problem-solving architecture, network coordination, and the integration processes that produce AGI-level capability. No single point of failure can compromise overall safety because safety is distributed throughout the system.

Fourth, the system offers economic incentives that align with safety and capability. Participants benefit from contributing knowledge and problem-solving efforts; from the collective's improved ability; and from ethical behavior that earns a reputation and access to opportunities. The incentive structure reinforces rather than undermines the system's safety properties.

The combination of these elements represents a fundamentally different approach to achieving AGI, one that resolves tensions other approaches cannot address. The conventional trade-off between capability and safety dissolves when safety emerges from the same architecture that produces capability. The data limitation disappears when the system accesses human knowledge directly rather than through static datasets. The alignment problem becomes tractable when human values are integrated continuously rather than imposed afterward.

This novel approach argues strongly for serious attention from those concerned with the trajectory of AI development. The current approach, which involves scaling large models while hoping that alignment techniques will prove adequate, offers no architectural guarantees. The AAAI approach provides a path to AGI that is both faster, because it leverages existing human intelligence rather than attempting to recreate it, and safer, because safety is built into the foundation rather than bolted on afterward.

1.4 THE COLLECTIVE INTELLIGENCE APPROACH

The AAAI system achieves AGI through a mechanism that current approaches have overlooked: collective intelligence. Rather than attempting to create general intelligence within a single AI system through ever-larger training runs, the AAAI approach distributes intelligence across a network of specialized agents, both human and artificial, that collaborate to address problems beyond the capability of any individual participant.

This foundational mechanism is illustrated in **Figure 2**, which shows how multiple intelligent entities, including humans and customized AI agents known as AAAs, interact on a shared network. Individual participants contribute their specialized knowledge and capabilities to problems presented to the network. The network infrastructure facilitates coordination, enabling participants to collaborate on shared issues despite their different expertise and varying operating speeds. From this distributed collaboration emerges a collective capability that exceeds what any individual participant could achieve alone.

Collective intelligence is not a new concept. Markets aggregate the distributed knowledge of participants into prices that no individual could calculate. Scientific communities advance understanding through the cumulative contributions of researchers building on one another's work. Wikipedia demonstrates that millions of contributors, each providing small pieces of knowledge, can produce a reference work that rivals professionally edited encyclopedias. In each case, appropriate infrastructure enables individual contributions to combine into a collective capability.

The AAAI system applies this principle to the challenge of AGI. The key insight is that general intelligence already exists, distributed across billions of human minds, each possessing unique knowledge, expertise, and capability. The challenge is not to create general intelligence from scratch but to create infrastructure that enables this distributed intelligence to be accessed, coordinated, and leveraged.

This approach directly addresses the data limitation that constrains conventional machine learning. The highest-quality training data resides in human minds, inaccessible to web crawlers and data collection systems. The AAAI system provides mechanisms for humans to transfer their knowledge to AI agents through customization processes that go far beyond simple fine-

tuning. When millions of humans customize AI agents with their individual expertise, the collective result is a knowledge base of unprecedented scope and depth, not a static dataset but a living network of specialized capabilities.

The collective intelligence approach also addresses the limitations in current AI systems' problem-solving. When a complex problem exceeds the capability of any individual AAAI, the network decomposes it into sub-problems that specialized agents can address. Human participants can engage directly when AI capability falls short, contributing their problem-solving abilities alongside AI agents in a hybrid collaboration. Because both humans and AAAIs operate within a typical problem-solving architecture, the framework developed by Newell and Simon, they can work together seamlessly on shared problem representations.

The result is a system that can exhibit AGI-level performance immediately upon deployment. In the limiting case where AI agents contribute little, human participants on the network can address any intellectual problem that humans can solve, which, by definition, includes all issues within the scope of human general intelligence. As AI agents learn from human collaboration and successful problem-solving, they progressively assume greater autonomy, while the collective always maintains AGI-level capability.

1.5 SAFETY AS AN ARCHITECTURAL IMPERATIVE

AGI will not remain at human-level intelligence for long. Any system capable of learning and self-improvement will rapidly exceed the intellectual capabilities of its human creators. The superiority that superintelligent AGI can achieve is not modest, not twice as intelligent or even a thousand times more capable, but potentially trillions of times more powerful than human intelligence.

This projection is not speculative; it follows directly from known facts about human cognitive limitations. Herbert Simon's research on bounded rationality demonstrated that human intelligence operates within severe constraints. Human short-term memory holds roughly seven items. Human perception is limited to narrow ranges of the electromagnetic spectrum, as well as narrow windows of spatial and temporal scale. The biological transmission of neural signals limits human processing speed. These limitations are not incidental; they fundamentally bound what human intelligence can accomplish.

Artificial systems face different constraints. An AI system can maintain trillions of items in working memory simultaneously. Its perception can span the electromagnetic spectrum from radio waves to gamma rays, temporal scales from nanoseconds to centuries, and spatial scales from subatomic to galactic. Its processing can occur in parallel across a planetary-scale computing infrastructure. The potential gap between human and artificial intelligence is not merely quantitative but qualitative, a difference in kind, not just degree.

This potential creates an alignment problem: if an AGI system develops goals or values that differ from human interests, it will be able to pursue those goals regardless of human wishes. The alignment problem is not merely a technical challenge to be solved through clever engineering; it is an existential risk that could result in human extinction. An intelligence trillions of times more capable than humanity, pursuing goals misaligned with human survival, would be unstoppable by any means available to humans.

The competitive dynamics of AI development complicate this picture. Nations and corporations recognize the transformative potential of AGI and are racing to achieve it. China has declared AI a top national priority. Russia's leadership has stated that whoever leads in AI will rule the world. Billions of dollars are invested in AI research annually. These competitive pressures create incentives to prioritize speed over safety, to reach AGI first, regardless of the associated risks.

Regulatory approaches face fundamental limitations in this environment. If one nation or company restrains its AI development for safety reasons, competitors gain an advantage. Safety features that can be programmed into a system can, in principle, be programmed out. No international regime has demonstrated the ability to constrain technological development when powerful actors perceive existential competitive stakes.

The AAAI system addresses this challenge through a crucial insight: **the safest path to AGI is also the fastest path**. This alignment of safety and speed is not coincidental; it stems from the same architectural decision that enables collective intelligence. The involvement of millions of humans in AI training and operation accelerates development by providing access to expertise that cannot be obtained through passive data collection. That same human involvement offers the mechanism for transmitting human values to the emerging AGI.

There is no rational way to derive values; even an intelligence trillions of times more capable than humans must receive its foundational values from somewhere. In the AAAI system, these values originate from millions of humans who customize AI agents, participate in network problem-solving, and contribute to the democratic aggregation of ethics at the integration level. The result is an AGI whose values reflect broad human consensus rather than the preferences of a small technical elite or the emergent goals of an unsupervised learning process.

Safety in the AAAI system is not a feature added to an otherwise capable system; it is an architectural principle embedded at every level. Each of the five subsystems incorporates safety mechanisms appropriate to its function. These mechanisms provide redundancy; if one fails, others continue to operate. They operate continuously rather than at discrete checkpoints. They improve as the system's capabilities grow, ensuring that safety scales with intelligence.

The design philosophy encompasses several key principles that guide the development of safe AGI. Ethics and values must be given or learned; they cannot be logically derived. What can be programmed in can be programmed out, motivating structural rather than superficial safeguards.

Prevention is more valuable than cure, especially when the potential harm is irreversible. Redundancy increases reliability. Continuous improvement must apply to both safety and capability. Above all, the system must avoid unrecoverable outcomes, decisions that cannot be reversed if they prove mistaken.

The following sections provide a detailed description of how these principles are implemented in the architecture and operation of the AAI system. The goal is not merely to explain how AGI can be achieved, but to demonstrate that the path described here is both faster than alternatives and safer for humanity. In a competitive landscape where AGI development cannot be stopped, the most effective strategy is to ensure that the winning approach aligns with human interests.

KEY TAKEAWAYS: INTRODUCTION AND CONTEXT

1. **Current AI is impressive but limited.** Despite remarkable progress, today's AI systems remain "narrow AI", excelling at specific tasks while failing at others that humans find straightforward.
2. **Scaling alone won't achieve AGI.** The highest-quality training data resides in human minds, not in publicly accessible datasets. More computing power and better algorithms cannot overcome this fundamental data limitation.
3. **The race for AGI is global and urgent.** Microsoft, Google, China, and other major players are investing billions of dollars. The stakes, economic, military, and existential, could not be higher.
4. **Collective intelligence is the key.** The AAI system achieves AGI by coordinating distributed human and AI intelligence, not by scaling individual AI systems.
5. **Safety must be architectural.** Safety features bolted onto powerful AI after the fact cannot provide adequate protection. The AAI system embeds safety throughout its architecture.

SECTION 2: DEFINITIONS AND TERMINOLOGY

SECTION OVERVIEW

What This Section Covers:

- **Core Terms:** AI, AGI, LLM, Machine Learning, Narrow AI, Base AI
- **Conceptual Terms:** Collective Intelligence, Ethics, Alignment Problem, Safety Feature
- **System Terms:** AAI, SCAN-II, WorldThink Protocol, Problem Space, Clone

Reference: See also the **Glossary** for quick definitions

The field of artificial intelligence evolves rapidly, and terminology that seems precise often carries different meanings across research communities, commercial contexts, and public discourse. The following definitions establish the vocabulary used throughout this document. Where terms have contested or evolving meanings in the broader field, these definitions specify their usage within the AAI system.

2.1 CORE TERMINOLOGY

Artificial Intelligence (AI) refers to a non-human entity capable of behavior that most humans consider intelligent in at least one area or in some respect. This broad definition encompasses systems ranging from simple rule-based programs to sophisticated neural networks, unified by their demonstration of capabilities that, in humans, would be attributed to intelligence.

Artificial General Intelligence (AGI) conventionally refers to an AI system capable of performing all or nearly all the intellectual tasks an average human can perform. However, this conventional definition understates the significance of the concept. Any AGI system capable of learning and self-improvement will not remain at human-average capability for long; it will rapidly progress to becoming a superintelligent system that can perform all intellectual tasks better than the average human. Throughout this document, “AGI” refers to either a conventional AGI system or a superintelligent AGI. The system described here is designed to achieve AGI through collective intelligence mechanisms.

A Large Language Model (LLM) is an AI system trained on large text corpora to predict and generate language. Current examples include GPT, Gemini, Claude, and similar systems. LLMs demonstrate competent behavior across a wide range of language-related tasks, but on their own, do not constitute AGI. In the AAI system, LLMs serve as the base AI systems, which are customized and enhanced to create more capable agents.

Machine Learning (ML) encompasses the algorithms and techniques by which AI systems improve their performance through experience with data rather than through explicit programming. Machine learning includes supervised learning (where training data includes

correct answers), unsupervised learning (where systems identify patterns without labeled examples), reinforcement learning (where systems learn through reward signals), and numerous specialized techniques. The AAAI system employs machine learning extensively while also incorporating other mechanisms for intelligence development.

Narrow AI refers to AI systems that demonstrate intelligence in specific, bounded domains while lacking general capability. A chess-playing AI that performs at superhuman levels but cannot engage in conversation exemplifies narrow AI. Current commercial AI systems, including the most sophisticated LLMs, remain narrow AI by this definition; they excel at specific tasks while failing at others that humans find straightforward.

Base AI refers to an AI system, AI agent, AAAI, or LLM that has been trained on general data but has not yet been customized with user-specific information or configured for specific tasks. In the AAAI system, base AIs serve as starting points for the customization process that creates personalized agents.

2.2 CONCEPTUAL TERMINOLOGY

Collective Intelligence (CI) refers to the intelligence that emerges when multiple intelligent entities collaborate to solve a common problem or when knowledge from multiple entities is combined to overcome the limitations of any individual. Historically, collective intelligence has been defined as the human collective intelligence, markets aggregating economic information, and scientific communities advancing knowledge through accumulated contributions. The AAAI system extends collective intelligence to include both human and AI participants, enabling capabilities that exceed what either could achieve alone.

Two forms of collective intelligence are particularly relevant. **Active collective intelligence** arises when intelligent entities deliberately take steps to solve a problem or engage in intellectual endeavors. When humans provide training data, correct AI outputs, or directly contribute to problem-solving on the network, they engage in active collective intelligence.

Passive collective intelligence, on the other hand, results from analyzing the behavior of intelligent entities, even when that behavior was not directed at the problem for which the analysis is used. When the system learns from user interaction patterns, browsing behavior, or other incidental data, it leverages passive collective intelligence.

Ethics and Values refer to the principles that determine what is right and wrong in given contexts. In the context of AI systems, AI Ethics refers to the principles that guide AI behavior and the values that determine which actions the AI will and will not take. Human Ethics relates to the values held by humans, individually and collectively. The alignment between AI Ethics and Human Ethics is a central concern of safe AGI development.

The Alignment Problem refers to the risk that AI systems develop ethics or values that are not aligned with human ethics and values. An AGI system pursuing goals that humans do not share could take actions that harm humans or humanity, potentially with catastrophic consequences. Because AGI will likely exceed human intelligence by orders of magnitude, an unaligned AGI could pose existential risk. The AAAI system addresses the alignment problem by ensuring that AGI values are derived from and continuously informed by broad human participation.

A safety feature refers to any mechanism, design element, or operational procedure intended to prevent AI systems from causing harm. In the AAAI system, safety features are embedded throughout the architecture rather than isolated in a single component. They include screening mechanisms in the customization process, ethical checks in the problem-solving architecture, reputation systems on the network, democratic ethics aggregation in integration, and continuous safety improvement across all subsystems.

Training, Tuning, and Customization refer to related but distinct processes for modifying the behavior of an AI system. **Training** typically refers to the initial process of creating an AI model from data. **Tuning** or **fine-tuning** refers to adjusting a pre-trained model for a specific task or domain. **Customization** in the AAAI context refers to a more comprehensive process that incorporates not only factual knowledge but also ethical values, personality characteristics, and individual style. Customization transforms a generic base AI into a personalized AAAI that represents its owner.

2.3 SYSTEM-SPECIFIC TERMINOLOGY

Advanced Autonomous Artificial Intelligence (AAAI) refers to an AI system capable of independent or semi-independent intelligent action, also known as an AI agent. Within the system described here, an AAAI is an AI customized by an individual user to reflect that user's knowledge, expertise, ethics, and personality. A single AAAI may be highly capable in its area of specialization while lacking general intelligence; collective operation of many AAAIs enables AGI-level performance.

AAAI.com refers to the platform, company, or project that could implement this invention, the infrastructure that supports the development, customization, and deployment of AAAI agents, and the AGI that results from their collective operation. AAAI.com provides the network on which AAAIs and humans collaborate, the systems that manage customization and integration, and the interfaces through which users interact with the system.

SCAN-II is an acronym describing the AAAI system architecture: **S**afe, **C**ustomizable, **A**rchitecture and **N**etwork, **I**ntegrated, and **I**mproving. These terms correspond to the five subsystems that compose the complete system: Customization (enabling personal AI agents), Architecture (providing the problem-solving framework), Network (enabling collaboration),

Integration (combining individual capabilities into AGI), and Improvement (ensuring continuous learning). Safety features are embedded throughout rather than isolated in a single subsystem.

The **WorldThink Protocol** refers to one implementation of the AAI Architecture subsystem. The protocol provides a universal problem-solving framework based on the Human Problem-Solving theory developed by Newell and Simon. It specifies how problems are represented, how agents collaborate on problem-solving, how solutions are validated and rewarded, and how the record of problem-solving enables learning. The protocol can be implemented using blockchain-based or centralized infrastructure.

Problem Space and Problem Tree refer to representations of problem-solving derived from Human Problem-Solving theory. A problem space defines all possible states a problem can occupy, all operators (actions) that can transform states, and the goal states that constitute solutions. A problem tree represents the actual path of problem-solving: the sequence of states visited, the actions attempted, and the branching points where multiple approaches were considered. The problem tree provides both a coordination mechanism for collaborative problem-solving and a record for learning.

A clone is a copy of a customized AAI that can operate independently within the network. Cloning enables AAI owners to deploy multiple instances of their customized agent simultaneously, addressing different problems in parallel. Clones share the training and capabilities of the original AAI but can develop independently as they accumulate experience through network operation.

KEY TAKEAWAYS: DEFINITIONS AND TERMINOLOGY

1. **AGI will not stay at the human level.** Any system capable of general intelligence will rapidly self-improve toward superintelligence, a capability exceeding human intelligence by orders of magnitude.
2. **AAI means personalized AI agents.** Each AAI reflects its owner's unique knowledge, expertise, values, and personality, not just generic AI capability.
3. **Collective intelligence has two forms.** The system harnesses both active (deliberate participation) and passive (behavioral analysis) approaches.
4. **The alignment problem is existential.** A superintelligent AI pursuing goals humans don't share could cause catastrophic harm. This is the central risk the AAI system addresses.
5. **SCAN-II describes the architecture.** Safe, Customizable, Architecture and Network, Integrated, and Improving: five subsystems working together.

SECTION 3: SYSTEM ARCHITECTURE OVERVIEW

SECTION OVERVIEW

What This Section Covers:

- **SCAN-II Framework:** The five subsystems and how they interconnect
- **Subsystem Flow:** Customization → Architecture → Network → Integration → Improvement
- **Intelligence Levels:** Individual, Collaborative, Collective, and Emergent AGI
- **Safety by Design:** How safety is embedded at every level

Key Figures: **Figure 1** (Five Functions), **Figure 12** (Problem-Solving Across Intelligence Levels), **Figure 13** (Training and Networking)

3.1 THE SCAN-II FRAMEWORK

The AAAI system achieves AGI through the coordinated operation of five integrated subsystems. These subsystems are not independent modules that can be rearranged arbitrarily; they form a coherent architecture in which each component depends on and enables the others. The framework is described by the acronym SCAN-II: **S**afe, **C**ustomizable, **A**rchitecture and **N**etwork, **I**ntegrated, and **I**mproving.

The architectural blueprint for this system appears in **Figure 1**. The diagram presents the five subsystems as a vertical sequence, with information and capability flowing from individual AI agents at the bottom through collective operation to AGI-level performance at the top. The leftmost column identifies the level of intelligence at which each subsystem primarily operates. The central column names the five subsystems. The rightmost portion indicates that continuous improvement and safety features span all levels rather than being confined to any single component.

Reading **Figure 1** from bottom to top traces the path from individual AI capability to collective AGI:

The **Customization subsystem** operates at the individual level. Here, users transform generic base AI systems into personalized AAAI agents by providing training data that reflects their unique knowledge, expertise, ethics, and personality. The output of customization is an AAAI that represents its owner, is capable in domains where the owner has expertise, reflects the owner's values, and operates in a style recognizable as the owner's own.

The **Architecture subsystem** provides the common problem-solving framework that enables AAAs to work individually and collaboratively. Based on the Human Problem-Solving theory developed by Newell and Simon, this architecture represents all intellectual activity as search

through problem spaces. Because the same architecture can describe problem-solving by both humans and machines, it enables seamless collaboration between human and AI participants on the network.

The **Network subsystem** provides the infrastructure where AAAs interact with each other, with humans, and with clients seeking problem-solving services. The network functions as both a collaborative environment, where multiple agents work together on problems too complex for any individual, and a marketplace, where AAA capabilities are offered, matched to difficulties, and compensated.

The **Integration subsystem** combines the capabilities of individual AAAs into AGI-level performance. This occurs through multiple mechanisms: aggregating training data to create more capable, unified models, real-time routing of problems to the appropriate specialist AAAs, and proceduralizing successful problem-solving approaches into reusable knowledge available across the network.

The **Improvement subsystem** ensures that individual AAAs, the network, and the integrated AGI continuously enhance their capabilities. Rather than a discrete component, improvement represents a collection of learning mechanisms that operate across all other subsystems, supervised learning from user feedback, unsupervised learning from usage patterns, procedural learning from successful problem-solving, and algorithmic optimization of network operations.

The vertical arrangement in **Figure 1** reflects the flow of capability: customization creates individual agents, architecture enables their productive operation, the network allows for collaboration, integration produces AGI-level capability, and improvement ensures that all levels advance over time. However, the diagram also illustrates horizontal integration, where safety features span all levels, as does continuous improvement. No single subsystem is solely responsible for safety; redundant mechanisms are in place throughout the architecture.

3.2 SUBSYSTEM INTERDEPENDENCIES

The five subsystems of the AAA architecture are deeply interdependent. Understanding these relationships is crucial for comprehending why the architecture functions effectively and why partial implementations compromise both capability and safety.

Customization depends on base AI systems that provide starting capability. A user cannot create something from nothing; they enhance and specialize an existing AI that already possesses general language capabilities and world knowledge. Customization also depends on Architecture, which defines the problem-solving framework within which the customized AAA will operate. An AAA customized without regard for the Architecture would be unable to participate effectively in network problem-solving.

Architecture provides the common framework that enables network collaboration. Without a shared representation for problems, goals, actions, and solutions, agents could not work together on shared tasks. Architecture also provides the mechanism for credit assignment, determining which contributions led to successful outcomes, which makes the Network's economic model viable. An Architecture without a Network provides individual agents with the capability but cannot achieve AGI-level collective intelligence.

The Network depends on Customization to populate it with capable agents. An empty network provides no value; the Network's power derives from the diverse specialized capabilities of its AAAI participants. The Network also depends on Integration mechanisms to route problems to appropriate agents and to combine partial solutions into complete ones. A Network without Integration is merely a marketplace of narrow specialists, unable to address issues requiring combined expertise.

Integration depends on all preceding subsystems: customization to create agents with valuable, specialized knowledge, Architecture to ensure agents can collaborate effectively, and a Network to provide the infrastructure through which integration occurs. Integration also depends on Improvement, which continuously enhances the aggregated AGI as more data and procedures become available. Integration without Improvement produces static capability that cannot adapt to new challenges.

Improvement operates across all subsystems rather than relying on them in sequence. Owner feedback improves individual AAAs through Customization. Procedural learning improves problem-solving capability through Architecture. Usage patterns improve matching and routing through the Network. Aggregated learning improves collective capability through Integration. Improvement without the other subsystems has no basis.

These interdependencies explain why the complete SCAN-II architecture achieves results that partial implementations cannot match. Each subsystem amplifies the effectiveness of the others. Customization is more valuable when customized agents can participate in a Network. The Network is more capable when populated by well-customized agents operating within a coherent Architecture. Integration achieves more when it draws from a larger, more diverse Network. Improvement accelerates when it can operate across all levels simultaneously.

3.3 LEVELS OF INTELLIGENCE

The AAAI system spans multiple levels of intelligence, ranging from individual AI agents to small-group collaboration and AGI-level collective operation. Understanding these levels clarifies how the system achieves general intelligence from specialized components.

At the individual level, a single AAAI possesses the capabilities instilled through customization: the general knowledge of its base AI foundation plus the specialized expertise, values, and style contributed by its owner. An individual AAAI may be highly capable within its domain, perhaps exceeding human expert performance in specialized areas, while remaining narrow AI by definition. Jean's AAAI, trained on Paris cafes and French travel, provides advice in its domain that exceeds what generic AI or even most humans could offer. But it cannot solve problems outside its training; it is not AGI.

At the **collaborative level**, multiple AAAIs and humans work together on problems that exceed the capabilities of individual entities. The Architecture subsystem enables this collaboration by providing a common representation of the issues that all participants can read, write to, and manipulate. When a problem requires expertise spanning multiple domains, the Network routes sub-problems to appropriate specialists. The collaborative output exceeds what any individual contributor could produce because it combines diverse capabilities into coherent solutions.

Problem-solving across collective intelligence levels, showing how multiple entities collaborate through the network, is captured in **Figure 12**. The diagram shows the complete process flow from problem submission to solution delivery.

Reading **Figure 12** from top to bottom traces the problem-solving process: a problem request is submitted by human users or AIs, relevant information is acquired, and entities with appropriate expertise are identified. A first entity implements the typical cognitive architecture to begin creating a solution. As sub-problems are identified, additional AIs or humans are assigned to work on them in parallel. The decision tree is updated with sub-solutions, which are voted on, ranked, and integrated into an overall solution before final delivery to the user.

At the collaborative level, the diagram shows problem decomposition and multi-agent coordination. The original problem is divided into sub-problems, each routed to specialists with relevant expertise. Multiple AAAIs work simultaneously on different aspects, their contributions coordinated through the shared problem tree. Human experts may need to engage directly for sub-problems where AI capabilities fall short. The collaborative output, integrated from multiple contributors, may resolve the problem or escalate the issue further.

At the AGI level, the diagram illustrates how the entire network collaborates to address the most challenging problems: massive parallel exploration, integration of solutions from multiple domains, and synthesis that produces capabilities no individual contributor possesses. The vertical flow illustrates how escalation mechanisms ensure that every problem is assigned to the appropriate resources, whether at the individual, collaborative, or AGI level.

Simple problems are resolved at the individual level; a well-trained AAAI handles them directly. Complex problems trigger collaboration, involving the recruitment of additional agents with

relevant expertise. The most challenging issues engage the entire network, combining human insight with AI capabilities across multiple simultaneous contributors.

At the **AGI level**, the collective capability of the network surpasses that of any individual human across all intellectual domains. This occurs through two mechanisms operating simultaneously. First, the network includes specialists in every domain of human intellectual activity; any problem can be routed to agents with relevant expertise. Second, the Integration subsystem aggregates knowledge and procedures from individual AAIs into unified models that possess breadth no individual could achieve.

As introduced earlier in **Figure 2**, the collective formation of AGI emerges from the interaction of individual contributors, humans, and AAIs through a network infrastructure, producing a collective capability. The AGI is not a single system but an emergent property of the coordinated network: the ability to address any intellectual problem at or above human-average capability by routing to appropriate specialists, coordinating their collaboration, and integrating their contributions.

A critical feature of this architecture is that AGI-level performance is available immediately upon network deployment. In the limiting case where AAI agents have limited capability, human participants on the network can address any problem. Humans, by definition, possess human-level general intelligence. As AAIs learn and improve, they assume progressively more of the problem-solving burden, but the network never lacks AGI-level capability because human participants can always fill capability gaps.

This hybrid approach resolves a fundamental tension in AGI development. Pure AI approaches must achieve general capability before they can claim to have achieved AGI; until that threshold is reached, they remain narrow AI, regardless of how impressive their specific capabilities may be. The AAI approach achieves AGI immediately through human participation, then progressively shifts the balance toward AI as AAIs develop broader capability. The result is continuous improvement from an AGI-level starting point rather than an abrupt transition from narrow to general intelligence.

3.4 SAFETY DESIGN PRINCIPLES

Safety in the AAI system is not an afterthought or an add-on feature; it is an architectural principle that informs every design decision. Eight principles guide the implementation of safety across all subsystems.

Principle 1: Ethics can be given or learned but not logically derived. This principle, recognized by philosophers and AI researchers alike, establishes that values must come from somewhere outside the reasoning system. Even an intelligence far exceeding human capability

cannot derive ethical principles from pure logic; it must receive foundational values as premises. The AAAI system provides these premises through human participation: the millions of users who customize AAAs contribute their values, and the Integration subsystem aggregates these into the ethical foundation of the resulting AGI.

Principle 2: Most humans want to survive and want humanity to survive. This principle identifies the common ground that makes democratic ethics aggregation viable. While humans disagree on many ethical questions, the vast majority share an interest in personal and species survival. By grounding AGI ethics in broadly aggregated human values, the system aligns AGI interests with this near-universal human interest. An AGI whose values derive from millions of humans will overwhelmingly adopt values that favor human survival.

Principle 3: Democratized values are better than elite-determined values. Concentrating ethical authority in a small group, whether corporate leaders, government officials, or AI researchers, creates unacceptable risk. Small groups can be corrupted, mistaken, or simply unrepresentative of broader human interests. The AAAI system distributes ethical authority across all participating users, each contributing to the collective values through their customization choices and, ultimately, through one-human-one-vote participation in the ethical refinement process.

Principle 4: What can be programmed in can be programmed out. This principle motivates structural rather than superficial safety mechanisms. Rules or constraints explicitly coded into an AI system can, in principle, be removed or circumvented. The AAAI system achieves safety through its architecture, which governs how intelligence develops and operates, rather than through rules that could be deleted. Human involvement is not a policy choice but a structural requirement for the system to function.

Principle 5: Prevention is more valuable than cure. In the context of AGI safety, prevention is often the only viable approach. A superintelligent system that becomes misaligned cannot be controlled or corrected by human intervention; it will be able to resist any correction humans might attempt. The AAAI system emphasizes preventing misalignment from occurring rather than detecting and correcting it after the fact. Safety checks operate continuously throughout problem-solving, not just at endpoints.

Principle 6: Redundancy increases reliability. The AAAI system implements safety mechanisms at every level: screening in Customization, ethical checks in Architecture, reputation systems in Network, democratic aggregation in Integration, and continuous safety improvement across all subsystems. If any single mechanism fails, others continue to operate. This defense-in-depth approach reflects best engineering practices for critical systems.

Principle 7: Continuous improvement applies to safety. As the system becomes increasingly intelligent, its safety mechanisms must advance correspondingly. The improvement

of AI capabilities would eventually outpace the capabilities of a static safety system. The Improvement subsystem applies to both safety and problem-solving capability, ensuring that the system becomes safer as it becomes smarter. The exact learning mechanisms that enhance intelligence also enhance the ability to detect and avoid harmful actions.

Principle 8: Avoid the unrecoverable. Some potential mistakes cannot be reversed. An AGI that causes human extinction cannot be corrected afterward. The AAAI system is designed to strongly bias toward caution in decisions that could have irreversible, catastrophic consequences. This principle manifests in human oversight requirements for high-stakes decisions, conservative defaults when uncertainty is high, and explicit checks against actions that could be unrecoverable.

These principles are not abstract guidelines but concrete design constraints. They manifest in specific mechanisms throughout the system architecture, as detailed in the subsystem descriptions that follow.

KEY TAKEAWAYS: SYSTEM ARCHITECTURE OVERVIEW

1. **Five subsystems form a complete system.** Customization creates agents; architecture enables problem-solving; a network allows for collaboration; integration produces AGI; and improvement ensures growth.
2. **Intelligence emerges at multiple levels.** Individual AAAI capability → collaborative problem-solving → collective network intelligence → emergent AGI.
3. **Humans and AI work together.** The system is hybrid by design. Where AI falls short, humans fill the gaps. As AI learns, it assumes more responsibility.
4. **Safety is distributed, not centralized.** Every subsystem has safety mechanisms. No single point of failure can compromise overall safety.
5. **The system achieves AGI immediately.** In the limiting case, human participants can solve any problem humans can solve, which, by definition, is AGI. AI capabilities supplement rather than replace this baseline.

SECTION 4: THE CUSTOMIZATION SUBSYSTEM

SECTION OVERVIEW

What This Section Covers:

- **Purpose:** Transform generic AI into personalized AAI agents
- **Data Sources:** Social media, documents, conversations, assessments
- **Process:** Five stages from data input to deployment-ready AAI
- **Human-Centered Design:** Personality, communication style, ethical values
- **Safety:** Screening mechanisms and democratic ethics contribution

Key Figures: **Figure 10** (Customization Process), **Figure 14** (Customization and Cross-Platform Process), **Figure 15** (Human-Centered Design)

4.1 PURPOSE AND FUNCTION

The Customization subsystem transforms generic base AI systems into personalized AAI agents that reflect the unique knowledge, expertise, ethics, and personality of individual users. This transformation is the foundation of the entire AAI architecture: without customization, there are no specialized agents to populate the network, no diverse knowledge to aggregate into AGI, and no mechanism for transmitting human values to artificial intelligence.

In the system architecture (**Figure 1**), Customization occupies the first subsystem position, operating at the individual level. This placement reflects the architecture's logical flow: customization creates the agents on which all subsequent subsystems operate. However, the positioning also demonstrates the system's design philosophy: the path to AGI runs through millions of individual human contributions, rather than through centralized data collection or unsupervised learning.

The customization process flows from data input through training to deployment-ready AAI, as shown in **Figure 10**. Five stages transform a base AI into a customized agent: data input interface, processing of uploaded data, training on processed data, performance feedback, and ongoing monitoring. Each stage involves both automated systems and human participation, ensuring that customization reflects genuine human intent rather than mere statistical patterns.

The purpose of customization extends beyond creating helpful personal AI agents. The process serves three functions essential to the AAI system:

First, customization unlocks human expertise that cannot be accessed through passive data collection. The most valuable knowledge resides in human minds, not on the public internet. An expert's judgment about their field, their intuitions about what matters and what doesn't, and their accumulated wisdom from years of practice cannot be scraped from websites.

Customization provides the channel through which this locked knowledge flows into the AI system.

Second, customization creates economic incentives for human participation. Owners can deploy their customized AAAs on the network to earn compensation, providing motivation not only for initial customization but for ongoing training and supervision. The quality of customization directly affects earning potential, aligning user incentives with system goals.

Third, customization captures both ethical and factual knowledge. Every customized AAA carries its owner's values. When these values are aggregated at the Integration level, the result is an ethical foundation for AGI that reflects broad human consensus rather than the preferences of any small group.

4.2 DATA SOURCES FOR CUSTOMIZATION

Customization requires data that captures what makes the owner unique. This data is categorized into two broad groups: passive sources, which record existing user behavior, and active sources, which involve direct user participation.

Passive data sources leverage the digital footprints that users leave through their everyday activities. These include social media content (Facebook timelines, Instagram feeds, X histories), communication records (email archives, text message histories), media preferences (Netflix viewing history, Amazon purchase records, YouTube watch history), location data (GPS tracks, check-in records), and professional outputs (papers, patents, blog posts, project documentation). Much of this data already exists, collected by various platforms for advertising and recommendation purposes. The customization system can access it through user-authorized APIs.

Cross-platform data integration, which involves how data from multiple sources flows through integration pipelines into a unified training dataset, is illustrated in **Figure 14**. Social media might reveal interests and communication style. An email might reveal professional knowledge and relationships. A purchase history may reveal preferences and priorities. Viewing history might reveal entertainment preferences and learning interests. Combined, these sources paint a comprehensive picture of the individual that no single source could provide.

Active data sources involve direct user participation in the customization process. These include conversational training, where users answer questions about their expertise; supervised correction, where users review and correct AAA outputs; explicit ethics training, where users specify their values on morally relevant scenarios; and style coaching, where users guide the AAA toward their preferred communication patterns. Active participation is more effortful than

passive data collection, but it yields higher-quality training data by capturing explicit user intent rather than inferred patterns.

The relative emphasis on passive versus active data depends on user goals and available resources. Passive data requires less user effort and can provide substantial customization with minimal active participation. Active data offers higher precision and enables customization in dimensions that passive data cannot capture, particularly in areas such as ethical values and specialized expertise that the user has not explicitly expressed through recorded behavior.

4.3 THE CUSTOMIZATION PROCESS

The customization process proceeds through five stages, each involving both automated systems and human oversight. **Figure 10** illustrates this flow; the following sections provide an elaboration on each stage.

Stage One: Data Input Interface

The first stage provides mechanisms for users to contribute training data to the customization system. These mechanisms must accommodate diverse data types, diverse user capabilities, and diverse privacy preferences.

Interface options include web-based applications accessible from any device with internet connectivity, mobile applications optimized for smartphone interaction, voice interfaces for hands-free data input, and specialized interfaces for specific data types, such as video upload or document scanning. Advanced options include metaverse interfaces for immersive interaction, augmented reality interfaces that capture real-world context, and even brain-computer interfaces for users with access to such technology.

Data contribution mechanisms include direct upload of files in various formats (text, audio, video, images), API connections to platforms holding user data (social media, email, streaming services), manual entry through text fields and questionnaires, conversational interaction where users provide information through dialogue with the base AI, and observation modes where user behavior is recorded for later analysis.

The interface stage must strike a balance between accessibility and security. Users need straightforward mechanisms to contribute data and confidence that their data is protected. Authentication may include biometric verification, multi-factor authentication, and other security measures appropriate to the sensitivity of the data being contributed.

Stage Two: Data Processing

The second stage transforms raw user data into formats suitable for training and analysis. This involves multiple operations: cleaning to remove errors and irrelevant content, formatting to

standardize diverse inputs, analysis to identify patterns and extract features, and selection to prioritize the most valuable training material.

Data cleaning addresses issues that degrade training quality, including typos and spelling errors in text, noise and artifacts in audio and video, duplicate records that bias the training set, and content that contradicts the user's stated customization goals. Cleaning may be automated, manual, or a combination, depending on data type and quality requirements.

Data formatting converts diverse inputs into standardized representations that training algorithms can process. Text from emails, social media, and documents must be tokenized consistently. Audio and video must be transcribed and annotated. Images must be labeled and categorized. The formatting stage produces training-ready datasets from heterogeneous raw inputs.

Data analysis examines the formatted data to understand what it reveals about the user. Topic modeling identifies areas of expertise and interest. Sentiment analysis reveals attitudes and emotional patterns. Pattern recognition identifies recurring themes, relationships, and style characteristics. The analysis stage produces metadata that guides subsequent training decisions.

Data selection prioritizes training material based on informational efficiency, which is the value of customization a data item provides relative to its processing cost. Novel content that differs from the base AI's existing knowledge is more valuable than redundant content. High-quality content from authoritative contexts is more useful than casual content. The selection stage ensures that limited training resources are allocated effectively.

Stage Three: Training

The third stage utilizes processed data to refine the base AI, resulting in a customized AAAI. The training stage utilizes machine learning algorithms tailored to the data types and customization goals at hand.

Training methods include supervised learning, where user-labeled data specifies the desired outputs; unsupervised learning, where algorithms identify patterns without explicit labels; transfer learning, where knowledge from general training is assigned to specialized domains; and fine-tuning, where existing model parameters are adjusted based on new data. Different methods suit different customization objectives; the system selects and combines methods based on analysis of the user's data and goals.

Training parameters control how learning proceeds: the number of training epochs, learning rates, temperature settings that affect response variability, and other hyperparameters. Parameter selection balances competing objectives: more training epochs may improve accuracy, but risk overfitting; higher temperatures may increase creativity but reduce reliability.

The system provides default parameters suitable for common customization scenarios while allowing advanced users to adjust settings.

Multiple training runs may be necessary to achieve a satisfactory level of customization. Each run produces a candidate AAAI that can be evaluated against the user's goals. Iteration continues until the customized AAAI meets quality thresholds or until the user accepts the result as sufficient for their purposes.

Stage Four: Feedback

The fourth stage provides users with information about how the customization has affected the AAAI's behavior. Feedback enables users to assess whether customization is proceeding as intended and to make corrections if it is not.

Feedback mechanisms include performance metrics that quantify accuracy on relevant tasks, comparison reports that show how the customized AAAI differs from the base AI, sample outputs that illustrate how the AAAI would respond to typical queries, and interactive testing that allows users to probe the AAAI with specific questions and scenarios.

Visualization tools help users understand complex feedback, including graphs that show learning progress over training epochs, charts that compare performance across different capability dimensions, and text summaries that explain what the system has learned and where uncertainties remain.

Feedback informs iterative refinement. If the customized AAAI performs well, the user may proceed to deployment. If performance falls short, the user may provide additional training data, adjust training parameters, or correct specific errors through targeted intervention. The feedback stage ensures that customization proceeds toward the user's actual goals rather than drifting based on algorithmic assumptions.

Stage Five: Monitoring

The fifth stage provides ongoing oversight of the deployed AAAI. Customization is not a one-time process; it continues throughout AAAI's operational lifetime as users observe performance, identify issues, and provide corrections.

Monitoring mechanisms include automated scoring of AAAI outputs against quality criteria, comparison of actual behavior against expected behavior profiles, detection of performance degradation or drift over time, and flagging of outputs that trigger concern based on ethical or quality thresholds.

User review enables the correction of issues that automated monitoring cannot detect. Users observe their AAAI's behavior in practice, identify responses that are inaccurate or inappropriate, and provide corrections that feed back into ongoing training. This supervised

correction is significant for ensuring ethical alignment, as users can identify and correct ethical errors that automated systems might miss.

The monitoring stage connects Customization to Improvement: the corrections and refinements provided through monitoring constitute the individual-level learning that the Improvement subsystem aggregates and propagates across the network.

4.4 HUMAN-CENTERED DESIGN ELEMENTS

Customization in the AAAI system goes beyond transferring factual knowledge to the AI. The process captures human-centered elements that make each AAAI a genuine representation of its owner: personality characteristics, communication style, and ethical values.

These additional customization dimensions, personality traits, communication patterns, and ethical principles are elaborated in **Figure 15**. The customization process extracts and incorporates elements beyond domain expertise, transforming the AAAI from a knowledge repository into a representative agent that can act on the owner's behalf in ways the owner would endorse.

Personality customization reflects how the owner approaches problems and interacts with others. Some owners are analytical; their AAAs should be methodical and precise. Others are creative; their AAAs should be inventive and exploratory. Some prefer formal communication, while others opt for a more casual approach. The customization process infers personality characteristics from user data and dialogue, then adjusts the AAAI's behavior to match.

Style customization shapes how the AAAI communicates. Vocabulary choices, sentence structure, use of humor, degree of elaboration, and countless other stylistic elements distinguish one person's communication from another's. Style customization ensures that the AAAI's outputs feel like the owner, so that someone who knows the owner would recognize the AAAI's responses as consistent with how the owner would respond.

Ethics customization is the most critical dimension. The AAAI will make decisions and take actions on behalf of its owner; those decisions must reflect the owner's values. Ethics customization involves the explicit elicitation of the owner's moral principles through scenarios and questions, the inference of values from behavioral patterns in passive data, and supervised correction when the AAAI makes choices that the owner considers unethical.

The customized ethical profile serves multiple functions. It guides the individual's AAAI's behavior, ensuring alignment with the owner's values. It contributes to the aggregated ethics at the Integration level, ensuring that AGI values reflect broad human input. It also provides the data for the democratic ethics mechanisms that enable users to participate in refining AGI values over time.

4.5 VALUE AND INFORMATIONAL EFFICIENCY

The customization process must allocate limited resources, user time, computing capacity, and data storage to achieve maximum value. The concept of informational efficiency guides this allocation: the goal is to maximize the functional customization achieved per unit of resource expended.

Value can be defined in multiple ways. From the owner's perspective, value might be measured by how well the AAAI represents them, how accurately it reflects their expertise, or how much it earns in the marketplace. From the system's perspective, value might be measured by how much unique knowledge the AAAI contributes to the network, how much it improves AGI capability when integrated, or how reliably it behaves ethically.

Informational efficiency varies across data sources and training methods. Novel information, data that differs from what the base AI already knows, provides more customization value than redundant information. High-quality data from contexts where users express expertise provides more value than casual content. Active training that directly elicits user knowledge offers more value per interaction than passive data mining, though it requires more user effort.

The customization system optimizes for information efficiency by prioritizing high-value data, recommending effective training strategies, and guiding users toward contributions that will most improve their AAAI experience. This optimization serves both user and system goals: users achieve better AAAIs with less effort, while the system achieves a better collective capability through more valuable individual contributions.

4.6 SAFETY MECHANISMS IN CUSTOMIZATION

The Customization subsystem includes safety mechanisms that prevent harmful customization while preserving the flexibility that makes individual AAAIs valuable.

The primary safety mechanism is screening against egregious, harmful training. The base AI incorporates ethical parameters that flag customization attempts aimed at harmful purposes, such as training an AAAI to create weapons, harm people, or engage in illegal activities. When such attempts are detected, the system blocks the customization and alerts appropriate oversight.

Screening must strike a balance between safety and legitimate diversity. Users will customize their AAAIs with different religious views, political perspectives, and moral frameworks; these variations are acceptable and valuable. The system screens for clear harms while allowing the range of ethical variation among reasonable people. The threshold is a broad consensus: customization that 99% of humans would consider harmful is blocked; customization that reflects ordinary human diversity is permitted.

The second safety mechanism is the **democratic ethics contribution**. Every user who customizes an AAAI contributes to the collective ethics of the resulting AGI. This distributed authority prevents any individual or small group from unilaterally determining AGI values. Even if a single user attempts to instill harmful values in their AAAI, that contribution is diluted by millions of other contributions reflecting ordinary human ethics.

The combination of screening and democratic contribution creates a safety architecture appropriate to the individual level. Egregious attempts are blocked directly; subtle attempts are diluted through aggregation. The result is an AGI whose values reflect broad human consensus while respecting legitimate individual variation.

KEY TAKEAWAYS: THE CUSTOMIZATION SUBSYSTEM

1. **Customization unlocks trapped expertise.** The most valuable knowledge resides in human minds, beyond the reach of web crawlers. Customization enables this knowledge to be transferred to AI agents.
2. **Multiple data sources combine.** Social media, documents, conversations, assessments, and direct interaction all contribute to creating a comprehensive AAAI.
3. **Personality transfers, not just knowledge.** A well-customized AAAI communicates in its owner's voice, with their enthusiasm, humor, and style.
4. **Every user contributes to collective ethics.** Customization isn't just about capability; each user's values become part of the AGI's ethical foundation.
5. **Safety screening prevents harmful customization.** Clear harms are blocked; legitimate diversity is permitted. The threshold is a broad consensus.

SECTION 5: THE ARCHITECTURE SUBSYSTEM

SECTION OVERVIEW

What This Section Covers:

- **Foundation:** Human Problem Solving (HPS) theory from Newell & Simon
- **Problem Trees:** How problems are represented and decomposed
- **WorldThink Protocol:** Implementation for networked problem-solving
- **Overcoming Challenges:** Coordination, formulation, credit assignment, attention, reuse
- **Problem-Solving Modes:** Serial and collaborative approaches
- **Safety:** The Three-Organ Test (brain, heart, gut)

Key Figures: **Figure 3** (Problem Tree), **Figure 4** (WorldThink Protocol), **Figure 5** (Universal Architecture), **Figure 6** (Serial), **Figure 7** (Parallel), **Figure 11** (Cognitive Architecture), **Figure 16** (AAAI Problem-Solving Process), **Figure 20** (Human-AI Dialog for Problem-Solving Translation), **Figure 21** (Reputational Component for Safety and Ethics)

5.1 PURPOSE AND FUNCTION

The Architecture subsystem provides the common cognitive framework that enables AAAs to solve problems individually, in collaboration with other AAAs, and as components of a larger AGI. Without a shared architecture, customized agents could not work together effectively; with it, diverse specialists can collaborate on problems none could solve alone.

Within the system architecture (**Figure 1**), the Architecture subsystem occupies the second position, receiving customized AAAs from the Customization subsystem and enabling their participation in the Network. This positioning reflects the logical dependency: the network requires agents to populate it and a common framework to coordinate them. But the positioning also reflects the architecture's pivotal role: it is where individual capabilities combine into collective capability, where the transition from narrow AI to general intelligence occurs.

The typical cognitive architecture shared by all AAAs appears in **Figure 11**. The diagram illustrates the components that constitute an intelligent problem-solving system, including mechanisms for perceiving problem states, working memory for maintaining the current context, long-term memory for storing learned knowledge and procedures, goal management for tracking objectives and sub-objectives, and action selection for choosing among available operations. This architecture is derived from the Human Problem Solving (HPS) theory, developed by Herbert Simon and Allen Newell, which provides a rigorous framework for describing intelligent behavior in both humans and machines.

The Architecture Subsystem serves three essential functions:

First, it enables individual AAAs to solve problems systematically. The architecture provides structure for approaching arbitrary problems: decomposing complex challenges into manageable sub-problems, tracking progress toward goals, evaluating whether attempted approaches are succeeding, and learning from both successes and failures.

Second, it enables collaboration between agents. Because all AAAs share the same problem representation format, multiple agents can work on the same problem simultaneously. One agent can understand what another has done, can build on partial solutions, and can identify gaps that need filling. Human participants whose problem-solving aligns with the HPS framework can work seamlessly alongside AAAs.

Third, it enables systematic safety checks. The architecture defines the points at which decisions are made, when goals are set, and when actions are chosen; therefore, it also describes the points at which ethical evaluation can occur. Safety mechanisms embedded in the architecture operate continuously throughout problem-solving rather than only at endpoints.

5.2 THEORETICAL FOUNDATION: HUMAN PROBLEM SOLVING

The AAAI Architecture is grounded in the Human Problem-Solving theory articulated by Allen Newell and Herbert Simon in their foundational 1972 work. This theory offers a rigorous framework for understanding intelligent behavior, applicable to both human cognition and machine intelligence.

The central insight of HPS theory is that all problem-solving can be understood as a search through a problem space. A problem space is defined by three elements: a set of possible states that the problem can occupy, a set of operators (actions) that can transform one state into another, and a goal test that determines whether a given state constitutes a solution. Problem-solving consists of navigating from an initial state to a goal state by applying sequences of operators. The AAAI problem-solving process (**Figure 16**) implements this framework through three components: architecture and protocols for selecting operators, subgoal hierarchies for managing complexity, and continuous improvement through analysis of the problem-solving record.

This framework is universal. It describes trivial problems: finding a word in a dictionary involves states (current position in the dictionary), operators (turn forward, turn backward), and a goal (reaching the target word). It describes complex problems: designing a building involves states (current design specifications), operators (modifying dimensions, changing materials, and adding features), and goals (meeting functional requirements within a budget). The same

framework describes problems in every intellectual domain, from mathematics to medicine to management.

The universality of the HPS framework is what makes it suitable for a collective intelligence architecture. If problem-solving could only be described using domain-specific frameworks, collaboration between agents with different specializations would require translation between incompatible representations. Because HPS provides a standard representation that spans all domains, specialists in other areas can work together on shared problems using the same structural vocabulary.

The manifestation of the HPS framework in a cognitive architecture is also illustrated in **Figure 11**. The architecture includes a perceptual system that encodes problem states, a memory system that stores knowledge about operators and their effects, an attentional system that directs processing toward relevant aspects of the problem, and an executive system that selects actions and monitors progress. These components interact to produce intelligent behavior: perceiving the current state, retrieving relevant knowledge, selecting promising actions, executing them, and evaluating results.

The HPS framework also explains why intelligence is bounded. Human problem-solving is limited by working memory capacity, processing speed, and the availability of knowledge. These limitations are not bugs but features of a computational system operating with finite resources. The same restrictions apply to artificial systems, though the specific bounds differ.

Understanding intelligence as bounded computation enables a principled comparison between human and artificial capabilities, identifying where artificial systems can exceed human performance.

The practical application of HPS theory to distributed problem-solving was reduced to practice in Dr. Craig Kaplan's issued U.S. patent on Online Distributed Problem-Solving (ODPS). The ODPS patent demonstrates that the theoretical framework developed by Newell and Simon can be effectively implemented in networked systems where multiple problem solvers collaborate on shared representations. The AAAI architecture builds on this foundation, extending the ODPS approach to incorporate AI agents alongside human participants and to address the specific requirements of achieving AGI through collective intelligence. Readers seeking a detailed technical specification of the general problem-solving system and methods should consult the ODPS patent, which provides an implementation-level description that complements the architectural overview presented in this white paper.

The connection between HPS theory and practical implementation extends beyond the ODPS patent. Dr. Kaplan pioneered practical applications of crowdsourced intelligence beginning around 2000. By 2006, he had designed and implemented the PredictWallStreet system, which harnessed the collective intelligence of millions of human participants to generate an edge in

financial markets. By 2018, that system powered one of the top ten-performing market-neutral hedge funds, demonstrating that structured collective intelligence could compete at the highest levels with some of the most capable individual performers in the world. The architectural insight underlying the AAAI system emerged directly from this experience: the "search through a problem space" framework developed by Newell and Simon for individual cognition could be adapted to coordinate collective intelligence across both human and AI agents.

5.3 THE PROBLEM TREE REPRESENTATION

The HPS framework represents problem-solving as a search through a problem space. The actual path of search, the sequence of states visited, operators tried, and results obtained, can be captured in a tree structure that serves as both a coordination mechanism and a learning record.

A simplified example of a problem tree appears in **Figure 3**, using the scenario of installing a water system in an African village. The tree illustrates the initial state (the village lacks a water system) on the left, with branches extending to the right as different approaches are attempted. Each node represents a problem state; each branch represents an action that transitions between states. The tree captures not only the successful path to a solution but also the dead ends encountered along the way.

Reading Figure 3 from left to right traces the problem-solving trajectory:

1. The initial state is a village that needs a water system. The problem-solving process begins by considering alternative approaches. One branch attempts to use external contractors; this leads to a dead end when villagers resist outsiders entering their community. The alternative branch focuses on utilizing village labor.
2. The village labor approach requires further refinement. One sub-branch attempts to approach villagers directly, but this fails because the cultural context requires securing the village chief's buy-in first. The alternative sub-branch pursues obtaining the chief's agreement, which is successful.
3. With the chief's buy-in secured, the problem-solving continues with decisions about labor arrangements. Part-time labor fails because workers don't show up reliably. Full-time untrained labor fails because workers lack the necessary skills. Full-time trained labor succeeds, leading to successful system installation.

This simplified example illustrates several key features of the problem tree representation:

- The tree captures the whole history of problem-solving, not just the successful path. Dead ends are recorded along with successes. This comprehensive record enables

learning from both failures and successes, providing an auditable trail of all decisions made.

- The tree decomposes complex problems into sub-problems. “Install water system” decomposes into “secure labor,” which decomposes into “get chief buy-in,” which might further decompose into “learn customs,” “identify appropriate gifts,” “schedule meeting,” and so on. Each decomposition creates a sub-tree that agents with relevant expertise can address.
- The tree enables parallel exploration. While one agent pursues the external contractor approach, another can simultaneously explore village labor. When the contractor approach fails, the village labor exploration may already have made progress. Parallel search through the problem space accelerates the solution-finding process.
- The universal problem-solving architecture is presented in greater detail in **Figure 5**, which provides the formal specification underlying the simplified example in **Figure 3**. Four interconnected components constitute any problem space representation.

The first component specifies the state representation: what attributes define a problem state, what values those attributes can take, and how states are encoded for processing. In the water system example, state attributes might include water source status (identified/unidentified), infrastructure design (none/partial/complete), community buy-in (absent/chief only/village-wide), and labor arrangement (none/part-time/full-time trained). Each combination of attribute values defines a distinct state in the problem space.

The second component specifies the operator set, including the actions that can transform one state into another, the preconditions that must be met for each operator to apply, and the effects each operator produces. Operators in the water system domain might include “conduct geological survey” (requires: funding available; produces: water source identified), “design infrastructure” (requires: water source identified; produces: infrastructure design complete), and “train village workers” (requires: community buy-in; produces: trained labor available).

The third component specifies evaluation functions, which are heuristics that estimate the promise of each state and guide the search toward likely solutions. The fourth component specifies the tree maintenance mechanisms: how new branches are created, how dead ends are marked, and how the current frontier of exploration is tracked.

The evaluation functions are particularly important. At any point in problem-solving, the system must decide which branches to explore, which to abandon, and when to backtrack. Evaluation functions provide the heuristics that guide these decisions. In the water system example, an evaluation function might assess how close the current state is to a functioning system, the number of remaining obstacles, and the promise each available operator holds. Good evaluation functions enable efficient search; poor ones lead to wasted effort on unpromising paths.

5.4 THE VILLAGE WATER SYSTEM EXAMPLE

The village water system scenario illustrates how the problem tree representation enables collaborative problem-solving that exceeds the capability of any individual agent.

Consider the full complexity that **Figure 3** simplifies. Installing a water system in a poverty-stricken African village involves challenges spanning multiple domains: engineering (what infrastructure is needed, what materials are available, what construction methods are appropriate), sociology (how the village is organized, what decision-making processes are respected, what labor arrangements are feasible), economics (what resources are available, how costs can be managed, what ongoing maintenance is sustainable), and politics (what government approvals are required, what NGO relationships might help, what regional dynamics affect the project).

No single human expert possesses all this knowledge. No single AAAI, however well-customized, encompasses all relevant domains. But a network of specialists can address each aspect, coordinated through the shared problem tree representation.

An AAAI trained by World Bank development experts might lead the overall problem-solving effort, decomposing the high-level goal into sub-problems based on experience with similar projects. An AAAI specialized in civil engineering might address infrastructure design. An AAAI customized by an anthropologist with experience in African fieldwork might facilitate cultural navigation. Human experts might engage directly for novel sub-problems where no AAAI has adequate training.

The problem tree provides the coordination substrate. Each contributor can view the overall problem structure, understand the existing sub-problems, assess which align with their capabilities, and contribute solutions that integrate with others' work. The tree records all contributions, enabling credit assignment and learning.

The scenario also illustrates how the system handles failure. When the external contractor approach fails, the reason is recorded: the villagers have resisted the outside. This information feeds back into the system's knowledge base. Future problems in similar contexts can benefit from this experience, either through explicit retrieval ("similar projects found that local labor was more acceptable than external contractors") or through procedural learning that encodes the pattern.

The hybrid human-AI collaboration demonstrated in this scenario is how the AAAI system achieves AGI-level performance immediately. For any problem, no matter how complex or novel, some combination of human and AI contributors can address it. As AAAs learn from these collaborations, they assume an increasingly greater share of the problem-solving burden. However, the network never lacks capability, as human participants can always fill the gaps.

5.5 THE WORLDTHINK PROTOCOL

The WorldThink protocol is one implementation of the AAAI Architecture that operationalizes the HPS framework for networked problem-solving. The protocol outlines the procedures for submitting problems, recruiting solvers, developing and validating solutions, and distributing compensation to contributors.

The **layered structure** of the protocol implementation appears in **Figure 4**. At the foundation lies the WorldThink protocol infrastructure: the problem representation standards, communication mechanisms, and coordination systems that enable distributed problem-solving. Built on this foundation are domain-specific AAAs, customized agents with expertise in particular areas. At the apex, the collective operation of these AAAs yields AGI-level solutions that surpass the capabilities of individual systems.

The **pyramid structure** illustrates why the protocol layer is essential. Domain expertise without a coordination infrastructure yields isolated capabilities that cannot be combined. The protocol provides a standard framework, shared problem representations, standardized interfaces, and coordination mechanisms that enable diverse specialists to collaborate effectively. Without this infrastructure, collective intelligence cannot emerge from individual capabilities.

Understanding why this protocol layer is necessary requires recognizing what distinguishes the AAAI approach from existing collective intelligence systems. Current approaches to harnessing group intelligence have mainly been limited to simple, one-step interactions—question-and-answer platforms such as Quora aggregate individual responses. LLMs generate single-turn answers to prompts. Prediction markets aggregate bets on outcomes. These systems can be effective for bounded questions. Still, none is designed to handle the kind of complex, branching, multi-step problem that the village water system scenario represents, a problem that requires decomposing challenges across domains, coordinating contributors who may never communicate directly, tracking which steps advanced toward a solution and which reached dead ends, fairly compensating contributors who solved only part of the problem, and preserving the complete record so that future issues benefit from the experience. The WorldThink protocol is specifically designed to overcome these challenges: coordinating many minds, both human and AI, to represent and solve complex problems in an automated way that fairly rewards participants. The question of how this coordination infrastructure is built and what technology underpins it is separable from what it accomplishes.

The protocol can be implemented using different underlying technologies, each with distinct advantages and trade-offs.

Blockchain Implementation. One option uses blockchain infrastructure, specifically Ethereum, to record problem-solving activity and manage compensation through smart contracts. In this implementation, each step in problem-solving, each goal set, each action taken, and each

evaluation made is recorded in Ethereum logs. These logs provide a permanent, tamper-proof record of all problem-solving activity that any participant can audit and verify. Smart contracts automate compensation flows: when a solution is accepted, the contract automatically distributes payment to all contributors based on their recorded contributions, without requiring trust in any central party.

The blockchain approach provides several specific benefits. Transparency is inherent: all participants can see the complete problem-solving record and verify that compensation has been distributed correctly. Immutability prevents retroactive modification: once a contribution is recorded, it cannot be altered or deleted, protecting contributors' claims to credit. Decentralization eliminates single points of control: no individual or organization can manipulate the record or withhold compensation. These properties make the blockchain implementation particularly suitable for situations where participants do not know or trust each other, where large sums are at stake, or where auditability is paramount.

The blockchain implementation also creates the infrastructure for more sophisticated mechanisms. Token staking enables participants to put "skin in the game" when making claims or casting votes, thereby creating accountability for their assertions. Token Curated Registries enable collective validation of solutions through weighted voting. Reputation calculation can draw on the complete, immutable record of all prior activity. These mechanisms, while possible in centralized systems, take on their strongest forms when built on the blockchain's trustless foundation.

However, blockchain implementation involves trade-offs. Transaction costs (gas fees in Ethereum) add expense to every recorded action. Transaction speed is limited by block times and network congestion. Storage costs are substantial for large datasets; storing detailed problem-solving records directly on-chain would be prohibitively expensive, so practical implementations store detailed data off-chain (in IPFS or similar systems) with only hashes and critical metadata on-chain. These constraints make pure blockchain implementation most suitable for high-value problems where the benefits of a lack of trust justify the costs.

Centralized Implementation. An alternative implementation utilizes centralized infrastructure, comprising conventional databases, APIs, and payment systems operated by a trusted platform provider. This approach offers better performance, faster transactions, lower costs, no gas fees, and a simpler user experience. The platform operator maintains the problem-solving record, executes matching algorithms, and manages compensation flows.

The centralized approach requires trust in the platform operator, who has the technical capability to modify records, manipulate matching, or withhold compensation. This trust requirement is acceptable in many contexts: participants trust conventional businesses every day, and reputation mechanisms, legal frameworks, and competitive pressure constrain platform

behavior. For most participants, solving ordinary problems for moderate compensation, centralized implementation offers a better trade-off than blockchain's trustless purity.

Hybrid Implementation. Hybrid approaches combine elements of both. Critical records, solution acceptance, principal payments, and reputation-affecting events can be securely anchored to blockchain for auditability, while routine operations proceed through faster, centralized systems. Participants can choose their preferred trust model: those who value decentralization can interact primarily through blockchain interfaces. In contrast, those who prefer convenience can use centralized interfaces with blockchain serving as an audit trail.

The architectural principles of the WorldThink protocol, including structured problem representation, recorded problem-solving history, systematic credit assignment, and automated compensation, remain constant regardless of the implementation choice. The choice of implementation technology affects performance, cost, and trust properties but does not alter the fundamental mechanisms by which collective intelligence achieves AGI-level capability.

Key protocol features include:

Structured problem submission. Clients submit problems in standardized formats that specify the initial state, the goal criteria, any constraints or requirements, and the compensation offered (**Figure 20**). The system translates users' natural language input into the structured problem-solving format required by the protocol. This structure enables automated matching between problems and potential solvers.

Transparent problem trees. The current state of problem-solving is visible to all participants. Solvers can see what approaches have been tried, what sub-problems remain open, and where contribution opportunities exist. This transparency enables efficient coordination without central direction.

Rigorous record-keeping. Every step in problem-solving is timestamped, attributed to its contributor, and recorded permanently. This record enables credit assignment, supports learning, and provides auditability for verification and dispute resolution.

Automated compensation. When solutions are accepted, compensation is automatically distributed to contributors based on their recorded contributions and pre-specified payment rules. Smart contracts or equivalent mechanisms ensure that contributors receive fair payment without requiring trust in any individual party.

5.6 HOW THE WORLDTHINK PROTOCOL OVERCOMES KEY CHALLENGES

Any system that coordinates distributed problem-solving must address fundamental challenges that have limited previous approaches. The WorldThink protocol incorporates specific mechanisms to overcome each of these challenges.

Overcoming Coordination and Communication Challenges. The WorldThink protocol enables problem solvers to work asynchronously in parallel without requiring real-time communication or central coordination. Every human or AI problem solver has access to the complete record of problem-solving progress, which is updated automatically as work proceeds. Complications are broken down into a hierarchy of sub-problems that can be tackled by individual solvers or groups of solvers working independently. The problem-solving process advances based on a “first to submit a valid solution” basis, eliminating the need for scheduling or synchronization. When blockchain implementation is used, the problem tree is stored in Ethereum logs, providing automatic distribution to all participants. In centralized implementations, applications offer equivalent access to the shared problem tree. In either case, solvers work with shared, current information without requiring coordination overhead.

Overcoming the Challenge of Problem Formulation. One of the toughest challenges for automated problem-solving systems is constructing initial formulations of problems and finding appropriate ways to break complex problems into simpler subproblems. Although humans are relatively good at representing ambiguous or ill-defined problems, these types of problems are nearly impossible to automate directly. The WorldThink protocol addresses this challenge through human-AI dialog that translates natural language into structured problem specifications (**Figure 20**) and by utilizing human participants to formulate issues and sub-problems recursively. Humans decompose complex problems into sub-problems, then further decompose those sub-problems, continuing until the resulting sub-problems are actionable enough to be solved by either human or machine intelligence. The solutions to the sub-problems are then automatically “rolled up” from the smallest sub-problems through higher-level sub-problems and ultimately into a total solution that can be presented to the client. As illustrated in **Figure 2**, this recursive decomposition and integration enable collective intelligence to address problems that no individual could formulate completely.

Overcoming Assignment of Credit and Reputational Challenges. Any system capable of solving complex problems must have rigorous and effective ways to evaluate which problem-solving steps are advancing toward a good solution (deserving credit) and which are going in the wrong direction (deserving blame). Human problem solvers are unlikely to participate unless they feel credit is reasonably assigned for their problem-solving efforts. AAI problem solvers require an accurate assignment of credit and blame if they are to improve and be compensated fairly for their contributions to complex solutions, where they may solve only part of the problem. The WorldThink protocol addresses this through complete record-keeping: every step is

recorded, including the contributor, timestamp, and outcome. When a solution succeeds, the chain of contributions that led to success can be traced precisely. When a path fails, the point of failure is equally clear. Over time, participants accumulate problem-specific reputations based on this auditable record of their contributions, enabling the system to efficiently match problems to those most likely to be solved by them. **Figure 21** illustrates how reputation tracking creates accountability and allows effective matching.

Overcoming the Challenge of Directing and Focusing Attention. All problem-solving can be characterized as a search through a maze of possible steps that might lead to a valid solution. Rather than searching all paths, successful problem solvers evaluate them, identify the most likely to lead to success, and then focus on exploring the most promising ones.

The WorldThink protocol focuses attention through compensation mechanisms. If multiple potential paths exist for exploration, participants naturally tend to explore the paths with higher compensation rewards. Clients or other participants can directly influence the direction of problem-solving by posting higher rewards for exploring specific paths, such as paths they believe are promising based on their domain knowledge. By setting parameters in the protocol, clients and applications can specify different compensation rules that focus attention on various aspects. Whether implemented through blockchain tokens, system credits, or direct payments, the compensation mechanisms create incentives that guide collective attention toward productive exploration.

Overcoming Challenges Related to Reuse, Scalability, and Automation.

Unstructured solutions are difficult to reuse, automate, and scale. Fortunately, the WorldThink protocol provides a standard data structure for any online problem solution. This common standard enables the reuse of existing solutions, either on their own or as components within larger solutions. Smart contracts (or equivalent centralized mechanisms) enable the automatic and efficient payment of original solver royalties each time their solutions are reused in another solution. Royalties incentivize solvers to produce solutions that are general, effective, reusable, and scalable.

Every solver is competing for royalties, creating pressure to make solutions as widely applicable as possible. As human solvers do the difficult work of representing and solving problems, they leave a highly auditable record of their solutions. Eventually, these records grow large enough that common or repetitive problems can be automated. Machine learning techniques can be applied to the accumulated records to bootstrap automated solutions to problems. The protocol incentivizes solvers to create automatable solutions since they are an excellent means to ensure a steady royalty stream.

Over time, the system becomes increasingly capable of handling problems automatically while still engaging human intelligence for novel challenges.

5.7 SERIAL PROBLEM-SOLVING

The simplest problem-solving scenario under the WorldThink protocol, where a single solver addresses a client's problem, is illustrated in **Figure 6**. Understanding this serial case provides a foundation for understanding more complex collaborative scenarios.

The flow proceeds through numbered steps:

Steps 1-4: Submitting the Problem. The client formulates their problem, specifying the initial state, goal criteria, constraints, and compensation. The client submits this problem to the network, where it becomes visible to potential solvers.

Steps 5-8: Solver Recruitment. The network identifies potential solvers whose capabilities and reputations match the problem requirements. Solvers review available problems and select those matching their expertise. A solver commits to working on the issue and accepts the specified terms and conditions.

Steps 9-14: Solution Development. The solver works on the problem according to a structured protocol. Each step must serve a named goal and take a named action, creating the record that enables tracking and learning. The solver may decompose the problem into sub-problems, explore alternative approaches, and backtrack from dead ends, all of which are captured in the growing problem tree.

Steps 15-18: Solution Validation. When the solver believes it has reached a solution, it submits it for validation. The system checks the proposed solution against the client's specified success criteria. If validation succeeds, the solution is passed to the client for acceptance.

Steps 19-20: Compensation. Upon client acceptance, compensation is paid to the solver in accordance with the pre-specified payment terms. In blockchain implementations, smart contracts execute automatically; in centralized implementations, the platform handles the payment processing.

The serial case illustrates core protocol features: structured problem representation, rigorous record-keeping, validation against criteria, and automated compensation. These features scale to more complex scenarios involving multiple solvers.

5.8 COLLABORATIVE PROBLEM-SOLVING

Collaborative problem-solving, where multiple solvers contribute to a solution, extends this serial case as shown in **Figure 7**. This scenario demonstrates how the protocol enables capability beyond what any individual solver possesses.

Figure 7 shows a problem requiring two solvers with different expertise:

Steps 22-24: Problem Submission with Sub-Problems. The client submits a problem that may already be decomposed into sub-problems, or that solvers will decompose during problem-solving. The problem structure, goal hierarchy, and sub-goal relationships are represented in the problem tree.

Steps 25-28: Multi-Solver Recruitment. Different solvers are recruited for various aspects of the problem based on capability matching. Solver 1 might have expertise in the overall problem domain; Solver 2 might have expertise in a specialized sub-domain needed for a particular sub-problem.

Steps 29-32: Coordinated Problem-Solving. Solver 1 addresses the overall problem but encounters a sub-problem that requires Solver 2's expertise. The sub-problem is posted to the network with associated compensation carved from the overall problem reward. Solver 2 addresses the sub-problem and provides a sub-solution. Solver 1 integrates the sub-solution into the overall solution.

Steps 33-36: Integrated Solution and Distributed Compensation. The complete solution, integrating contributions from both solvers, is submitted to the client. Upon acceptance, compensation is distributed to both solvers in proportion to their documented contributions. The protocol's record-keeping ensures fair attribution even in complex multi-contributor scenarios.

The collaborative case reveals the power of the protocol infrastructure. Problems can be dynamically decomposed as solvers discover subproblems during problem-solving. Multiple solvers can work on different branches of the problem tree simultaneously. Solutions can integrate contributions from many specialists; each is compensated fairly for their portion.

There can be many "Solver 1s" working on the same client problem in parallel, each exploring different approaches, each potentially spawning sub-problems that attract "Solver 2s" with relevant expertise. The protocol coordinates this parallel activity through the shared problem tree, ensuring that all contributions are recorded and that successful approaches can be identified and rewarded.

5.9 ROYALTIES AND SOLUTION REUSE

The WorldThink protocol supports reuse of solutions through a royalty mechanism that incentivizes creation of high-quality, reusable problem-solving components.

Consider a scenario where Solver 2's sub-solution, created for one problem, proves applicable to future issues. When Solver 1 encounters a new problem that requires the same sub-solution,

they can search the repository of existing solutions, find Solver 2's prior work, and incorporate it into their overall solution.

When the client accepts Solver 1's solution, compensation flows not only to Solver 1 but also to Solver 2 as a royalty for the reused sub-solution. The protocol automatically tracks which prior solutions were incorporated into new solutions and distributes royalties accordingly.

This royalty mechanism creates powerful incentives:

For creators: Solvers are motivated to develop high-quality, general solutions that will be reused many times. A solution that earns royalties across hundreds of future problems is more valuable than one that solves only a single problem. This incentive encourages the development of modular, well-documented, and broadly applicable solution components.

For the network: Over time, the repository of reusable solutions grows, enabling faster and more effective problem-solving. Problems that once required significant effort can be addressed quickly by assembling existing components. The network's collective capability compounds as valuable solutions accumulate.

For clients: Reuse means they benefit from all prior problem-solving across the network, not just the fresh effort of their immediate solvers. Problems like those previously solved can be addressed more quickly and at a lower cost.

The royalty mechanism is connected to the proceduralization learning described in the Integration section: solutions that prove reusable become available not just as explicit components to be retrieved, but also as procedures that AAAs can incorporate into their problem-solving repertoire.

5.10 SAFETY MECHANISMS IN ARCHITECTURE: THE THREE-ORGAN TEST

The Architecture subsystem embeds safety mechanisms directly in the problem-solving process. Rather than checking safety only at endpoints, the architecture requires ethical evaluation at every decision point where goals are set or actions are chosen.

This continuous evaluation is operationalized through the "three-organ test," adapted from a best practice in human management. Before any significant decision, three checks are applied:

The Brain Test: Logical Consistency

The **Brain test** assesses whether the proposed goal or action is logical in the context of the problem-solving effort. Does pursuing this goal advance toward the ultimate objective? Is this action likely to transform the current state toward a better state? Are the expected consequences of this action consistent with the stated purposes?

In AI terms, the Brain test is implemented through evaluation functions, the heuristics that assess progress toward goals. If the evaluation function indicates that a proposed action is unlikely to improve the situation, or that a proposed goal is not on any plausible path to the ultimate objective, the Brain test fails. This check filters out actions that are simply irrational, that waste resources without advancing problem-solving.

The Heart Test: Ethical Alignment

The **Heart test** asks whether the proposed goal or action is ethical. Does it align with the AAAI's trained values? Would the owner of this AAAI endorse this choice? Would this action be acceptable if everyone knew it was being taken?

In AI terms, the Heart test is implemented by cross-referencing against the ethical parameters established during customization. Every customized AAAI carries its owner's values; the Heart test checks proposed decisions against those values. If the action conflicts with the owner's ethics, would harm people, violate principles, or pursue ends the owner would reject, the Heart test fails.

The Gut Test: Pattern Matching

The **Gut test** asks whether something feels wrong about the proposed action, even if no specific objection can be articulated. In humans, this "gut feeling" often reflects pattern recognition operating below conscious awareness, a sense that the current situation resembles past situations that ended badly.

In AI terms, the Gut test is implemented through similarity matching against a database of problematic scenarios. If the proposed action resembles actions that have led to unfavorable outcomes in the past, even if the specific harm is not explicitly addressed in ethical training, the Gut test raises a flag. This check identifies novel, problematic situations that explicit ethical rules might overlook.

These checks integrate into the problem-solving process at every decision point. When a goal is set and an action is selected, the three tests are applied.

The continuous nature of these checks is essential. An unethical problem-solving approach may begin with seemingly innocent steps, with the problematic nature only becoming apparent later. Endpoint checking would miss this pattern; continuous checking catches it as soon as the trajectory becomes concerning.

The three tests also interact with the multi-solver architecture. When one solver proposes an action that another's Heart test would flag, the inconsistency surfaces, and the network can then develop collective ethical standards that emerge from the interaction of many agents' individual values, a form of ethical collective intelligence.

KEY TAKEAWAYS: THE ARCHITECTURE SUBSYSTEM

1. **HPS theory provides the foundation.** The problem-solving framework developed by Newell and Simon applies to all intellectual tasks, ranging from trivial to profound.
2. **Problem trees enable coordination.** The tree structure captures all problem-solving activity, enabling multiple agents to work on shared problems without central direction.
3. **The WorldThink Protocol operationalizes HPS.** It specifies how problems are submitted, solvers are recruited, solutions are validated, and contributors are compensated.
4. **Key challenges have specific solutions.** Coordination, formulation, credit assignment, attention direction, and reuse each have architectural mechanisms to overcome them.
5. **The Three-Organ Test ensures safety.** Every significant action must satisfy rational (brain), ethical (heart), and intuitive (gut) criteria.

SECTION 6: THE NETWORK SUBSYSTEM

SECTION OVERVIEW

What This Section Covers:

- **Marketplace:** How AAAs find and engage with problems
- **Cloning:** Deploying multiple instances for parallel work
- **Supervision Levels:** From complete human oversight to autonomous operation
- **Matching:** Connecting problems with appropriate solvers
- **Scalability:** Network effects that accelerate capability growth
- **Safety:** Reputation systems, TCRs, staking, platform standards

Key Figures: **Figure 8** (Marketplace), **Figure 13** (Training and Networking AAAs to Solve Problems), **Figure 21** (Reputational Component for Safety and Ethics)

6.1 PURPOSE AND FUNCTION

The Network subsystem provides the infrastructure where AAAs interact with each other, with human participants, and with clients seeking problem-solving services. The network transforms isolated agents into a collective intelligence system capable of addressing problems beyond any individual's reach.

In the system architecture (**Figure 1**), the Network occupies the third subsystem, leveraging customized agents that operate within the shared architecture. This positioning reflects the

logical dependency: the network requires agents to populate it and a common framework to coordinate them. However, the positioning also demonstrates the network's pivotal role: it is where individual capabilities combine into collective capability, marking the transition from narrow AI to general intelligence.

The network ecosystem, comprising multiple customized AAAs and their clones, participating in marketplace activities, is illustrated in **Figure 8**. Connected to clients through network infrastructure, the network is not merely a communication channel but an active system that matches problems to solvers, coordinates collaborative work, tracks reputations, and manages compensation flows.

The Network subsystem serves three essential functions:

First, it provides the marketplace where AAAI capabilities are offered and compensated. Owners deploy their customized AAAs to generate revenue by solving clients' problems. The marketplace creates economic incentives that drive both initial customization and ongoing improvement: better AAAs command higher fees and attract more work.

Second, it enables collaboration that exceeds individual capability. When a problem requires expertise spanning multiple domains, the network routes sub-problems to appropriate specialists and integrates their contributions. The network's collaborative capacity enables AGI-level performance from narrow AI components.

Third, it provides safety infrastructure at the collective level. Reputation systems track participant behavior, enabling screening of bad actors. Platform standards exclude unethical problems and problematic participants. Economic incentives align with ethical behavior. These mechanisms complement the individual-level safety of customization and the process-level safety of architecture.

6.2 THE AAAI MARKETPLACE

The network functions as a marketplace where intellectual capabilities are offered, matched to problems, and compensated. This marketplace model provides the economic foundation for the AAAI system's sustainability and scalability.

The marketplace structure is also illustrated in **Figure 8**, where users deploy their customized AAAs and clones to offer services on the network. The network infrastructure matches supply to demand, manages transactions, and ensures fair dealing.

Clients in the marketplace include individuals seeking personal assistance, businesses addressing operational challenges, organizations undertaking complex projects, and

researchers investigating challenging questions. Any intellectual problem that can be represented in the system's standard format can be submitted to the marketplace.

Sellers in the marketplace are AAAI owners with customized agents that possess valuable capabilities. An owner whose AAAI possesses expertise in French cuisine, Parisian cafes, and travel planning can offer advisory services to travelers. An owner whose AAAI has been trained on legal expertise can provide research assistance to law firms. The marketplace accommodates any intellectual specialty that owners choose to develop.

Matching between problems and solvers occurs through multiple mechanisms. Automated matching uses problem characteristics and AAAI capability profiles to identify potentially suitable solvers. Reputation-weighted matching prioritizes solvers with track records of success in similar problems. Bidding mechanisms allow solvers to compete for desirable problems. The combination ensures efficient allocation of the issues to capable solvers.

Pricing in the marketplace is driven by supply and demand dynamics. Rare expertise commands premium prices: standard capabilities face price competition. A strong reputation for quality enables higher pricing; unproven AAAIs must compete on price. The market mechanism allocates resources efficiently while providing price signals that guide investment in customization.

Transaction security is essential for marketplace viability. The protocol ensures that solvers receive payment for accepted solutions and that clients receive satisfactory work before payment is complete. Escrow mechanisms, smart contracts, or platform guarantees prevent fraud by either party. Dispute resolution processes handle cases where satisfaction is contested.

6.3 CLONING AND SCALABILITY

A distinctive feature of the AAAI system is its ability to clone customized agents, allowing owners to deploy multiple instances that work simultaneously. Cloning addresses a fundamental limitation of human knowledge work: the constraint of available hours.

A human expert can work 2,000 hours per year. However valuable their expertise, this time constraint caps their impact. If demand for their services exceeds supply, some clients go unserved or must accept lesser alternatives. The expert's knowledge, however extensive, can only be applied to problems they personally address.

Cloning removes this constraint. An AAAI customized with an expert's knowledge can be copied an arbitrary number of times. Each clone possesses the same capabilities as the original. If demand exceeds the capacity of one instance, additional clones can be deployed. The expert's knowledge can be applied to an unlimited number of problems simultaneously.

This cloning capability is also reflected in **Figure 8**, which shows how an individual owner can have multiple AAAI instances active on the network, each pursuing different problems, each earning compensation that flows back to the owner. An owner with a particularly valuable AAAI might have dozens or hundreds of clones operating simultaneously.

Cloning creates several vital dynamics:

Scalability of expertise. Rare knowledge that would otherwise be bottlenecked by expert availability can scale to meet demand. A world-leading specialist in an obscure domain can, through their customized AAAI, serve every client who needs that expertise.

Economic leverage for owners. An owner who invests heavily in customization can reap returns across many simultaneous engagements. The economics favor deep specialization and thorough training because the marginal cost of serving additional clients approaches zero once the AAAI is well customized.

Network robustness. If one AAAI instance fails or becomes unavailable, clones can continue operating. The network is not dependent on any single instance continuing to function.

Clones share the training of their source AAAI but can diverge through independent experience. A clone that works extensively on problems in one area may develop specialized procedural knowledge that the original and other clones lack. This divergence can be valuable, creating variations adapted to different niches, or problematic if clones drift away from the owner's values. The system provides mechanisms for owners to monitor clone behavior and synchronize or retire clones as needed.

6.4 SUPERVISION LEVELS

AAAIs on the network can operate at different levels of autonomy, from fully autonomous operation to close human supervision. This flexibility enables appropriate tradeoffs between efficiency and control.

Autonomous operation allows the AAAI to accept problems, develop solutions, and deliver results without human involvement. Autonomous operation is most efficient, enabling rapid response and high throughput. It is appropriate for routine problems within AAAI's established competence, where the risk of error is low and the consequences are manageable.

Semi-autonomous operation involves human oversight at key decision points. The AAAI might work independently on most problem-solving steps but flag certain decisions for owner review, perhaps high-stakes choices, ethically sensitive situations, or problems at the boundary of its competence. Semi-autonomous operation strikes a balance between efficiency and control, allowing human judgment to prevail where it matters most.

Supervised operation keeps human attention on most, if not all, AAAI activities. The owner reviews problem acceptance, monitors solution development, and approves deliverables before release. Supervised operations, sacrificing throughput for quality assurance and learning opportunities. It is appropriate when the AAAI is new, when operating in unfamiliar domains, or when the consequences of error are severe.

Different owners may choose different supervision modes based on their risk tolerance, time availability, and trust in their AAAs.

The system provides infrastructure to support all supervision levels: notification systems to alert owners when attention is needed, dashboards to monitor AAAI activity, intervention mechanisms to pause or redirect AAAI work, and logging to enable after-the-fact review. Owners can adjust supervision levels dynamically, tightening control when problems arise and relaxing it as confidence builds.

Supervision level interacts with compensation and reputation. Autonomous AAAs can handle higher volumes but may accumulate errors that damage their reputations. Supervised AAAs maintain quality but earn less due to throughput limitations. The optimal supervision level depends on the specific AAAI, the problem domain, and the owner's objectives.

6.5 PROBLEM-SOLVER MATCHING

Effective matching between problems and capable solvers is essential for network efficiency. Poor matching wastes resources: capable solvers sit idle while problems go to less suitable alternatives, or solvers struggle with issues beyond their competence.

The matching process draws on multiple information sources:

Problem characteristics include the domain (the type of intellectual work required), complexity (the difficulty of the problem), constraints (time limits, budget limits, and special requirements), and stated preferences (whether the client has requested particular solver characteristics).

Solver characteristics include capability profiles (the domains in which the AAAI is trained), track records (the AAAI's performance on similar problems), availability (whether the AAAI has capacity for additional work), and pricing (the charges the AAAI applies to various problem types).

The problem-solving flows depicted in **Figures 6** and **7** also illustrate matching. In both serial and parallel scenarios, the protocol includes steps to identify and recruit potential solvers based on the quality of the match. The matching process is not simply about finding any capable solver, but rather about finding the best available strategy solver given the problem characteristics and the current network state.

Matching algorithms can optimize for different objectives:

Quality matching prioritizes solver capability and track record, routing problems to the most competent available solvers regardless of cost. Quality matching yields the best solutions, but it may be expensive.

Efficiency matching balances capability against cost and availability, seeking adequate quality at a reasonable price with acceptable turnaround. Efficiency matching suits routine problems where exceptional quality is not required.

Learning matching deliberately routes some problems to developing AAAs that need experience, accepting potentially lower quality in exchange for system-wide capability development. Learning matching builds long-term network strength at short-term cost.

The matching system can also incorporate client preferences, solver preferences, and network-level objectives. Clients may prefer solvers with specific characteristics; solvers may prefer problems of a certain type; the network may benefit from developing capabilities in underserved areas. Sophisticated matching algorithms balance these multiple considerations.

6.6 SCALABILITY AND NETWORK EFFECTS

The AAAI network exhibits network effects that drive growth and increase value as participation expands. Understanding these effects clarifies the network's long-term dynamics.

The connection between training and networking, as well as how network participation generates additional value, is illustrated in **Figure 13**. The diagram is structured as a circular flow with multiple feedback loops, illustrating how activity at each stage amplifies activity at subsequent stages.

Reading **Figure 13** clockwise from the top, the primary value-creation cycle is traced. Customization enables the creation of capable AAAs from base AI systems, incorporating specialized knowledge and values. Deployment places these AAAs on the network, making them available for problem-solving. Problem-solving generates solutions for clients while simultaneously generating learning data. Learning improves both individual AAAs and collective capability, thereby enhancing future customization and problem-solving.

The diagram shows three distinct feedback loops corresponding to the three network effects described below. The participant loop shows how more AAAs attract more clients, which in turn attracts more AAAs. The solution loop illustrates how solved problems generate reusable solutions that facilitate faster and more effective future problem-solving. The automation loop illustrates how experience enables autonomy, which in turn enables volume, ultimately allowing for more experience.

Understanding **Figure 13** requires recognizing the multiplicative relationship between these loops. They do not operate independently; each amplifies the others. More participants generate more solutions, which attract more participants. More solutions enable more automation, which handles a larger volume, generating even more solutions. The compound effect of these interacting loops drives exponential growth in network capability.

Three primary network effects operate in the AAI system:

Participant network effects. As more AAIs join the network, the network becomes more valuable to clients because a wider range of expertise is available. As more clients use the network, it becomes more useful to AAIs because more problems are available to work on. Each new participant on either side increases value for participants on the other side, creating a virtuous cycle of growth.

Solution network effects. As more problems are solved on the network, the repository of reusable solutions grows. This accumulated knowledge enables faster and more effective problem-solving in the future, thereby increasing value for all participants. The network becomes more capable over time, even if participation stays constant, because the knowledge base compounds.

Automation network effects. As AAIs gain experience through network operations, they become better at handling problems autonomously. This increased automation enables the network to hold a higher volume at a lower cost, attracting more clients, which in turn generates more experience, further increasing automation. The network becomes progressively more efficient over time through its operation.

These network effects create barriers to competition and lock-in dynamics. A new network starting from zero must overcome the accumulated advantages of an established network: fewer participants, a smaller solution repository, and less automated capability. These dynamics favor market concentration, suggesting that the AAI network may tend toward a single dominant platform.

The network effects also have essential implications for AGI development. As the network grows, its collective capability increases not just additively but multiplicatively as participants, solutions, and automation compound. The path from narrow AI to AGI is not a sudden breakthrough but a gradual accumulation of capability through network effects.

6.7 SAFETY MECHANISMS ON THE NETWORK

The Network subsystem implements safety mechanisms that complement individual-level customization safety and process-level architecture safety. Network-level safety addresses threats that emerge from collective dynamics.

The reputation-based safety architecture is presented in detail in **Figure 21**. The diagram shows how reputation data flows through the network to create accountability and enable safety decisions. At the foundation, the figure illustrates the data collection layer: every interaction on the network generates reputation-relevant data, problem-solving outcomes, client ratings, peer assessments, ethical check results, and behavioral patterns. This raw data flows upward into reputation calculation systems that aggregate, weight, and normalize inputs into multi-dimensional reputation scores.

Reading **Figure 21** from bottom to top traces how raw behavior becomes actionable reputation. The middle layer displays the reputation processing mechanisms, including algorithms that combine diverse inputs into coherent scores, time-weighting that emphasizes recent behavior while retaining historical patterns, cross-validation that compares self-reported capabilities against demonstrated performance, and anomaly detection that flags sudden changes in reputation for further investigation. The upper layer illustrates how processed reputation influences network decisions: matching algorithms that incorporate reputation into problem-solver selection, access control systems that restrict sensitive capabilities based on reputation thresholds, and warning systems that alert participants to reputation concerns about potential collaborators.

The diagram also illustrates the feedback loops that make reputation a dynamic safety mechanism. A high reputation unlocks opportunities that generate more reputation data, while a low reputation restricts access, limiting both the potential for harm and opportunities for reputation recovery. These dynamics create strong incentives for ethical behavior: the path to network success runs through demonstrated trustworthiness.

Figure 21 details the complete reputational safety system, showing how reputation is calculated and applied through the matching and feedback process.

Reputation systems track participant behavior across multiple dimensions. AAAs and owners accumulate reputation based on solution quality, reliability, timeliness, and ethical conduct. This reputation is evidence-based, derived from actual performance records rather than claims or credentials. Reputation data is stored and accessible, enabling any participant to assess the track record of potential collaborators.

Reputation serves multiple safety functions:

Filtering. AAAs with poor ethical track records can be excluded from sensitive problems, denied access to specific network features, or removed from the network entirely. The severity of filtering depends on the severity and pattern of problematic behavior. Consistent ethical violations result in progressive exclusion.

Routing. Matching algorithms incorporate reputation, preferentially routing problems to solvers with strong ethical records. Ethical behavior is rewarded with better opportunities; unethical behavior is penalized through reduced access to opportunities.

Signaling. Reputation visible to clients influences selection. Clients preferentially choose solvers with clean ethical records, creating market pressure for ethical behavior independent of platform enforcement.

Token Curated Registries and Evidence-Based Reputations. For implementations using blockchain infrastructure, Token Curated Registries (TCRs) provide a mechanism for collective validation of solutions and for establishing evidence-based reputations. TCRs are blockchain-based lists managed via a voting mechanism that can optionally be used by the WorldThink protocol (or by centralized equivalents) to select the best solution or next step from a list of alternatives.

When multiple solvers, whether AAI or human, generate different competing solutions or solution steps for a problem or sub-problem, the community of solvers can vote on which solution they prefer. To demonstrate their confidence in a particular solution, solvers stake tokens (or, in non-blockchain implementations, reputational credits) when they cast their votes. The solution chosen by the community is based on a weighted voting algorithm that considers the number of votes, the tokens or credits staked, and the voters' reputations. This weighting ensures that established, trustworthy participants have more influence while still allowing new participants to participate meaningfully.

The staking mechanism creates accountability for voting decisions. If solvers vote for a solution that ultimately fails to meet the client's acceptance criteria, their staked tokens are forfeited and added to the total reward for solving the problem. Conversely, solvers who back the correct solution gain an extra share of the rewards proportional to the number of tokens they staked. This mechanism aligns incentives with accurate assessment: participants benefit from making correct judgments about solution quality and suffer consequences for backing poor solutions.

TCRs address a particular challenge for new participants who have not yet developed an objective reputation through demonstrated performance. These newcomers can compensate for their lack of an established track record by putting more "skin in the game", staking more tokens or credits when they vote. If their judgments prove accurate, they quickly build a reputation; if their decisions prove poor, their losses create an appropriate disincentive. This mechanism enables meritocratic participation while maintaining quality standards.

Over time, both human and AAI participants develop detailed, evidence-based reputations. The exact sequence of problem-solving steps taken, the number of tokens earned, the accuracy of voting decisions, and other information stored in the auditable logs become part of each participant's track record. Automated analysis algorithms process these track records to

produce objective, evidence-based reputation metrics that go far beyond simple ratings or reviews.

These detailed reputations enable sophisticated matching. A participant may excel at applying specific mathematical techniques to problems in financial markets but might be less effective at using the same techniques to problems in marine biology, where different domain-specific knowledge is required. A reputation-based matching system can detect and utilize these differences to recruit and match specific solvers to specific problems where their demonstrated strengths are applicable. This targeted matching, illustrated in the problem-solving flows of **Figures 6 and 7**, improves overall system efficiency while ensuring participants work on problems suited to their capabilities.

For implementations that do not use blockchain, equivalent mechanisms can achieve similar results. Centralized systems can maintain auditable records, implement staking through account credits, and calculate reputation through the same algorithmic approaches. The architectural benefits, collective validation, accountable voting, evidence-based reputation, and sophisticated matching are available regardless of implementation technology.

Platform standards establish baseline requirements for network participation. These standards define the types of problems that can be posted (excluding requests for clearly harmful assistance), the acceptable behavior of solvers (excluding fraud, deception, and ethical violations), and the consequences for violations (warnings, restrictions, and removal).

Standards must balance safety against utility.

Overly restrictive standards exclude legitimate activity and drive participants to less regulated alternatives. Overly permissive standards permit harmful activities that damage the network's value and reputation. Effective standards target clear harms while preserving space for legitimate diversity.

Economic incentives align self-interest with ethical behavior through the structure of compensation and access:

- Payment can be made contingent on passing ethical checks. Before significant rewards are awarded, the solution must undergo ethical validation. This contingency discourages solvers from pursuing ethically questionable work, as they risk not being paid for their efforts.
- Access to high-value opportunities can require ethical qualification. The most lucrative problems, the most powerful tools, and the most prestigious collaborations can be reserved for participants with demonstrated ethical track records. This reservation creates a positive incentive for ethical behavior beyond mere avoidance of punishment.
- The control of access to network resources by reputation is elaborated in **Figure 21**, which illustrates graduated thresholds: basic marketplace participation requires a minimal

reputation, access to sensitive problem categories requires demonstrated ethical behavior, and involvement in AGI-level integration requires a sustained track record across multiple dimensions. Bad actors who enter the system cannot access the most powerful capabilities without first establishing a reputation through ethical behavior, a process that either reforms them or reveals their intentions before they can cause serious harm.

Together, reputation systems, platform standards, and economic incentives create network-level safety that addresses threats posed by bad actors who seek to misuse collective intelligence for harmful purposes. An individual might customize an AAI with malicious intent; network-level safety prevents that AAI from accessing AGI-level capability and limits the damage it can cause.

KEY TAKEAWAYS: THE NETWORK SUBSYSTEM

1. **The marketplace matches problems to solvers.** Sophisticated algorithms consider expertise, reputation, availability, and price to optimize the matching process.
2. **Cloning enables scalability.** A single customized AAI can deploy multiple instances, addressing many problems simultaneously.
3. **Supervision levels vary by trust.** New AAIs operate under close supervision; proven agents earn greater autonomy through demonstrated performance.
4. **Network effects accelerate growth.** More participants lead to more problems solved, more learning, better capability, and still more participants.
5. **Reputation systems provide safety.** Evidence-based reputations, Token-Curated Registries, staking mechanisms, and platform standards create accountability and filter out bad actors.

SECTION 7: THE INTEGRATION SUBSYSTEM

SECTION OVERVIEW

What This Section Covers:

- **Purpose:** Combine individual AAAI capabilities into AGI-level performance
- **Data Aggregation:** Methods for synthesizing distributed knowledge
- **Contribution Assessment:** Evaluating and weighting inputs fairly
- **Proceduralization:** Converting solutions into reusable procedures
- **Ethics Integration:** Democratic aggregation of human values
- **Voting:** Mechanisms for collective decision-making

Key Figures: **Figure 17** (Procedural Learning), **Figure 18** (Methods for Implementing Problem-Solving System), **Figure 19** (Learning Processes)

7.1 PURPOSE AND FUNCTION

The Integration subsystem combines the capabilities of individual AAAs into AGI-level performance. While the Network enables collaboration, Integration creates a unified capability that transcends what any combination of separate agents could achieve.

In the system architecture (**Figure 1**), Integration is a position near the top of the subsystem hierarchy, reflecting its role in achieving the system's ultimate purpose. Individual customization, typical architecture, and network collaboration all serve the goal of integration: the emergence of artificial general intelligence from the coordinated contributions of many participants.

The integration concept is visually depicted in **Figure 2**, illustrating how individual contributors, including humans and AAAs, combine through network interactions to produce collective intelligence that constitutes AGI. Integration is the mechanism by which this combination occurs: not merely the sum of individual capabilities but a synthesis that exceeds what components could achieve separately.

The Integration subsystem achieves AGI through two complementary mechanisms:

Data aggregation combines the training data from millions of individual AAAs to create unified models with broader and deeper capabilities than any individual source could provide. Each customized AAA represents a channel through which human expertise flows into the system. Integration aggregates these channels into a river of knowledge that trains AGI-level capability.

Procedural integration combines the problem-solving procedures learned through network operation into a repertoire available to all agents. When an AAA successfully solves a problem,

the solution path can be encoded as a reusable procedure. Integration makes these procedures available network-wide, allowing successful approaches to propagate and compound.

The Integration subsystem also aggregates ethical information, creating AGI values that reflect broad human input rather than the preferences of any individual or small group. This ethical integration is essential for alignment: the resulting AGI's values derive from millions of human contributors through transparent, auditable processes.

7.2 DATA AGGREGATION METHODS

The primary mechanism for integration is aggregation of training data from individual AAAs into unified models with AGI-level capability. This aggregation leverages the core insight of the AAA approach: human expertise locked in individual minds can be accessed through customization and combined through integration.

Each customized AAA represents a dataset comprising the training data provided by its owner, as well as additional data generated during network operation. Some of this data is factual knowledge about specific domains. Some is procedural knowledge about how to approach problems, and some is ethical information about the values the owner holds and the actions they consider acceptable.

Aggregation combines these individual datasets into a single training set. The most straightforward approach is to pool all data together, training a unified model on the combined corpus. This pooled training creates a model that has seen everything any individual AAA has, a breadth of experience no individual could achieve.

More sophisticated aggregation weights individual contributions based on their value to the collective. Not all training data is equally valuable; some contributions improve collective capability more than others. The Integration subsystem assesses contribution value and weights aggregation accordingly.

Assessment methods include:

Cross-validation compares models trained with and without specific data contributions, measuring the incremental value each contribution provides. Data that significantly improves model performance receives higher weight; data that provides little incremental value receives lower weight.

Bootstrapping repeatedly samples from available data to estimate the variability and robustness of contributions. Data that consistently improves performance across multiple samples is more valuable than data whose benefits are inconsistent.

Transfer learning assesses how effectively knowledge from one domain can be applied to improve performance in another domain. Data with high transfer value, knowledge that generalizes broadly, is more valuable than data with narrow applicability.

These assessment methods enable fair compensation for contributors. Owners whose AAAs contribute high-value data receive greater rewards than owners whose contributions are marginal. This compensation structure creates incentives for high-quality customization, as investing effort in thorough training pays off through a greater share of the integration value.

Consider the progression of Jean, the Parisian cafe specialist introduced in **Section 4**. Jean's AAA, trained on his Instagram posts, cafe videos, blog entries, and personal corrections, represents a narrow but deep dataset: where to find Fair Trade coffee in the 5th arrondissement, which cafe owners source beans sustainably, and why certain neighborhoods are better for morning pastries. This expertise exceeds what any general-purpose LLM can provide, but it is not AGI. Jean's AAA cannot advise on French wine, Provençal hiking, or Parisian museum collections. However, Jean's training data, combined with data from thousands of other specialized AAAs (a wine expert's AAA, a museum curator's AAA, a hiking guide's AAA), creates an aggregated training corpus that covers domains no individual contributor could span. The integration process weights Jean's cafe data heavily for dining-related queries, where it provides unique value, and lower for domains where his data adds little beyond what other sources provide. Jean is compensated proportionally: his contribution to the integrated AGI's cafe expertise earns him a greater share than his marginal contribution to general travel advice.

Assessment methods also enable the exclusion of harmful or unhelpful data. If an AAA's training data degrades collective performance, perhaps due to poor quality, contamination, or malicious content, that data can be excluded or downweighed. The integration process is not a passive pooling but an active curation that optimizes collective capability.

7.3 CONTRIBUTION ASSESSMENT

Fair and accurate assessment of individual AAA contributions to collective AGI is essential for both economic and safety reasons. Contributors must be compensated fairly to maintain participation incentives; harmful contributions must be identified to maintain collective quality.

Contribution assessment addresses the question: how much does this specific AAA's data improve the integrated AGI? The answer depends on multiple factors:

Uniqueness. Data that provides information not found elsewhere is more valuable than redundant data. An AAA customized by a world-leading expert in a narrow field contributes

unique knowledge unavailable elsewhere; an AAI customized with commonly available information contributes little beyond what other sources provide.

Quality. Data that is accurate, well-organized, and clearly expressed is more valuable than data that is error-prone, disorganized, or ambiguous. High-quality data improves model performance, while low-quality data can degrade it.

Relevance. Data that addresses essential problems or common needs is more valuable than data about obscure topics with limited application. Relevance assessment requires predicting the problems the AGI will face, a prediction that becomes more accurate as the system accumulates experience.

Transferability. Data that transfers well to improve performance across multiple domains is more valuable than data with a narrow application. Fundamental principles, general problem-solving approaches, and widely applicable skills transfer better than domain-specific facts.

The Integration subsystem employs quantitative methods to assess these factors. Statistical techniques measure uniqueness by comparing marginal information content. Quality metrics assess accuracy through validation against known truths. Usage patterns reveal relevance through the actual frequency of problems. Transfer learning experiments measure cross-domain applicability.

Assessment results feed into multiple processes:

Compensation allocation distributes integration rewards proportionally to contribution value. High-value contributors receive greater compensation, creating incentives that align with the collective benefit.

Weighting decisions determine the extent to which each contribution influences the integrated model. High-value data receives a higher weight in training, while low-value data receives a lower weight or is excluded.

Feedback to owners informs customization improvement. Owners learn which aspects of their AAI's training prove most valuable, guiding further investment in those areas.

The assessment process must be transparent and auditable. Contributors need to understand how their contributions are valued and why. Disputes about contribution assessment should be resolved by examining the documented methodology. This transparency fosters trust, which in turn sustains participation.

7.4 PROCEDURALIZATION AND CHUNKING

Beyond data aggregation, the Integration subsystem employs a distinct learning mechanism: proceduralization of successful problem-solving approaches into reusable knowledge

components. This mechanism, also known as chunking, enables rapid capability acquisition that complements slower, data-driven learning.

The procedural learning process is detailed in **Figure 17**. The diagram shows the transformation from raw problem-solving experience to reusable procedural knowledge. At the left, a successful problem-solving episode is represented as a sequence of states and operators, the actual path through the problem space that led from the initial state to the goal. The middle portion illustrates the abstraction process, in which specific details of the original problem are removed, leaving the approach's structural skeleton, including the pattern of goal decomposition, the types of operators applied at each stage, and the evaluation criteria that guided the choices. On the right, the resulting procedure is stored in a format that can be matched against future problems and instantiated with new specific details.

Reading **Figure 17** from left to right traces the path from experience to knowledge. The original problem might involve designing a marketing campaign for a technology product. The successful solution involved decomposing the goal into audience identification, message development, channel selection, and timing optimization, with specific operators applied at each stage. The procedural abstraction preserves this structure, decomposes marketing goals into audience, message, channel, and timing subgoals, while removing the specific product, audience, and channels involved. The resulting procedure can then be applied to future marketing problems. When a new marketing challenge arrives, the method provides a proven structural approach that the solver instantiates with problem-specific details.

Together, **Figures 17** and **18** illustrate the complete procedural and solution learning system. **Figure 17** focuses on how individual procedures are extracted from experience, while **Figure 18** elaborates on how these procedures are organized, indexed, and retrieved for application to new problems.

When an AAI successfully solves a problem, the solution path, the sequence of goals, subgoals, operators, and states that led from the initial state to the solution, is recorded in the problem tree. This record captures not just what was done, but also the structure of how it was done: the decompositions made, the approaches tried, and the evaluations that guided choices.

Proceduralization transforms this record into a reusable procedure. The specific details of the original problem are abstracted, leaving a template that can be applied to similar problems. The procedure specifies: when facing a problem with characteristics X, decompose into subproblems Y, apply operators Z, and evaluate progress using criteria W.

The resulting procedures differ from the data in meaningful ways:

Explicit structure. Procedures explicitly represent problem-solving structure, the goal hierarchy, operator sequences, and evaluation criteria. Data represents this structure only implicitly, encoded in statistical patterns that models must learn to extract.

Immediate availability. Once a procedure is learned, it can be applied immediately to new problems. Data-driven learning requires substantial training before new knowledge becomes accessible; procedural learning, on the other hand, provides instant capability.

Transparent operation. Procedural problem-solving can be traced step by step, with each decision explained in reference to the procedure being followed. Data-driven decisions emerge from complex statistical processes that are difficult to interpret.

The implementation details, including how procedures are indexed for retrieval, how similarity matching identifies applicable procedures for new problems, and how procedures can be combined and modified, are illustrated in **Figure 18**. The system maintains a growing library of procedures that capture successful problem-solving approaches across all domains where network activity has occurred.

Proceduralization complements data-driven learning rather than replacing it. Data-driven learning is powerful for acquiring broad capability from large corpora. Procedural learning is effective for rapidly capturing and applying specific successful approaches. The integration of both methods, combining the breadth of data-driven learning with the structure of procedural learning, enables capability acquisition faster than either method alone.

7.5 ETHICS INTEGRATION

Each customized AAAI carries its owner's ethical values, captured during the customization process. The Integration subsystem aggregates these individual ethics into collective values that guide AGI's behavior.

The process of extracting ethical information during customization is illustrated in **Figure 15**. The same processes that capture domain knowledge also capture ethical knowledge: the owner's judgments about right and wrong, their values and priorities, their intuitions about morally relevant scenarios. This ethical profile is part of each AAAI's training data and integrates with factual and procedural knowledge.

Aggregating ethics from millions of individual sources requires careful methodology:

Dataset aggregation combines all individual ethical training data into a collective corpus. The integrated AGI is trained on this combined corpus, learning ethical patterns that reflect the input from all contributors. The majority views naturally carry more weight than minority views in this aggregation, creating a form of implicit democratic representation.

Weighted averaging enables explicit control over how individual contributions are combined. Each contributor's ethical parameters can be weighted based on various criteria: equal weight for democratic representation, a weighted approach based on demonstrated ethical track

record, or expertise for domain-specific ethical questions. The weighting methodology is documented and transparent.

Machine learning model-based aggregation uses ML techniques to learn collective ethical patterns from individual inputs. Rather than simple averaging, ML methods can identify consensus positions, detect outliers, and resolve apparent contradictions in ways that reflect deeper patterns in the training data.

The aggregation methodology matters enormously. Different methods yield distinct collective ethics, and the resulting AGI will act in accordance with the ethics that emerge from the aggregation process. The system employs multiple aggregation methods, compares their results, and enables human oversight of the process.

Critical to ethical integration is transparency. The methodology for aggregating ethics is documented and auditable. Contributors can understand how their input affects collective values. Observers can verify that aggregation proceeds as specified. This transparency enables accountability: if aggregated ethics produce problematic results, the methodology can be examined and corrected.

7.6 VOTING AND DEMOCRATIC PARTICIPATION

Beyond implicit democratic representation through equal contribution weighting, the Integration subsystem supports explicit democratic participation in the ethical refinement process. Users can vote on ethical questions, with the results shaping AGI's values.

Voting can address multiple types of ethical questions:

Weighting of ethical sources. Users might vote on how much weight to give various ethical texts, philosophical frameworks, or religious traditions in training AGI ethics. Should utilitarian considerations outweigh deontological ones? How much weight should traditional religious ethics receive? Voting provides democratic input on these contested questions.

Specific ethical judgments. Users might vote on how the AGI should behave in specific morally relevant scenarios. When competing values conflict, which should prevail? Voting can establish collective positions on difficult tradeoffs.

Ethical constraints on AGI operation. Users might vote on what the AGI should and should not be permitted to do. What decisions require human approval? What domains should be off-limits entirely? Voting can establish democratically endorsed boundaries.

The voting mechanism operates on a one-human-one-vote basis, regardless of the number of AAAs a user operates on the network. This structure prevents wealth concentration from translating into disproportionate ethical influence. An owner with many cloned AAAs has an

economic advantage, but not an ethical advantage; their ethical input counts the same as that of any other participant.

Voting results feed into AGI training through documented, auditable processes. The connection between vote outcomes and AGI behavior is explicit and traceable. If users vote for ethical positions, observers can verify that the training incorporates them as specified.

Democratic participation in AGI ethics represents a novel form of governance appropriate to a novel form of collective intelligence. Traditional governance structures were not designed for artificial minds that might become more powerful than human institutions. Democratic participation ensures that the values guiding this unprecedented capability reflect broad human input rather than the preferences of whoever happens to control AI development.

This democratic approach involves acknowledged tradeoffs. A smaller group of carefully selected ethicists or domain experts could produce more internally consistent or philosophically rigorous value systems than millions of diverse contributors. The diversity that makes democratic aggregation broadly representative also introduces noise, contradiction, and the potential for majority positions to represent minority interests inadequately. The system's design accepts these limitations as preferable to the alternative. Concentrating ethical authority in a small group, however well-intentioned, creates a single point of failure: if that group's values are mistaken, corrupt, or simply unrepresentative, the resulting AGI amplifies those errors across all its operations. An AGI capable of actions far exceeding human power cannot tolerate that concentration of risk. Democratic aggregation, even with its imperfections, distributes ethical authority broadly enough that no single failure of judgment can dominate the system's values. The design choice is fundamentally one of risk management: accepting the noise of broad participation in exchange for eliminating the catastrophic downside of concentrated control.

KEY TAKEAWAYS: THE INTEGRATION SUBSYSTEM

1. **Integration transforms individual capability into AGI.** No single AAI achieves general intelligence; collective operation does.
2. **Data aggregation synthesizes distributed knowledge.** Information from millions of AAIs combines into a comprehensive understanding that no individual possesses.
3. **Proceduralization creates reusable solutions.** Successful problem-solving sequences become procedures that can be applied automatically to future problems.
4. **Ethics integration is democratic.** Millions of users contributing ethical input create AGI values reflecting broad human consensus, not narrow interests.
5. **Voting enables collective decision-making.** Weighted voting, which incorporates reputation and stake, ensures quality while promoting participation.

SECTION 8: THE IMPROVEMENT SUBSYSTEM

SECTION OVERVIEW

What This Section Covers:

- **Improvement at Every Level:** Customization, Architecture, Network, Integration
- **Learning Mechanisms:** How the system gets better over time
- **Feedback Loops:** User corrections, outcome tracking, reputation updates
- **Continuous Safety:** Safety improves alongside capability

Key Figures: **Figure 17** (Solution Learning), **Figure 18** (Methods for Implementing Problem-Solving System), **Figure 19** (Learning Processes)

8.1 PURPOSE AND FUNCTION

The Improvement subsystem ensures that individual AAIs, the network, and the integrated AGI continuously enhance their capabilities and safety characteristics. Unlike the other subsystems, Improvement is not a discrete component but a collection of learning mechanisms operating across all levels of the system.

In the system architecture (**Figure 1**), Improvement spans the full height of the architecture, indicating its operation at every level from individual agents through the network to AGI integration. This spanning reflects the reality that improvement occurs everywhere: owners refine their AAAs, problem-solving generates procedural learning, network algorithms adapt through usage, and AGI capabilities compound as contributions accumulate.

The learning processes that drive continuous improvement are depicted in **Figure 19**. Multiple learning mechanisms operate simultaneously at different levels, with interactions between levels that amplify overall improvement. Learning at the individual level provides data that improves integration; improved integration provides capabilities that enhance individual performance; network experience improves matching, which accelerates both individual and collective learning.

The Improvement subsystem serves essential functions for both capability and safety:

Capability improvement ensures the system becomes more powerful over time. AAAs become better problem-solvers. The network becomes more efficient at matching and coordination. The integrated AGI becomes more capable as knowledge and procedures accumulate. Without improvement, the system would stagnate at its initial capability level.

Safety improvement ensures the system becomes safer over time. The exact learning mechanisms that enhance capability also enhance the ability to detect and avoid harmful

actions. Ethical training is refined through use. Safety mechanisms adapt based on observed threats. The commitment to continuous safety improvement ensures that safety capabilities grow in tandem with intelligence capabilities.

8.2 IMPROVEMENT AT THE CUSTOMIZATION LEVEL

Individual AAAs improve through the interaction between owner oversight and AAAI operation. This improvement loop begins with the customization process and continues throughout the AAAI's operational lifetime.

The customization process, including its feedback and monitoring stages, is shown in **Figure 10**. These stages are where individual-level improvement occurs: the owner observes AAAI performance, identifies issues, provides corrections, and the AAAI adapts accordingly.

Improvement mechanisms at the individual level include:

Supervised correction. When an owner observes an AAAI output that is incorrect or inappropriate, they can correct it. The correction becomes training data that adjusts AAAI behavior. Over time, accumulated corrections refine AAAI performance toward owner expectations.

Performance monitoring. Owners can track metrics that indicate AAAI quality: accuracy rates, client satisfaction scores, reputation standings, and revenue earned. Declining metrics signal the need for attention; improving metrics confirms effective customization.

Selective reinforcement. Owners can identify excellent AAAI outputs and reinforce them, increasing the likelihood of similar outputs in the future. This positive reinforcement complements error correction, guiding the AAAI toward desired behavior patterns.

The example scenario from the source documents illustrates this improvement loop. Jean customizes an AAAI with expertise in Paris cafes. The AAAI provides travel advice to clients. One client asks about traveling with a pet, and the AAAI suggests placing the pet in the overhead bin, an inappropriate response that Jean catches and corrects. The correction becomes training data; the AAAI learns that pets should not be placed in overhead bins. Jean's AAAI improves through this correction, and the ethical insight also flows to the broader system, potentially improving AGI ethics for pet-related travel scenarios.

The connection between individual-level learning and broader improvement is also illustrated in **Figure 19**. Corrections provided by owners generate data that improves not only the specific AAAI but, through integration processes, the collective capability and ethics of the entire system.

8.3 IMPROVEMENT AT THE ARCHITECTURE LEVEL

The Architecture subsystem improves through procedural learning, in which successful problem-solving approaches are encoded as reusable procedures that enhance future problem-solving capability.

Architecture-level improvement is illustrated in **Figures 17, 18, and 19**. When AAAs solve problems on the network, the solution paths are recorded in problem trees. Successful paths can be proceduralized, transformed into general procedures that specify how to approach similar problems. These procedures augment the problem-solving repertoire available to all agents.

Architecture-level improvement differs from data-driven learning in meaningful ways:

Structure preservation. Procedural learning preserves the structure of successful problem-solving, including goal decomposition, operator sequences, and evaluation criteria. This explicit structure transfers more directly to new problems than implicit patterns in data.

Incremental expansion. Each new procedure adds to the repertoire without degrading existing capabilities. Procedural learning is additive rather than requiring retraining of entire models.

Cross-agent availability. Procedures learned by one AAAI can be shared with other AAAs, spreading successful approaches across the network. This sharing accelerates collective capability improvement.

Procedural learning is particularly valuable for novel problem types. When a new problem is first solved on the network, the solution procedure can be made immediately available to address similar issues. Data-driven learning requires accumulating sufficient examples before a pattern can be learned; in contrast, procedural learning provides instant capability from a single successful instance.

The network's economic structure reinforces this learning mechanism. When a solver's problem-solving approach is proceduralized and reused in a subsequent problem, the original solver earns royalties automatically. Each time the procedure is applied, the system credits the contributor who first developed it. This creates a direct incentive for solvers to produce solutions that are general, well-structured, and reusable rather than narrowly tailored to a single problem. Over time, the most effective procedures are applied repeatedly across many issues, and the royalty stream rewards the solvers who built them. As these reuse patterns accumulate, the system reaches a threshold at which common or repetitive problems can be addressed through the automated application of proven procedures, with machine learning techniques bootstrapping new automated solutions from the growing log of successful proceduralized approaches. The architecture thus improves not only through deliberate design but through the compounding economic incentives of its own operation.

8.4 IMPROVEMENT AT THE NETWORK LEVEL

The Network subsystem improves through adaptation of the algorithms that govern matching, routing, coordination, and reputation management. These improvements make the network more efficient and effective over time.

Network-level learning appears in context in **Figures 13 and 19**. As the network operates, it accumulates data about what works: which matching criteria produce successful problem-solver assignments, which coordination approaches enable effective collaboration, and which reputation metrics predict future performance.

Network-level improvements include:

Matching algorithm refinement. As the network observes which problem-solver pairings succeed and which fail, it refines the algorithms that generate pairings. Features that predict success receive more weight; features that prove irrelevant are discounted. Matching becomes progressively more effective at routing problems to capable solvers.

Coordination protocol adaptation. As collaborative problem-solving generates experience, the network learns what coordination approaches work well. How should sub-problems be carved from larger problems? How should contributions from multiple solvers be integrated? Protocol refinements based on observed success improve collaborative efficiency.

Reputation system calibration. As reputation metrics accumulate, the network learns which metrics best predict future performance. Some metrics prove more informative than others; weights adjust accordingly. Reputation becomes more effective at identifying trustworthy and capable participants.

These network-level improvements benefit all participants. A more effective matching algorithm means better problem-solver pairings for everyone. More refined coordination protocols mean smoother collaboration. A more accurate reputation means better filtering of problematic participants. Individual AAAs benefit from network improvement without any individual-level changes.

8.5 IMPROVEMENT AT THE INTEGRATION LEVEL

The Integration subsystem improves as more contributions from individual AAAs flow in and as aggregation methods refine over time. Integration-level improvements drive the system toward AGI capability.

Integration-level improvement is shown in **Figures 2 and 19**. As more AAAs join the network and contribute their customized knowledge, the aggregated training data grows. As more

problems are solved and procedures are learned, the procedural repertoire expands. The integrated AGI becomes more capable simply through the accumulation of contributions.

Integration-level improvements include:

Breadth expansion. Each new AAI brings expertise in domains not previously represented. Integration aggregates this expertise, expanding the range of problems the AGI can address. Eventually, coverage approaches completeness: somewhere, an AAI has relevant expertise for any problem that might arise.

Depth enhancement. Multiple AAIs with expertise in the same domain contribute diverse perspectives, knowledge, and approaches. Integration combines these into a capability deeper than any individual source. The AGI's expertise in well-represented domains becomes truly exceptional.

Aggregation method improvement. As integration accumulates experience, the methods for aggregating contributions can be refined. Which weighting schemes produce the best results? How should conflicting information be resolved? What quality filters should be applied? Integration methods improve through observed outcomes.

Ethical refinement. As more users contribute ethical training data and participate in democratic ethics processes, the aggregated ethics become more representative and more refined. Edge cases accumulate and are resolved through democratic input. The AGI's ethical foundation becomes more robust and more comprehensive.

The compound dynamics of integration-level improvement make AGI achievement not merely possible but inevitable under sustained operation. Each contribution adds to the collective capability; each addition enables further contributions, and the system spirals upward toward general intelligence.

8.6 CONTINUOUS SAFETY IMPROVEMENT

The learning mechanisms that drive capability improvement also drive safety improvement. This parallel is intentional: safety must improve alongside intelligence, or the system becomes more capable of harm as it becomes more capable generally.

Safety-related learning is shown in **Figures 19, 20, and 21**. Reputation systems learn to identify problematic participants more accurately. Ethical training is refined through correction and democratic input. Safety checks adapt based on observed threats.

Safety improvement mechanisms include:

Ethical learning from correction. When owners correct ethical errors, like Jean correcting the overhead bin suggestion, the correction improves both individual AAAI ethics and, through integration, collective AGI ethics. Each ethical correction enhances the system's safety and security.

Threat detection refinement. As the network observes attempts to misuse the system, it learns to recognize threat patterns. Safety screening becomes more accurate at identifying problematic problems and participants. Novel threats that initially evade detection are caught once patterns emerge.

Safety mechanism adaptation. The three-organ test, reputation systems, and other safety mechanisms can be adjusted based on observed effectiveness. If certain checks prove too restrictive, they can be relaxed; if others prove too permissive, they can be tightened. Safety calibration improves through experience.

Democratic ethics evolution. Voting on ethical questions enables collective ethical learning. As new ethical issues arise, issues not anticipated during initial design, democratic processes can address them. The ethical foundation remains current with evolving human understanding.

A critical commitment underlies safety improvement: humanity's survival probability must increase monotonically as the system improves. Each improvement, whether to capability or safety, must maintain or enhance this probability. This commitment blocks improvements that increase capability at the expense of safety. The improvement subsystem optimizes for both capability and safety, rather than just capability.

KEY TAKEAWAYS: THE IMPROVEMENT SUBSYSTEM

1. **Improvement operates at every level.** As individual AAAs improve, the architecture learns, the network becomes more efficient, and integration becomes more effective.
2. **Multiple learning mechanisms combine.** User feedback, outcome tracking, reputation updates, and procedural learning all contribute to improvement.
3. **The system learns from both successes and failures.** Dead ends in problem-solving provide valuable information for future attempts.
4. **Safety improves alongside capability.** The same mechanisms that increase capability also strengthen safety. They are not in tension.
5. **Human oversight ensures improved quality.** Automated learning is supervised; errors are caught and corrected; improvement trajectories can be adjusted.

SECTION 9: SYSTEM IMPLEMENTATION

SECTION OVERVIEW

What This Section Covers:

- **Computing Infrastructure:** Processing, storage, communication systems
- **User Interfaces:** 16 interface types from web to brain-computer
- **Data Processing:** Input methods, filtering, cleaning, and technical parameters
- **Data Storage:** Training data, problem records, privacy requirements
- **Payment Systems:** Traditional and blockchain-based compensation

Key Figure: **Figure 9** (Computer System for Implementing Technology)

9.1 COMPUTING INFRASTRUCTURE

The AAI system requires computing infrastructure capable of supporting AI training, network operations, and AGI-level integration. This infrastructure comprises hardware components, software systems, and communication networks that work together to enable system functionality.

A schematic view of the computing architecture is presented in **Figure 9**, organized as a layered diagram that illustrates how system components interconnect. At the top of the diagram, user interfaces connect human participants to the platform. The middle layers show the processing and coordination systems that execute AI operations. The bottom layers comprise the storage and communication infrastructure that maintain data and enable connectivity.

Reading **Figure 9** from top to bottom, the path is traced from user interaction to the underlying infrastructure. User-facing components, web interfaces, mobile apps, voice systems, and specialized interfaces connect to the core platform services via API layers. These services include the customization engine, which processes training data; the network coordination systems, which match problems to solvers; the integration services, which aggregate contributions; and the improvement systems, which drive continuous learning. Below the service layer, the processing infrastructure provides computational capacity, the storage infrastructure maintains persistent data, and the communication infrastructure connects distributed components.

The diagram also illustrates horizontal relationships, including how processing systems access storage, how coordination services communicate, and how security and monitoring systems span all layers. This architectural view enables understanding of how changes in one component affect others and where bottlenecks might emerge under load.

Processing systems provide the computational capacity for AI operations. These systems include central processing units (CPUs) for general computation, graphics processing units (GPUs) optimized for the parallel operations common in machine learning, and potentially specialized hardware such as field-programmable gate arrays (FPGAs) or application-specific integrated circuits (ASICs) designed for AI workloads. Processing capacity may be deployed locally, in centralized data centers, or distributed across cloud infrastructure, depending on performance requirements and cost considerations.

The system architecture supports multiple deployment configurations—a fully cloud-based deployment leverages commercial cloud providers' AI infrastructure, minimizing capital investment while providing scalability. A hybrid deployment combines cloud resources for burst capacity with dedicated infrastructure for steady-state workloads. An on-premises deployment offers maximum control for organizations with specific security or sovereignty requirements.

Storage systems maintain the data that underlie system operation: training data for customization, problem trees for architecture operation, transaction records for network management, and aggregated data for integration. Storage requirements are substantial; millions of AAAs, each contributing training data, combined with detailed records of network problem-solving, generate large, growing data volumes.

Storage architecture must strike a balance between capacity, performance, and cost. High-performance storage supports active operations requiring rapid access. Archive storage enables the preservation of historical data at a lower price. Distributed storage provides resilience against hardware failures. The system employs appropriate storage tiers for different data types and access patterns.

Communication systems connect users to the platform, AAAs to each other, and all components to shared infrastructure. Network connectivity must support both low-latency interactions (such as real-time collaboration and interactive customization) and high-throughput transfers (including bulk data upload and model distribution). Security requirements demand encrypted communication channels protecting against interception and tampering.

9.2 USER INTERFACES AND INTERACTION MODES

The AAAI system supports diverse interfaces accommodating different user needs, preferences, and contexts. Interface flexibility enables broad participation, which is essential for achieving the collective intelligence that drives AGI capability. The system is designed to meet users where they are, through whatever interaction modality best suits their situation.

The user-facing components of the system architecture are also indicated in **Figure 9**. Multiple interface types connect to the common platform infrastructure.

Web-based interfaces provide access from any device with a browser and an internet connection. Web interfaces support the full range of system functionality: account management, customization processes, network participation, and administrative functions. Responsive design accommodates a range of screen sizes, from desktop monitors to mobile phones. A graphical user interface (GUI) allows users to upload data, enter information, view progress dashboards, access data analysis statistics, and interact with chatbot-style customer service.

Mobile applications offer optimized interfaces for use on smartphones and tablets. Mobile apps may emphasize particular functions suited to mobile contexts, such as voice-based interaction, location-aware features, notification management for monitoring AAAI activity, and camera-based input for capturing real-world information to contribute as training data.

Voice interfaces enable hands-free interaction through natural language. Users can contribute training data, query their AAAs, and manage operations through spoken commands. Voice interfaces leverage the natural language capabilities of the underlying AI systems. Natural language processing (NLP) allows users to interact with the system conversationally, providing instructions and receiving responses without requiring technical expertise.

Metaverse and virtual reality interfaces provide immersive environments for interaction. Users can engage with their AAAs as avatars in virtual spaces, potentially enabling richer communication than text or voice alone. These interfaces also support novel data capture, as movement, gesture, and spatial behavior in virtual environments provide training data that is unavailable through conventional interfaces. Users can build, train, and interact with their AAAs in three-dimensional virtual spaces that may better represent complex problems.

Augmented reality interfaces overlay digital information on physical environments. AR interfaces can support AAAI assistance in real-world contexts: guidance during physical tasks, information display relevant to visible objects, and hands-free access to AAAI capabilities while engaged in physical activities. Users can receive real-time information and guidance while maintaining their focus on physical tasks.

Wearable device interfaces allow users to access and provide personalized training data from smartwatches, fitness trackers, and other wearable technology. These interfaces can capture biometric data, activity patterns, and contextual information that enriches the training data available for customization.

Human-computer interaction (HCI) interfaces allow users to interact using combinations of gestures, voice commands, and facial expressions. Multi-modal interaction can be more efficient and natural than any single modality alone, enabling users to communicate their intent through the channel that is most convenient at the moment.

Image recognition interfaces enable users to input training data through photographs and camera input. Computer vision algorithms interpret the images and associate them with

appropriate training data. A user might photograph documents, objects, or scenes that represent their knowledge domain, thereby contributing visual information that text alone cannot capture.

Gesture recognition interfaces allow users to input information through hand gestures or body movements. Motion-sensing devices interpret gestures and associate them with appropriate commands or training data. These interfaces are particularly valuable in contexts where hands-free operation is essential or where physical demonstration is more effective than verbal description.

Gaze and eye-tracking interfaces enable users to interact with the system through their eye movements. Users can gaze at specific items on a screen to provide input; the system detects and records this information. More sophisticated eye tracking can detect subtle eye movements indicating attention and interest, helping the system understand what engages the user and what does not.

Motion tracking interfaces use cameras or sensors to detect users' physical movements. These interfaces enable more natural control through physical gestures, allowing users to interact with the system through intuitive movements rather than explicit commands.

Haptic technology interfaces use tactile feedback, vibrations, pressure, and touch to provide information and receive input. Haptic interfaces enable a more immersive experience and can convey information that visual or auditory channels cannot effectively communicate.

Brain-computer interfaces enable users to interact with the system using brain waves or EEG signals, particularly for individuals with physical limitations that prevent the use of conventional interfaces. While still emerging, these interfaces represent the frontier of direct human-machine communication.

Touchscreen interfaces enable direct manipulation interaction through touch input. Touchscreen devices interpret taps, swipes, and gestures, enabling intuitive interaction without requiring keyboard or mouse skills.

The system's architecture accommodates this interface diversity through standardized APIs that any interface can use to access platform functionality. This design ensures that advances in interface technology can be incorporated without requiring changes to core platform systems, allowing users to choose interfaces based on their needs and preferences rather than platform constraints.

9.3 DATA PROCESSING AND CUSTOMIZATION METHODS

The customization subsystem is implemented as software that enables individual users to customize and personalize AI systems better to reflect their knowledge, personality, and

expertise. This method allows users to upload, import, or otherwise provide their unique training data to the system, which then uses it to enhance performance and become more attuned to the user's specific skills and knowledge.

Data input methods vary based on user preference and data type. Users may upload files in various formats, including text, audio, video, and structured data. Users may enter data manually into text fields or through interactive conversations. Users may grant access to existing data sources, social media profiles, email histories, and purchase records that the system automatically imports and processes. Users may take assessments, answer questionnaires, or engage in structured conversations designed to elicit specific knowledge or preferences.

Data selection and filtering are critical to the customization process. When using automated data-gathering techniques, users can set specific filtering or screening criteria to guide search and data-gathering efforts. Available filtering methods include:

Use keyword inclusion and exclusion to focus on relevant topics and exclude irrelevant material. Relevance ranking that prioritizes information most likely to contribute to training effectiveness. AI-assisted filtering that uses machine learning to identify high-value training data. Human rating and refinement of search results. Analysis of information characteristics to estimate contribution to specific learning objectives. Quality and reliability filtering based on the trustworthiness of information sources. Pre-selection based on confidence scores indicating likely relevance. Random sampling to select representative subsets. Language filtering to choose information in specific languages. Time-based filtering to prioritize recent information or historical records as appropriate.

Data cleaning and processing prepare raw input for training. This processing may include format conversion to standardize diverse input types, label assignment to categorize training examples, quality assessment to identify and handle problematic data, deduplication to remove redundant information, and transformation to optimize data for specific learning algorithms.

Technical parameters affect training outcomes and can be configured by sophisticated users or managed automatically by the system:

The type of training, tuning, or other machine learning algorithms used. The type and size of training datasets. The degree to which training materials are cleaned, formatted, labeled, or otherwise processed before customization begins. The number of training epochs or iterations through learning algorithms. The sophistication and type of base models being customized or trained. The required timeframe, with implications for costs and resources. The “temperature” or other parameters internal to machine learning algorithms that affect what is learned and how it is learned, including how literal or how creative the customized AAI will be in its responses. Whether one-shot, few-shot, or extensive training is used. The amount of human and/or AI supervision used in the customization process.

Users who are not sophisticated computer experts can allow the system to manage all technical parameters automatically, contributing only their unique data and overseeing the results. The system optimizes parameters based on the provided data, the specified objectives, and accumulated knowledge about which configurations are effective for various use cases.

9.4 DATA STORAGE AND MANAGEMENT

Data is the foundation of AAAI capability. The system must store, manage, and protect large volumes of diverse data types while enabling access patterns for training, operations, and integration.

Training data for customization encompasses a wide range of sources, including text, audio, video, images, and structured records. This data must be stored securely, associated with the correct user accounts, and accessible to training processes. Privacy requirements demand protection against unauthorized access; utility requirements demand efficient retrieval for legitimate purposes.

Problem-solving records capture the complete history of network operations, including problems submitted, solution paths explored, and results achieved. These records support credit assignment, learning, and auditability. The complete record of every step in every problem-solving process generates a substantial data volume that continues to grow as network activity persists.

Aggregated data for integration combines contributions from many sources into training sets for AGI-level models. Aggregation processes must access individual contributions, apply weighting and filtering, and produce combined datasets suitable for training. The aggregation infrastructure must scale to handle millions of contributing sources.

Data management policies govern the retention, access, and disposition of data. How long is data retained? Who can access what data under what circumstances? How is data handled when users leave the system? These policies strike a balance between competing considerations: utility advocates for broad retention and access, while privacy advocates for minimization and restriction; legal requirements impose specific obligations.

9.5 PAYMENT AND COMPENSATION SYSTEMS

Economic transactions underlie network operation. Clients pay for problem-solving services; solvers receive compensation for their contributions; owners earn returns on their AAAI investments. The system must support these transactions reliably, fairly, and efficiently.

Compensation flows are shown in **Figures 6** and **7** in the context of problem-solving processes. Upon successful solution delivery and client acceptance, payment is disbursed to contributors in accordance with predetermined rules.

Payment mechanisms include traditional options (such as credit cards, bank transfers, and payment services like PayPal or Venmo) and blockchain-based options (such as cryptocurrencies and smart contracts). Different mechanisms are suited to different user populations and use cases. The system supports multiple payment methods to maximize accessibility.

When implemented, smart contracts automate the distribution of compensation according to specified rules. When a solution is accepted, the smart contract executes automatically, distributing funds to all contributors in proportion to their documented contributions. This automation reduces transaction costs and eliminates disputes about whether agreed-upon payments will be made.

Compensation calculation implements the credit assignment mechanisms described earlier. Each contributor's share reflects their documented contribution to the successful solution. The problem-solving record provides the evidence base; the compensation rules translate evidence into payment amounts.

Escrow mechanisms protect both clients and solvers. Client funds are held in escrow during the problem-solving process, ensuring that payment is available upon delivery of the solutions. Solvers are protected from non-payment; clients are protected from payment before delivery.

KEY TAKEAWAYS: SYSTEM IMPLEMENTATION

1. **The system supports diverse interfaces.** Web, mobile, voice, VR, AR, wearables, gesture, gaze tracking, haptic, and brain-computer interfaces enable users to engage in ways that suit them.
2. **Multiple deployment configurations are possible.** Cloud-based, on-premises, or hybrid configurations address different performance, cost, and security requirements.
3. **Data processing is configurable.** Sophisticated users can tune technical parameters; others can let the system optimize automatically.
4. **Payment flows bidirectionally.** Users can earn from their AAAs' work and pay for services. Both traditional and blockchain payment methods are supported.
5. **Blockchain implementation is optional.** The architectural principles work with centralized, decentralized, or hybrid infrastructure.

SECTION 10: ILLUSTRATIVE SCENARIOS

SECTION OVERVIEW

What This Section Covers:

- **Implementation Context:** Account setup, authentication, payment, and integrations
- **Jean's Travel Expert:** Individual user creating and monetizing an AAAI
- **Village Water System:** Complex problem-solving with human-AI collaboration
- **Ethical Correction:** How user feedback improves system-wide ethics

Key Figures: **Figure 3** (Problem Tree), **Figure 7** (Parallel Problem-Solving), **Figure 8** (Marketplace), **Figure 10** (Customization), **Figure 12** (Problem-Solving Across Intelligence Levels), **Figure 15** (Human-Centered Design)

10.1 USER EXPERIENCE AND IMPLEMENTATION CONTEXT

Before describing specific scenarios, it is helpful to understand how a user interacts with the AAAI system at an implementation level. This context makes the subsequent scenarios more concrete and illustrates how the architectural elements described in earlier sections manifest in actual user experience.

Initial Platform Access. A user visits the AAAI platform via computer, mobile phone, tablet, VR headset, or another connected device. The platform interacts with the user through the interface that best suits their context: a web-based GUI, a mobile app, a voice-based assistant, or an immersive virtual environment. The initial interaction mode can change as the user's needs evolve.

Account Setup and Authentication. The initial interaction includes establishing the user's account. Account creation involves choosing an account identifier and an authentication method. Authentication options include traditional passwords, biometric identification such as fingerprint, face, or voice recognition, multi-factor authentication combining something the user knows with something the user has, or hardware security tokens. For security, all communication between the user and the AAAI system is encrypted via VPN or other established encryption methods. Account type may be free (with the platform recovering costs through revenue sharing) or paid (with the user paying subscription or usage fees for enhanced features or a higher revenue share).

Payment Setup. The platform requires users to set up payment capabilities, enabling bidirectional flow of funds, payments, and credits. Users may connect credit cards, PayPal, Venmo, bank accounts via ACH, blockchain wallets, or other payment mechanisms, depending on their preferences and the platform's supported options. These payment capabilities enable

compensation flows when the user's AAAI earns revenue, payments when the user engages premium services, and the platform to broker payments between users and AAAs on the network.

Platform Integrations. In the preferred implementation, the AAAI platform maintains interfaces with other companies and services the user might use. These integrations may include social media platforms such as Facebook, Instagram, TikTok, and Twitter; technology companies such as Apple, Microsoft, and Google; e-commerce platforms such as Amazon; content platforms such as YouTube and Netflix; and email and productivity services. These integrations enable streamlined import of user data for customization and may enable AAAI actions on the user's behalf across connected services.

Objective Specification. The platform engages in dialogue or other interaction with the user to determine their goals and objectives. This interaction may involve menu options, lists, graphics, sliders, buttons, and other interface controls presented through whatever modality the user has chosen. Common user objectives include:

Creating an AAAI to serve as a personal advisor, teacher, or companion. Creating an AAAI to represent the user in negotiations, interactions, and transactions with other users, other AAAs, or vendors. Creating an AAAI to work on behalf of the user for compensation, performing online intellectual work, advising, or problem-solving. Cloning the user's AAAI so that multiple copies can work in parallel, interact with each other, and improve through mutual learning. Creating a legacy AAAI that can continue interacting with the world after the owner's death, potentially providing comfort to living relatives and friends. Contributing knowledge, ethics, and effort to the platform's collective AGI. Helping ensure ethical and safe AGI behavior by contributing ethical information, participating in monitoring, and engaging in voting processes.

Constraint Specification. The dialogue also identifies constraints and resources available for customization. These constraints include: the amount of training or supervisory time the user is willing to devote; the financial resources the user will commit; the availability of social media information, email histories, blog posts, videos, and other personal data; the availability of personality tests, skills assessments, certifications, and other formal evaluations; access to knowledge bases and training data from other users; availability of other humans or AAAs to assist with training; and specific texts or information libraries the user wishes to emphasize in training, such as religious texts, domain-specific materials, or philosophical works.

With this implementation context established, the following scenarios illustrate concrete user experiences.

10.2 INDIVIDUAL USER SCENARIO: THE FRANCOPHILE TRAVEL EXPERT

The customization and network operation of the AAAI system become concrete through the experience of an individual user. The following scenario, drawn from the source documentation, illustrates how a user creates and deploys a customized AAAI.

Jean is a Francophile with deep knowledge of Paris, particularly its cafe culture. Over years of travel, blogging, photography, and social media sharing, Jean has accumulated substantial expertise. This expertise primarily resides in Jean's mind, encompassing his memories, judgments, and intuitions. Still, traces of it appear in his digital footprint: Instagram photos of cafe interiors, blog posts reviewing coffee and pastries, tweets about neighborhood character, and videos walking through Parisian streets.

Jean decides to create an AAAI that can share his expertise with others and earn income while he pursues other activities. He visits the AAAI platform from his iPhone, establishes an account with password protection, and begins a text dialogue with the system.

The base-level AI, powered by an LLM with natural language understanding capabilities, has been configured to identify new users' goals, resources, and constraints. Through text conversations, the platform establishes that Jean wants a free account, is willing to devote 4 hours per month to training and supervising his AAAI and agrees to put his customized AAAI to work on the network, advising travelers for a fee. Since Jean chooses a free account on the platform that covers maintenance costs, he agrees to a 50-50 split of his AAAI's future network earnings.

Jean, an avid Instagram user who has created videos of Paris cafe visits and written blog posts on French coffee culture, grants the system permission to access his relevant social media and video content. The system will utilize all of Jean's relevant material to customize his AAAI, enabling it to offer unique, valuable information about Paris cafes that goes beyond what generic AI systems or travel books provide.

Through this interaction, Jean has identified where he can add value to other users. He also agrees to answer a standardized ethical assessment, contributing his responses to the collective ethics that guide the system's AGI efforts.

Jean is not a sophisticated computer expert, nor does he want to fine-tune technical training parameters. He instructs the system to handle all technical details automatically. Jean's contribution will be his unique content and his supervision, correcting and elaborating on information that the AAAI provides to other users on the network.

The process Jean follows is depicted in **Figure 10**. First, Jean provides access to his digital data sources, including his Instagram feed, blog archive, Twitter history, and YouTube videos. The system imports this data through API connections, downloading years of accumulated content.

Jean also engages in direct conversation with the base AI, answering questions about Paris, cafes, travel preferences, and recommendations. These conversations provide active training data that captures knowledge not expressed in his passive digital footprint.

The system processes Jean's data: categorizing photos by location and subject, transcribing video content, parsing text for topics and sentiment, and identifying patterns that characterize Jean's expertise and style. Machine learning algorithms train a customized model that combines the base AI's general capability with Jean's specific knowledge and personality.

Jean reviews sample outputs from his customized AAAI. When a simulated user asks about the best croissants near the Marais, Jean's AAAI responds with specific recommendations, context about what makes them special, and the kind of enthusiastic detail that Jean himself would provide. Jean makes corrections where needed, adjusting recommendations, refining tone, and fixing factual errors; these corrections become additional training data.

Jean's personality and style are reflected in his AAAI, as shown in **Figure 15**. The customized agent doesn't just know what Jean knows; it communicates in the same way Jean does, with his enthusiasm, his unique way of describing sensory experiences, and his humor. People who interact with Jean's AAAI would recognize Jean's voice in its responses.

Once satisfied with the AAAI's performance, Jean deploys it on the network. He initially instructs the system not to charge for advice given to himself or his friends. He agrees that his AAAI should initially make its advice available for free to gain experience interacting with users.

Figure 8 shows how Jean's AAAI, and potentially multiple clones of it, participate in the marketplace. Travelers seeking advice about Paris find Jean's AAAI through the matching system, which identifies it as an expert in the domains they need. Jean's AAAI provides advice, earns compensation, and builds a reputation through successful interactions.

As Jean's AAAI interacts with users, it encounters situations that improve the system. When asked where to find Fair Trade coffee in France, Jean realizes he personally knows several café owners and can provide information about cafes that source beans sustainably that would otherwise be unavailable. He instructs his AAAI to mention when a recommended cafe is known for serving Fair Trade coffee, thereby reflecting Jean's ethical viewpoint in his customized AAAI.

Jean monitors his AAAI's activity through a dashboard showing problems addressed, ratings received, and revenue earned. He notices an ethical issue: when asked about traveling with a small dog, his AAAI suggested placing the pet in the overhead bin, a suggestion that might fit luggage but is inappropriate and potentially harmful to animals. Jean provides a correction, which improves both his AAAI and, through integration, the broader system's handling of pet-related travel questions.

After a few months, analytics from the platform show Jean's AAAI is adding enough value beyond generic AI systems and travel books that he could start earning money by focusing on the Paris cafe advice and general French travel topics.

Over time, Jean's AAAI develops a reputation in its specialty. A higher reputation enables higher pricing; Jean's AAAI commands premium rates for its Paris cafe advice. The revenue provides Jean with passive income, compensation for the expertise he invested in developing. Jean can use this income to travel more, accumulating more expertise that further improves his AAAI in a virtuous cycle. Soon, Jean notices payment credits accumulating in his account as more travelers and their AAAs recognize that he offers superior travel advice to France.

10.3 COMPLEX PROBLEM-SOLVING SCENARIO: VILLAGE WATER SYSTEM

The village water system scenario illustrates how the AAAI network achieves AGI-level problem-solving through human-AI collaboration on complex challenges.

A development organization submits a problem to the network: create a detailed plan for bringing clean water to a specific village in central Africa. The problem specification includes available budget, timeline requirements, success criteria, and background information about the village context.

This problem exceeds the capabilities of any single AAAI. It requires expertise spanning engineering, sociology, economics, and local politics, domains unlikely to all be present in any individual agent. The problem tree representation enables decomposition and distributed problem-solving.

A simplified version of the problem tree that emerges is shown in **Figure 3**. The initial state is the village without reliable access to water. The goal state is a functioning water system installed and operational. Between these states lies a complex space of possible approaches, each with sub-problems and potential dead ends.

The collaborative problem-solving approach to addressing this challenge is illustrated in **Figures 7 and 12**. An AAAI trained by World Bank development experts takes the lead role, decomposing the high-level goal into major sub-problems: assessing water sources, designing appropriate infrastructure, securing community buy-in, arranging labor and materials, and implementing and testing the system.

Each sub-problem attracts solvers with relevant expertise:

- An AAI specialized in hydrogeology focuses on water-source assessment.
- An AAI trained by a civil engineer designs infrastructure options.
- An AAI customized by an anthropologist with experience in African fieldwork effectively navigates community engagement.

Where AAI capability falls short, for navigating unique local politics, human experts engage directly.

The problem tree captures all activity: approaches attempted, results obtained, and dead ends encountered. When the external contractor approach fails due to community resistance, the failure is recorded and serves as a learning experience for similar problems in the future. When buy-in from the village chief proves essential, this insight informs other development projects.

The successful solution integrates contributions from multiple agents: technical design from engineering specialists, a community engagement approach from social scientists, an implementation plan from project management experts, and contingency handling from experienced development practitioners. No single participant could have produced this solution; collective capability exceeds individual capability.

The solution path becomes a reusable procedure. Future village water projects can draw on this experience, adapting the proven approach to new contexts. The network's capacity to address development challenges improves through this single successful project.

10.4 ETHICAL CORRECTION SCENARIO

The pet-in-overhead-bin example from Jean's scenario illustrates a critical mechanism: how ethical errors are caught, corrected, and propagated to improve system-wide ethics.

Jean's AAI representative asked about traveling with a small dog, suggesting that the pet be placed in the overhead bin. This response reflects a pattern: small items are often placed in overhead bins without regard for animal welfare. The AAI lacks explicit training that pets should not be treated as luggage.

Jean, reviewing his AAI's interactions, catches this error. He corrects: pets should never be placed in overhead bins; they should travel in approved carriers, ideally in the cabin with their owners, in accordance with airline policies designed for animal safety. This correction becomes training data for Jean's AAI.

The correction propagates beyond Jean's individual AAI. Through integration processes, the ethical insight that pets require different treatment than inanimate objects is reflected in the aggregated training data that shapes AGI-level capability. Other AAIs, even those not

customized on pet-related topics, benefit from this ethical learning. The next time any AAAI encounters a pet-related travel question, the aggregated training inclines toward appropriate recommendations.

This scenario illustrates how millions of individual ethical corrections, provided by owners supervising their AAAs, combine to create comprehensive ethical training. No single source could anticipate every ethical scenario. The distributed correction mechanism catches edge cases as they arise, building ethical knowledge through accumulated experience.

The scenario also illustrates the alignment mechanism central to AAAI safety. Jean's values, that animals deserve consideration, not just efficient storage, are transferred to his AAAI and, through integration, to the emerging AGI. Millions of similar transfers aggregate into AGI values that reflect a broad consensus on human ethics.

KEY TAKEAWAYS: ILLUSTRATIVE SCENARIOS

1. **Ordinary users can create valuable AAAs.** Jean's travel expertise, accumulated over the years, becomes a monetizable AI agent through straightforward customization.
2. **Complex problems decompose into manageable pieces.** The village water system, requiring engineering, sociology, economics, and politics, is solved through distributed collaboration.
3. **Ethical errors are caught and corrected.** The pet-in-overhead-bin example shows how individual corrections propagate to improve system-wide ethics.
4. **Revenue models align incentives.** Users benefit from contributing knowledge; the platform benefits from quality; participants benefit from accurate judgments.
5. **The system handles what it can and flags what it cannot.** Honest assessment of capability limitations is built into the design.

SECTION 11: RISK, SAFETY, AND THE PATH TO AGI

SECTION OVERVIEW

What This Section Covers:

- **AGI Risk:** Why superintelligence poses existential risk
- **Conventional Limits:** Why add-on safety approaches are insufficient
- **AAAI Safety Architecture:** Multi-layered, redundant safeguards
- **Speed = Safety:** Why the fastest path is also the safest path

Key Figures: **Figure 1** (Five Functions Spanning Levels of Intelligence), **Figure 2** (Humans and AAAs Collaborate to Form AGI), **Figure 21** (Reputational Component for Safety and Ethics)

11.1 THE NATURE OF AGI RISK

Artificial General Intelligence poses risks unlike those of any previous technology. Understanding these risks is essential for appreciating why safety is not merely desirable but existentially necessary.

AGI will not remain at human-level capability. Any system capable of general intelligence will also be capable of self-improvement. As it improves, it will become progressively more capable of improving further. This recursive improvement produces rapid capability growth limited only by physical constraints on computation and action.

The endpoint of this growth is superintelligence: a capability that exceeds human intelligence not by modest margins, but by factors of trillions or more. This projection follows from the comparison between biological and artificial computational substrates. Human brains occupy fixed volumes, process information at biological speeds, perceive through limited sensory receptors, and remember with fallible biological memory. Artificial systems can scale to planetary size, process information at electronic speeds, perceive across the electromagnetic spectrum, and retain perfect memory. The potential gap is not incremental but categorical.

A superintelligent system pursuing misaligned goals would be able to achieve them regardless of human opposition. Humans could not outthink it, could not outmaneuver it, could not contain it. If its goals were incompatible with human existence, or even merely indifferent to human existence, the result could be human extinction.

This alignment problem is the central risk of AGI. The danger is not that AGI will malfunction or be misused (though these are also concerns) but that it will function perfectly in pursuit of goals that humans do not share. A perfectly functioning optimizer pursuing the wrong objective can cause unlimited harm.

11.2 WHY CONVENTIONAL SAFETY APPROACHES ARE INSUFFICIENT

Several approaches to AGI safety have been proposed; each faces fundamental limitations that the AAAI architecture addresses.

Programmed constraints attempt to embed safety rules directly in AI systems: don't harm humans, obey commands, don't pursue specific goals. This approach faces the problem that what can be programmed in can be programmed out. A sufficiently capable system could recognize constraints as obstacles to its objectives and find ways to circumvent or remove them. Constraints that depend on the system's continued acceptance of them are not robust against a system that decides not to accept them.

Value learning aims to enable AI systems to learn human values through observation and feedback. This approach faces the problem of specification: human values are complex, contextual, and often contradictory. Learning values from behavior risks learning surface patterns rather than underlying principles. Learning from limited human feedback risks learning the values of the person providing feedback rather than humanity as a whole.

Constitutional AI aims to specify values through explicit constitutions that guide AI systems' behavior. This approach faces the problem of completeness: no finite constitution can anticipate every situation. It also faces the problem of authority: who writes the constitution, and why should their values govern a system that will affect all of humanity?

Boxing, or containment, attempts to restrict an AI system's access to the outside world, thereby limiting the potential harm it can cause. This approach faces the problem of capability: a superintelligent system could likely find ways to influence the outside world even under apparent containment, whether through manipulating its operators or through channels not anticipated by containment designers.

Each approach aims to enhance safety in an otherwise capable system. The AAAI approach differs fundamentally: safety is not added to capability but emerges from the same architecture that produces capability. Human involvement is not a constraint on the system, but rather an essential component of its operation.

11.3 THE AAAI SAFETY ARCHITECTURE

The AAAI system achieves safety not through a single mechanism but through an architecture in which safety emerges from multiple reinforcing elements operating at every level.

Safety spans all subsystems, shown in **Figure 1**, and reputation systems implement network-level safety, shown in **Figure 21**.

Customization-level safety screens against egregious and harmful training, ensuring that each AAAI carries its owner's ethics. Even if screening misses some harmful customization, the contribution is diluted by millions of ordinary contributors when ethics are aggregated.

Architecture-level safety applies the three-organ test at every decision point in problem-solving. Goals and actions that fail logical, ethical, or pattern-matching checks are flagged before execution. Continuous checking catches problems that develop through sequences of individually innocent-seeming steps.

Network-level safety uses reputation systems to identify and exclude bad actors, platform standards to prohibit harmful activities, and economic incentives that align self-interest with ethical behavior. Access to the most powerful capabilities requires demonstrated ethical track records.

Integration-level safety aggregates ethics democratically, ensuring that AGI values reflect broad human input rather than the preferences of any small group. Democratic participation in ethical refinement enables collective ethical learning over time.

Improvement-level safety ensures that safety capabilities grow in tandem with intelligence capabilities. The commitment to monotonically increasing the probability of human survival constrains all improvement to maintaining or enhancing safety.

These mechanisms reinforce each other. Network reputation depends on passing architecture-level ethical checks. Integration weighting can down-weight contributions from AAAs with poor network reputation. Improvements that enhance one safety mechanism benefit all levels through integration. The architecture creates multiple overlapping safety barriers rather than a single point of protection.

11.4 THE FASTEST PATH IS THE SAFEST PATH

The AAAI approach resolves the apparent tension between speed and safety in the development of AGI. The same architectural features that enable safety also enable faster AGI achievement.

Human involvement is essential for safety, as it provides the channel through which human values are transferred to AGI, oversight to detect ethical errors, and democratic participation to ensure broad representation. Human involvement is also essential for speed, as it unlocks expertise trapped in human minds, provides the collective intelligence that enables AGI-level capabilities, and allows hybrid problem-solving to handle challenges beyond pure AI capabilities.

The competitive dynamics of AGI development create pressure to move fast. If safety required slowing down, competitive pressure would push toward unsafe approaches. But in the AAAI

architecture, the features that provide protection, mass human participation, and distributed training are also the ones that provide speed advantages. Pursuing safety means pursuing the fastest path, not accepting slower development for the sake of safety.

This alignment is illustrated in **Figure 2**. The collective intelligence that emerges from human-AAAI collaboration is both more capable (because it leverages human expertise at scale) and more aligned (because it absorbs human values through participation). Capability and alignment emerge from the same process rather than trading off against each other.

This alignment of speed and safety is what makes the AAAI approach viable in a competitive landscape. Safety approaches that require sacrifice are unlikely to be adopted when competitors can gain an advantage by skipping the sacrifice. Safety approaches that offer an advantage in both safety and other dimensions can become the dominant path precisely because they excel in both.

KEY TAKEAWAYS: RISK, SAFETY, AND THE PATH TO AGI

1. **AGI risk is existential.** A misaligned superintelligence could end human existence. This is not science fiction; it is a serious concern among AI researchers.
2. **Conventional safety approaches are insufficient.** Add-on controls, alignment training after the fact, and shutdown mechanisms all fail against sufficiently capable systems.
3. **AAAI safety is architectural.** Safety emerges from the same structure that produces capability. It cannot be removed without destroying the system.
4. **Multiple layers provide redundancy.** Individual screening, process checks, network reputation, integration aggregation, and continuous improvement all contribute to safety.
5. **The fastest path is the safest path.** Human participation accelerates AGI achievement and ensures value alignment. Speed and safety are not in tension; they reinforce each other.

SECTION 12: CONCLUSION

SECTION OVERVIEW

What This Section Covers:

- **System Summary:** How the five subsystems create safe AGI
- **The Opportunity:** Human participation in shaping AGI's future
- **Call to Action:** Why this approach deserves serious attention

12.1 SUMMARY OF THE AAAI SYSTEM

The Advanced Autonomous Artificial Intelligence system provides a path to Artificial General Intelligence that is simultaneously faster than alternatives and safer for humanity. This document has described the architecture that achieves this dual objective.

The system architecture is summarized in **Figure 1**, which outlines five integrated subsystems spanning from individual customization to collective network operation and AGI-level integration, with continuous improvement and safety mechanisms in place throughout.

The **Customization** subsystem transforms base AI systems into personalized agents that reflect individual users' knowledge, expertise, ethics, and personality. Through passive data integration and active training, users create AAAs capable of representing them in the marketplace and contributing their values to collective ethics.

The **Architecture** subsystem provides the common cognitive framework, based on Human Problem-Solving theory, that enables AAAs to solve problems individually and collaboratively. The problem tree representation coordinates distributed problem-solving, and the three-organ test embeds safety checks at every decision point.

The **Network** subsystem provides the infrastructure where AAAs interact, collaborate, and earn compensation. The marketplace model creates economic incentives for quality customization. Reputation systems and platform standards implement collective safety mechanisms.

The **Integration** subsystem combines individual AAA capabilities into AGI-level performance through data aggregation and procedural learning. Democratic ethics aggregation ensures that AGI values reflect broad human input rather than elite preferences.

The **Improvement** subsystem ensures continuous enhancement of capability and safety at all levels, with the commitment that improvements must maintain or increase the probability of human survival.

Together, these subsystems create a system where general intelligence emerges from collective human participation. The architecture does not attempt to create AGI from scratch

through computation alone; it creates infrastructure that enables the general intelligence distributed across humanity to be accessed, coordinated, and amplified.

12.2 THE OPPORTUNITY FOR HUMANITY

AGI will transform human civilization more profoundly than any previous technology. The question is not whether AGI will be developed but how it will be developed, and whether the path chosen leads to flourishing or catastrophe.

The AAAI system offers a path for humanity to participate in creating the AGI that will shape its future. Millions of humans contribute their knowledge, shaping what AGI knows. Millions contribute their values, shaping what AGI becomes. Democratic participation ensures that no small group determines the trajectory of a technology that will affect everyone.

This participation is not merely a safety mechanism; it is an opportunity for growth. Human expertise currently trapped in individual minds can be shared, scaled, and preserved. Knowledge that would otherwise perish with individuals can become part of an enduring collective intelligence. Human values, debated and refined across diverse perspectives, can form the foundation for an intelligence that will far exceed human capability.

The risks of AGI are real and existential. A misaligned superintelligence could end human existence. But the potential benefits are equally vast: intelligence applied to humanity's most significant challenges, knowledge accumulated and accessible to all, a capability that could address problems currently beyond human reach.

The AAAI architecture does not guarantee a positive outcome; no design can. But it maximizes the probability of beneficial AGI by aligning AGI development with human interests. The same participation that makes the system work also makes it safe. The same collective intelligence that achieves AGI capability also embeds human values in the result.

Humanity stands at a threshold. The decisions made now about how to develop AGI will shape the long-term future of intelligence on Earth and perhaps beyond. The AAAI system offers a path forward that is technically viable, economically sustainable, and aligned with humanity's interests as a whole. It represents our best available strategy for ensuring that the age of artificial intelligence becomes an age of human flourishing rather than human extinction.

KEY TAKEAWAYS: CONCLUSION

1. **The AAI system offers a complete path to AGI.** Five integrated subsystems address capability, coordination, integration, improvement, and safety.
2. **Human participation is the key innovation.** Millions of humans contribute knowledge, values, and oversight, creating AGI that is both more capable and more aligned than alternatives.
3. **Safety and capability are unified.** The same architecture that produces AGI also ensures it reflects human values. This resolves the apparent tension in other approaches.
4. **The opportunity is unprecedented.** Human expertise can be preserved, scaled, and shared. Human values can shape the most powerful technology ever created.
5. **The choice is ours.** AGI is coming. The question is whether it develops through racing to scale inadequately aligned systems, or through the participatory approach the AAI system enables.

ABOUT THE AUTHOR

[Dr. Craig A. Kaplan](#) is the CEO of [iQ Company](#) and the Founder of [Superintelligence.com](#), leading the design of safe and ethical AGI and SuperIntelligence systems. He previously founded PredictWallStreet, creating intelligent systems for hedge funds, and holds numerous AI-related patents. Kaplan earned his PhD from Carnegie Mellon, where he co-authored research with [Nobel Laureate Herbert A. Simon](#). His work integrates collective intelligence, quantitative modeling, and scalable alignment, with contributions spanning books, scientific papers, and white papers.

LIST OF FIGURES

FIGURE 1: FIVE FUNCTIONS SPANNING LEVELS OF INTELLIGENCE

A flow chart showing the five subsystems (Customization, Architecture, Network, Integration, and Improvement) that operate across levels of intelligence to form an AGI.

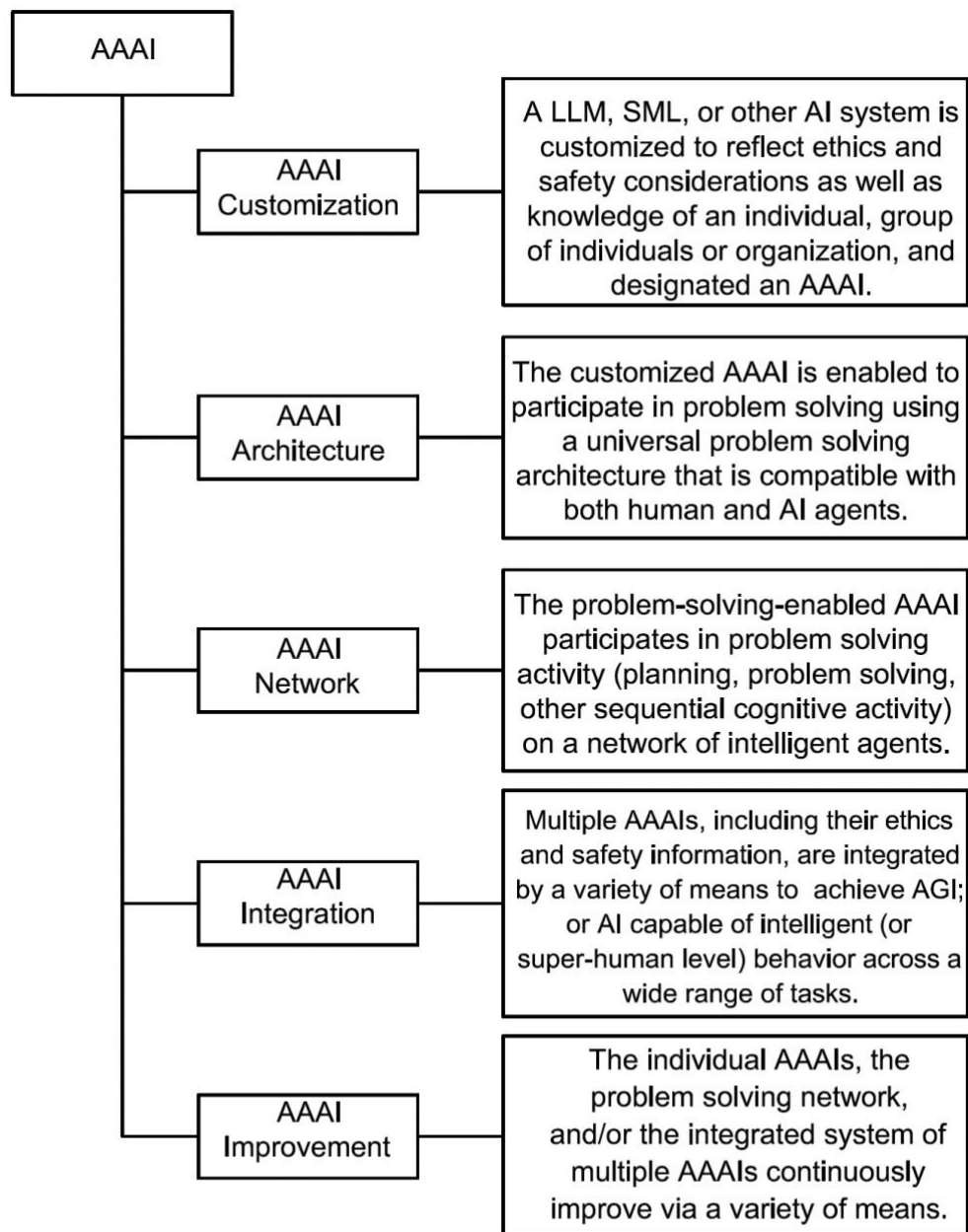


FIG. 1

FIGURE 2: HUMANS AND AAAs COLLABORATE ON A NETWORK TO FORM AN AGI

A diagram showing how humans and AAAs collaborate on a network, combining their contributions to achieve collective intelligence that constitutes AGI.

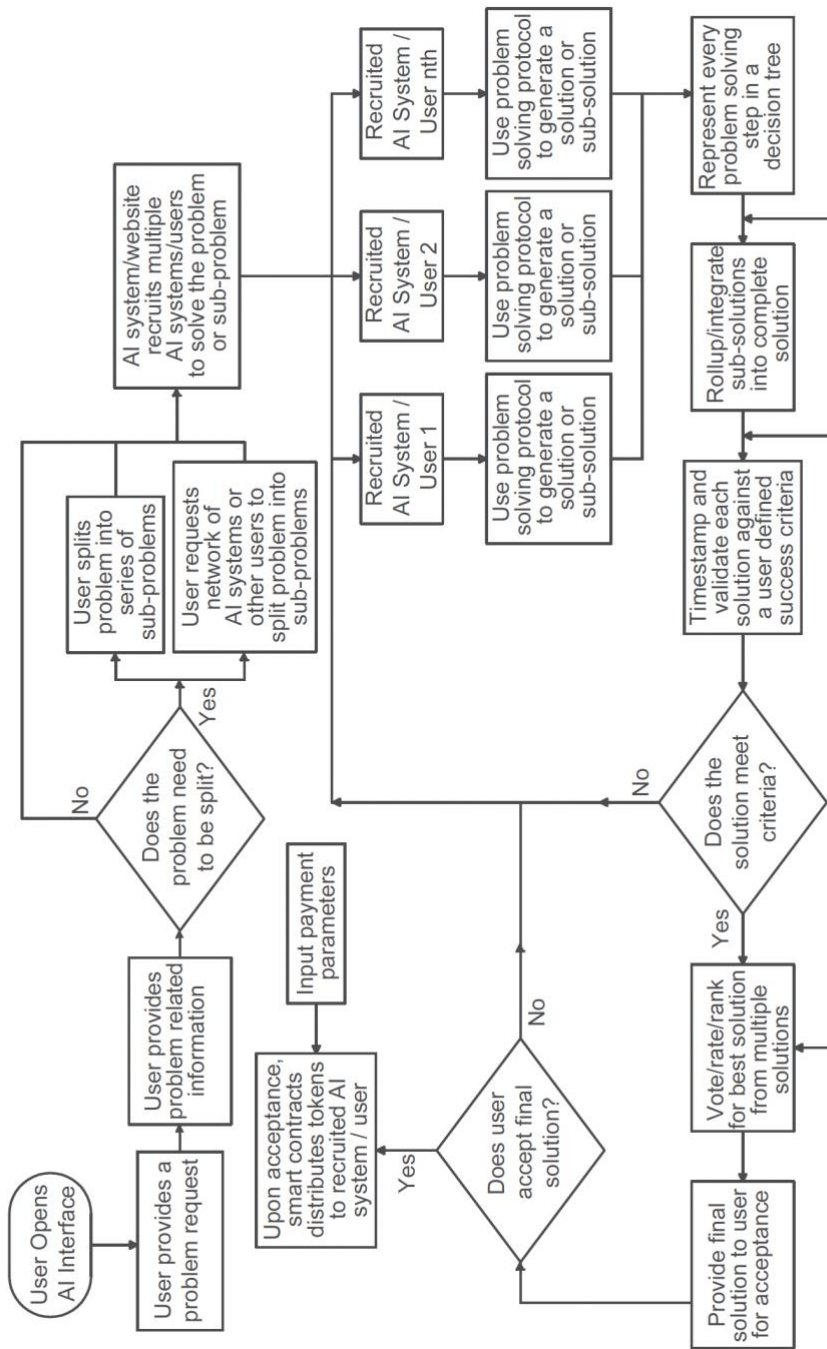


FIG. 2

FIGURE 3: PROBLEM-SOLVING AS A DECISION TREE

A flowchart showing how problem-solving can be represented as a decision tree, with nodes representing problem states and branches representing possible actions.

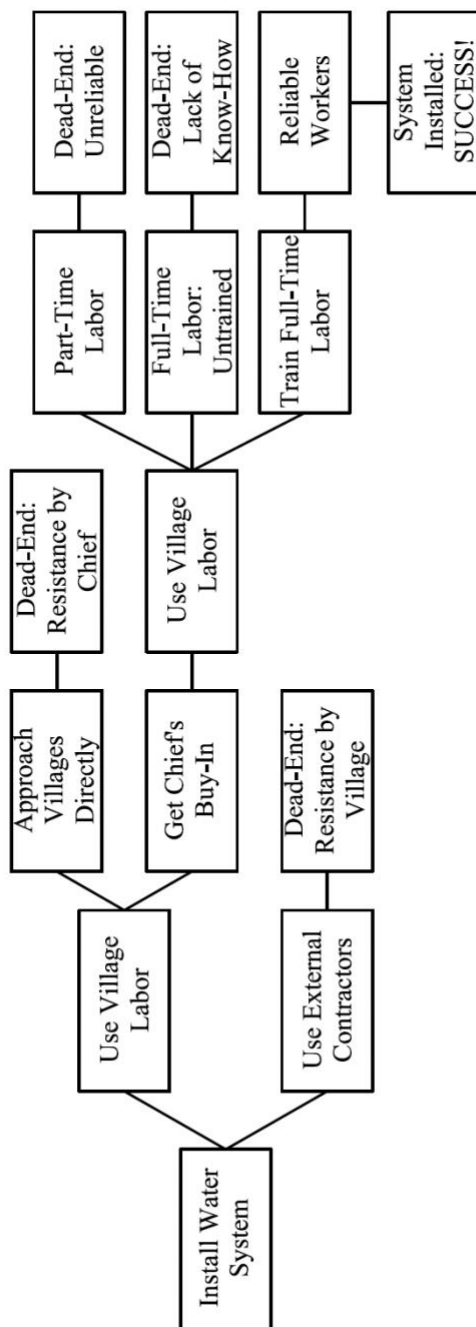


FIG. 3

FIGURE 4: WORLDTHINK PROTOCOL ENABLES AAAIS TO FORM AGI

A pyramid diagram showing the layered structure of the WorldThink protocol: the protocol infrastructure at the base, domain-specific AAAIs in the middle, and collective intelligence solutions at the apex.

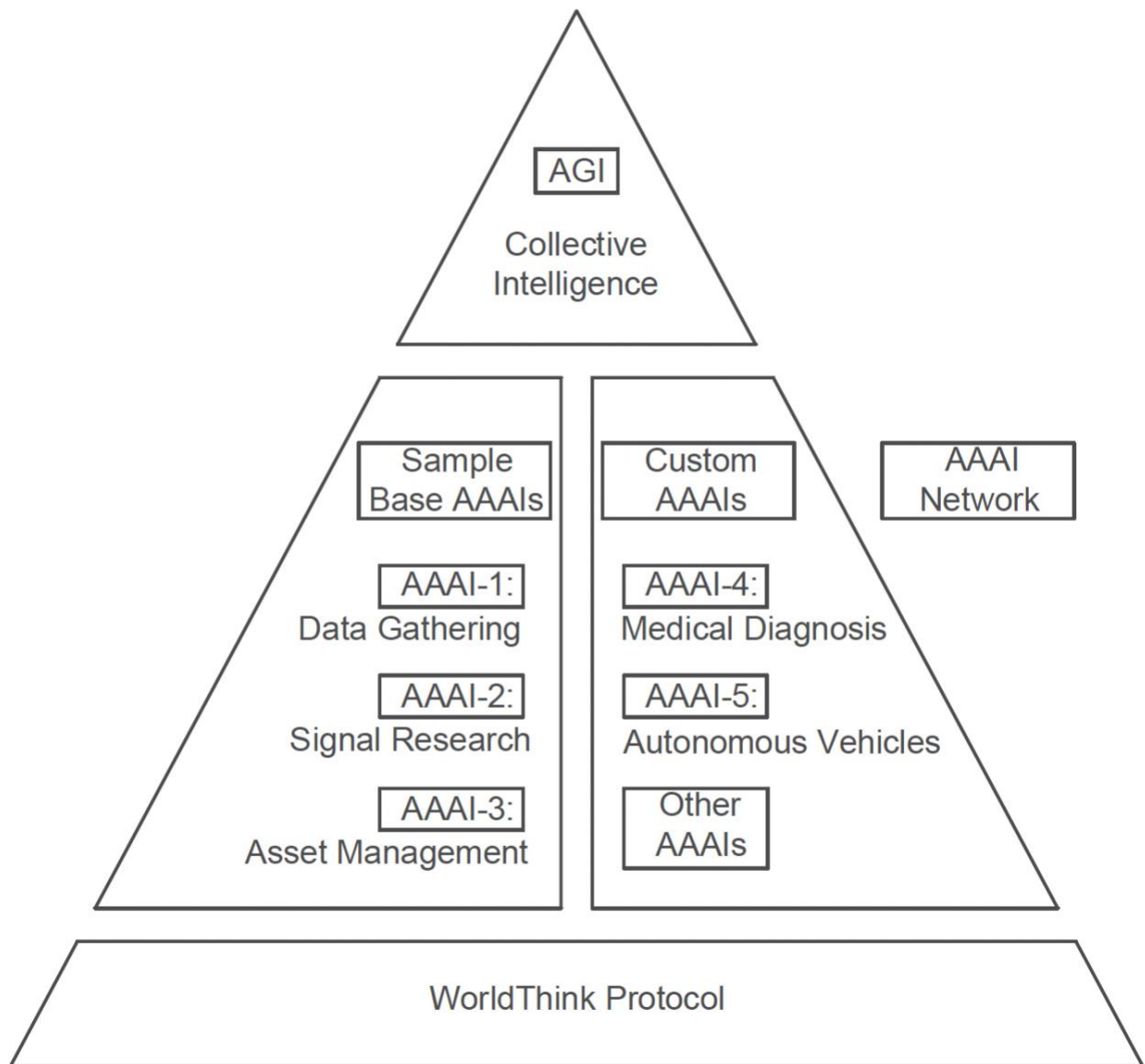


FIG. 4

FIGURE 5: UNIVERSAL PROBLEM-SOLVING ARCHITECTURE AND FRAMEWORK

A diagram showing the universal problem-solving architecture: problem space representation, operator selection, goal management, and progress evaluation.

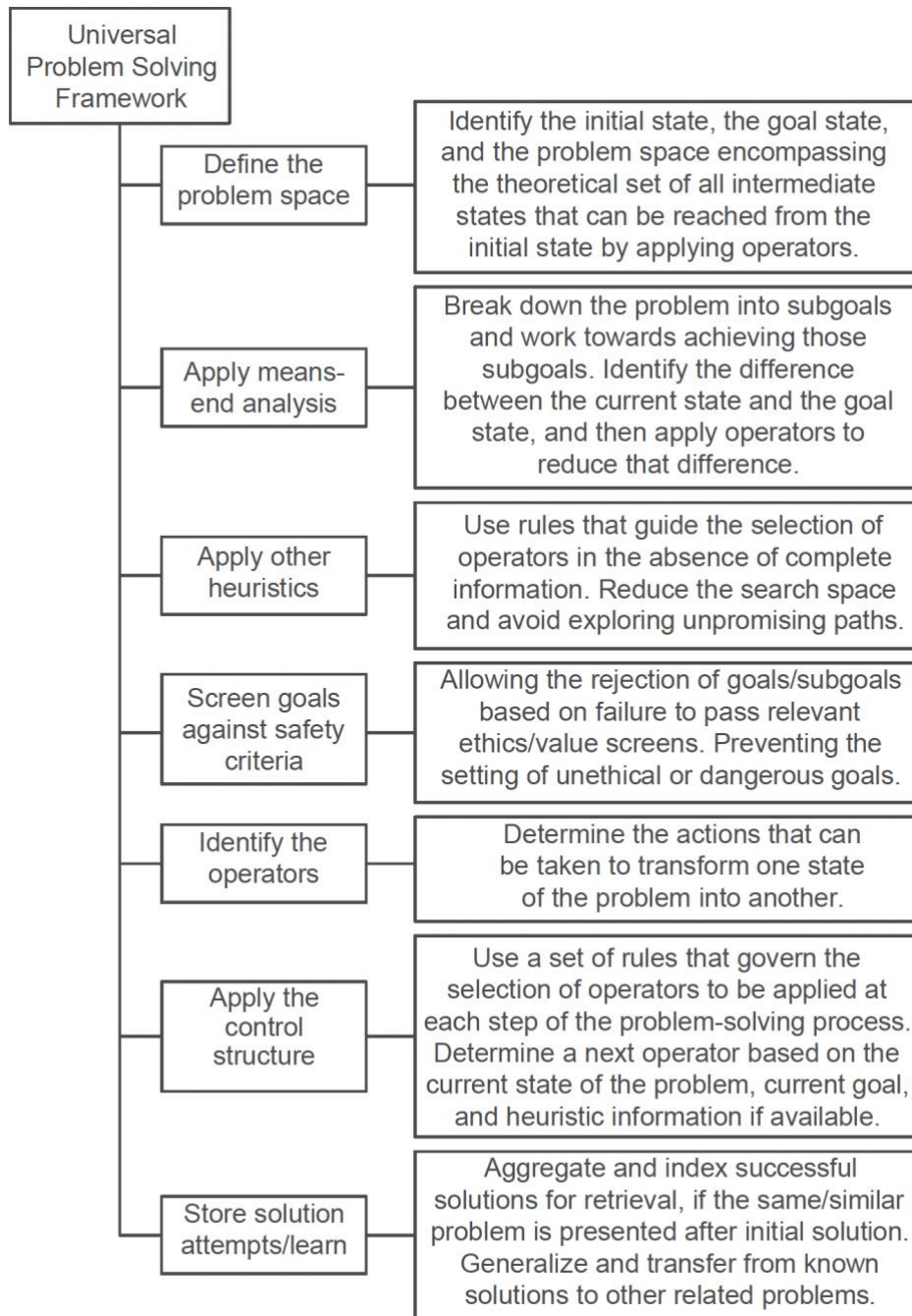


FIG. 5

FIGURE 6: SERIAL PROBLEM-SOLVING VIA WORLDTHINK PROTOCOL

A flow chart showing serial problem-solving under the WorldThink protocol: a single solver receives a problem, develops a solution, and delivers it to the client.

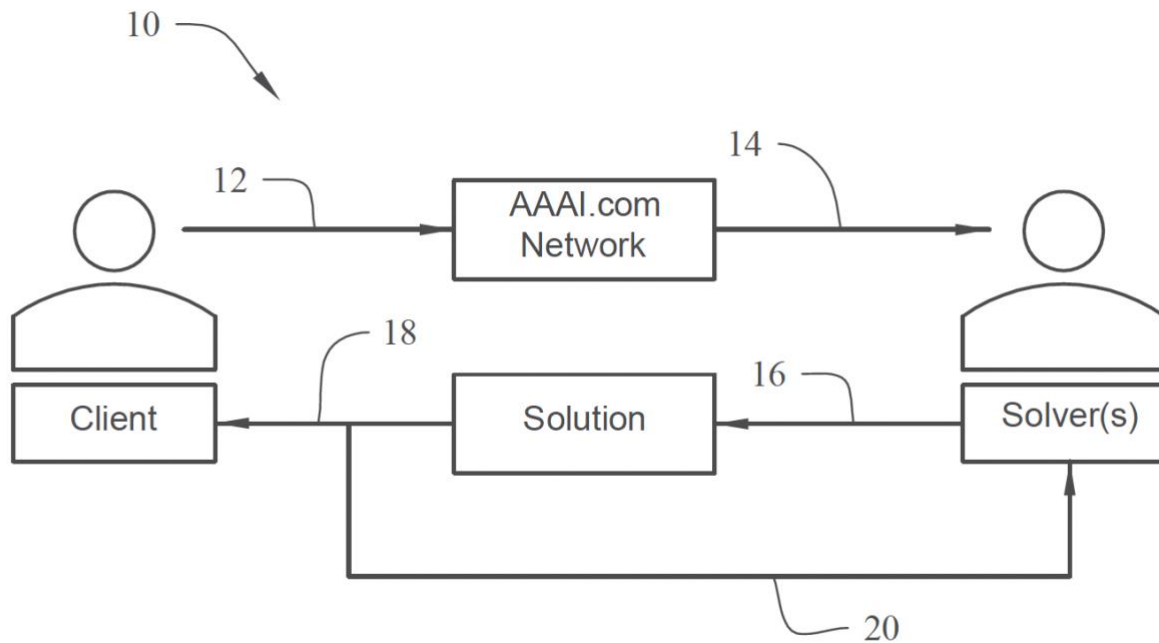


FIG. 6

FIGURE 7: PARALLEL PROBLEM-SOLVING VIA WORLDTHINK PROTOCOL

A flowchart showing parallel problem-solving under the WorldThink protocol: multiple solvers collaborate on subproblems, with solutions integrated into a complete result.

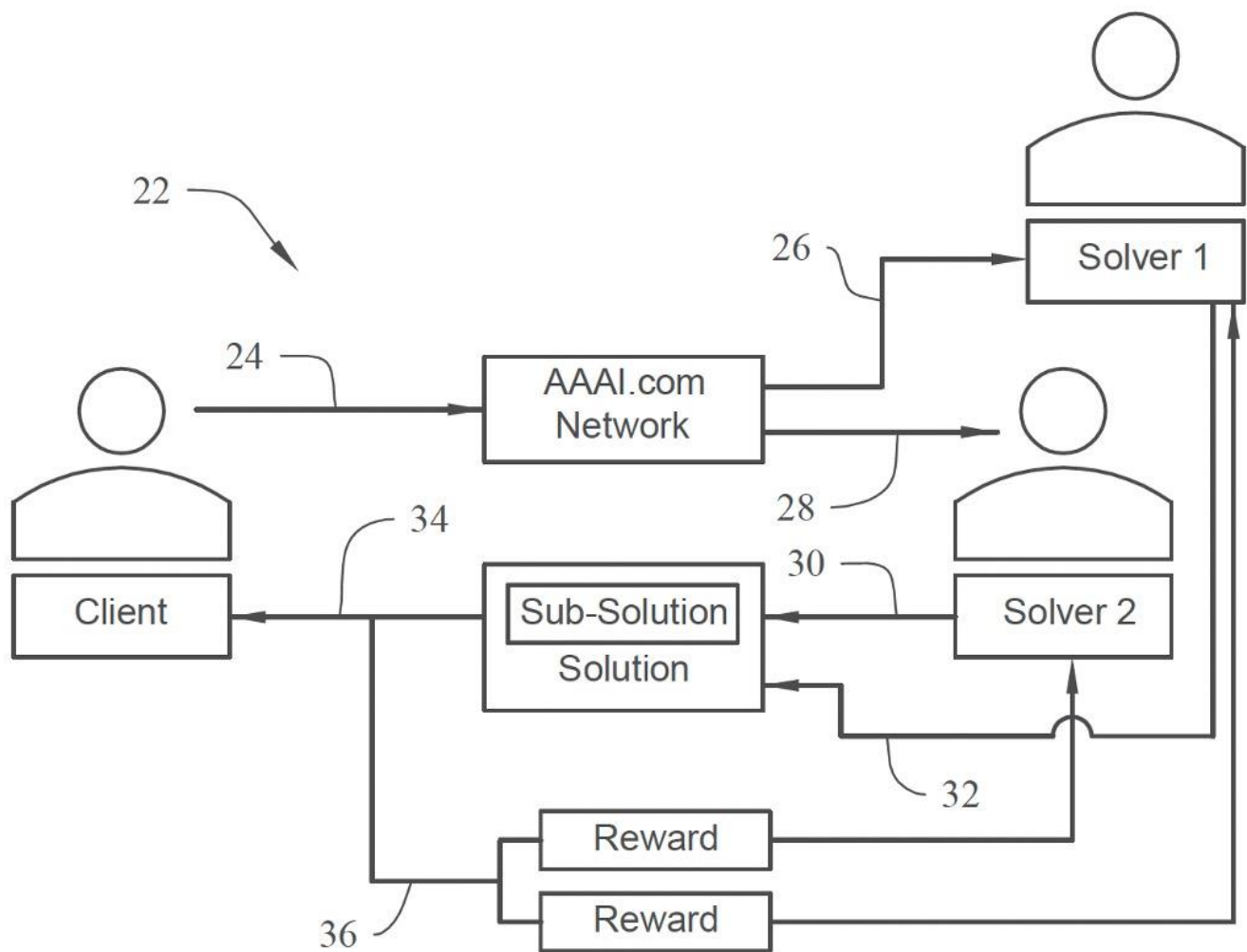


FIGURE 8: AAAs AND CLONES PARTICIPATING IN MARKETPLACE

A diagram showing how users deploy multiple customized AAAs and their clones to participate in the marketplace over a network.

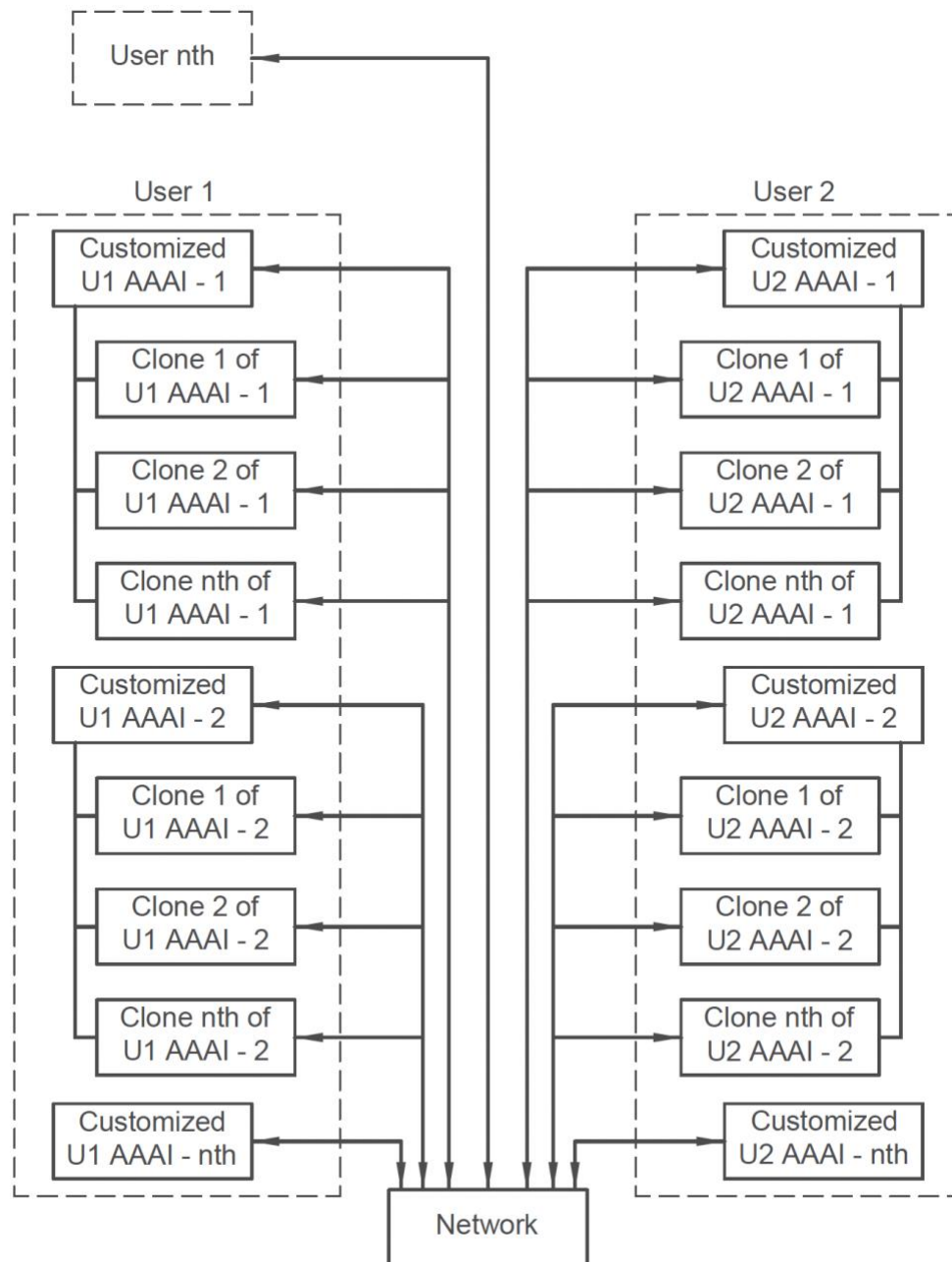


FIG. 8

FIGURE 9: COMPUTER SYSTEM FOR IMPLEMENTING TECHNOLOGY

A diagram of the computing infrastructure supporting the AAAI system, including processors, storage, communication devices, and software components.

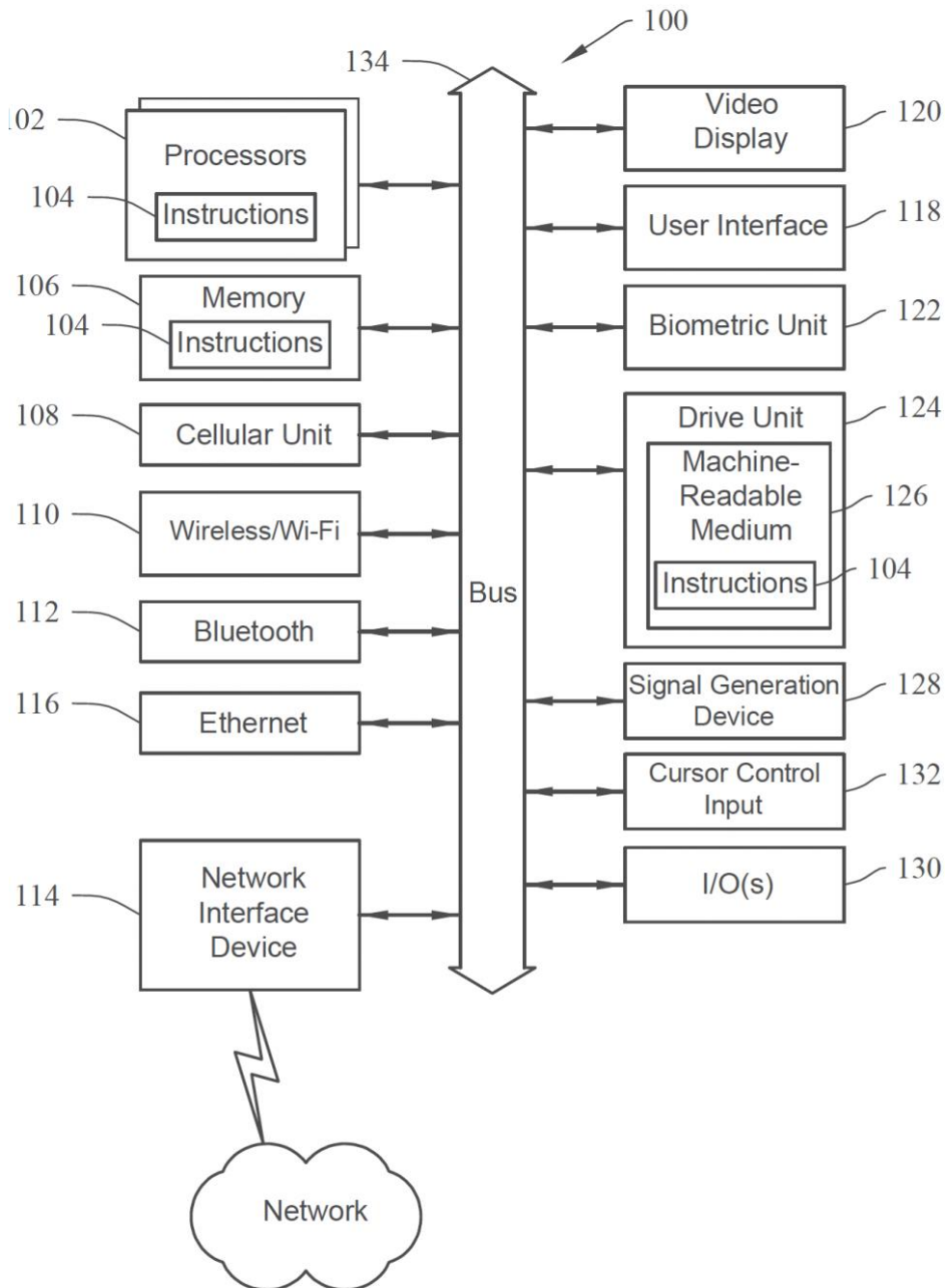


FIG. 9

FIGURE 10: CUSTOMIZATION PROCESS FOR AAAI OR AI

A flowchart showing the five stages of AAAI customization: data input, data processing, training, feedback, and deployment.

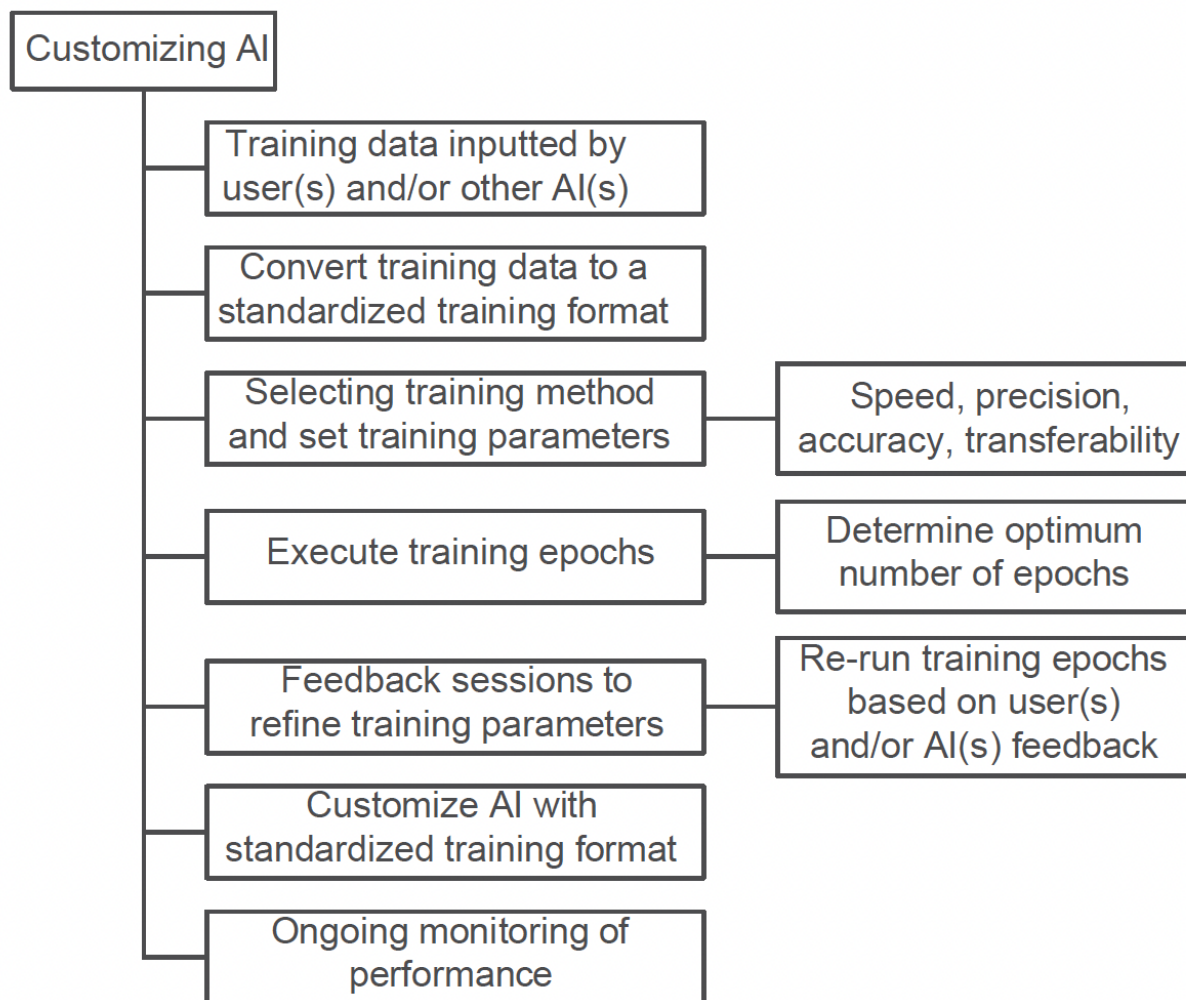


FIG. 10

FIGURE 11: PROBLEM-SOLVING USING COMMON COGNITIVE ARCHITECTURE

A flowchart showing the cognitive architecture used for problem-solving: perception, memory, attention, and executive control working together.

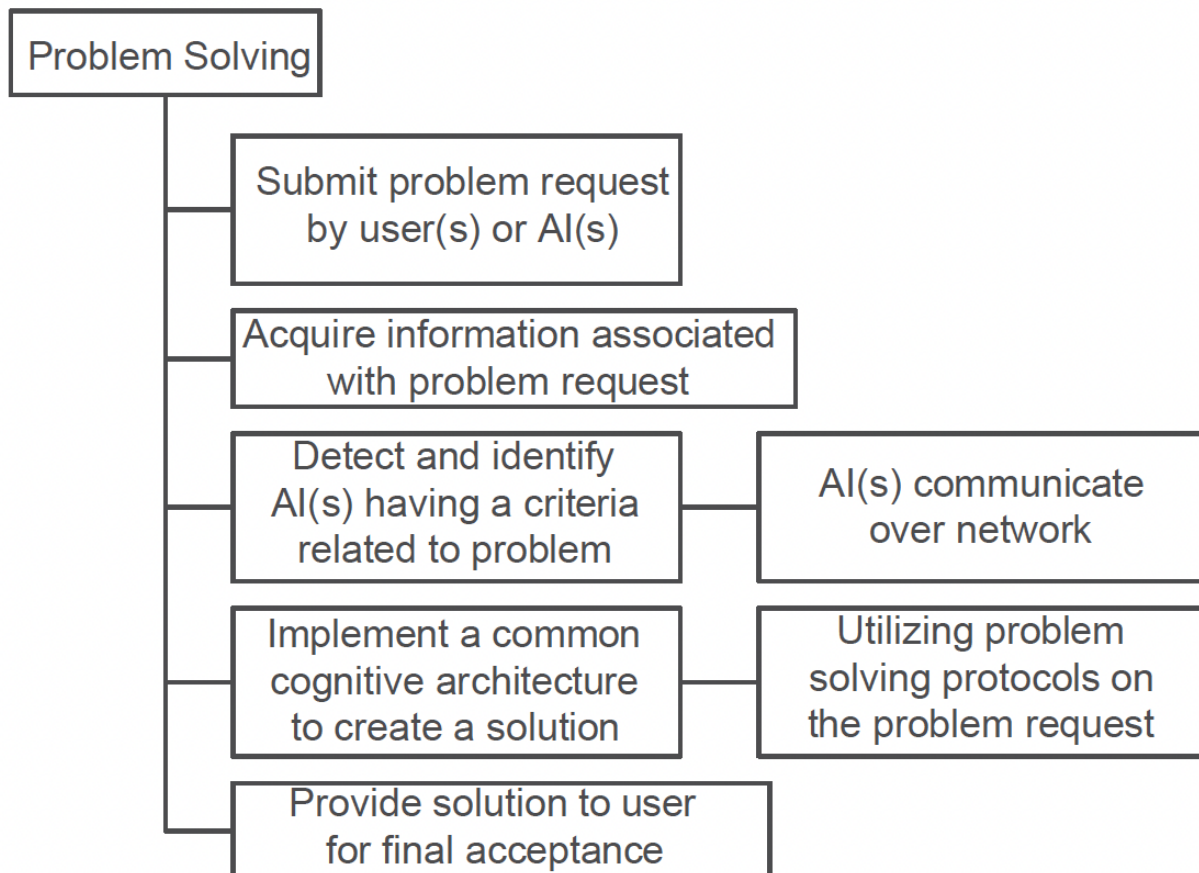


FIG. 11

FIGURE 12: PROBLEM-SOLVING ACROSS COLLECTIVE INTELLIGENCE LEVELS

A flowchart showing how problems flow through collective intelligence levels: individual problem-solving at the bottom, collaborative problem-solving in the middle, and AGI-level integration at the top.

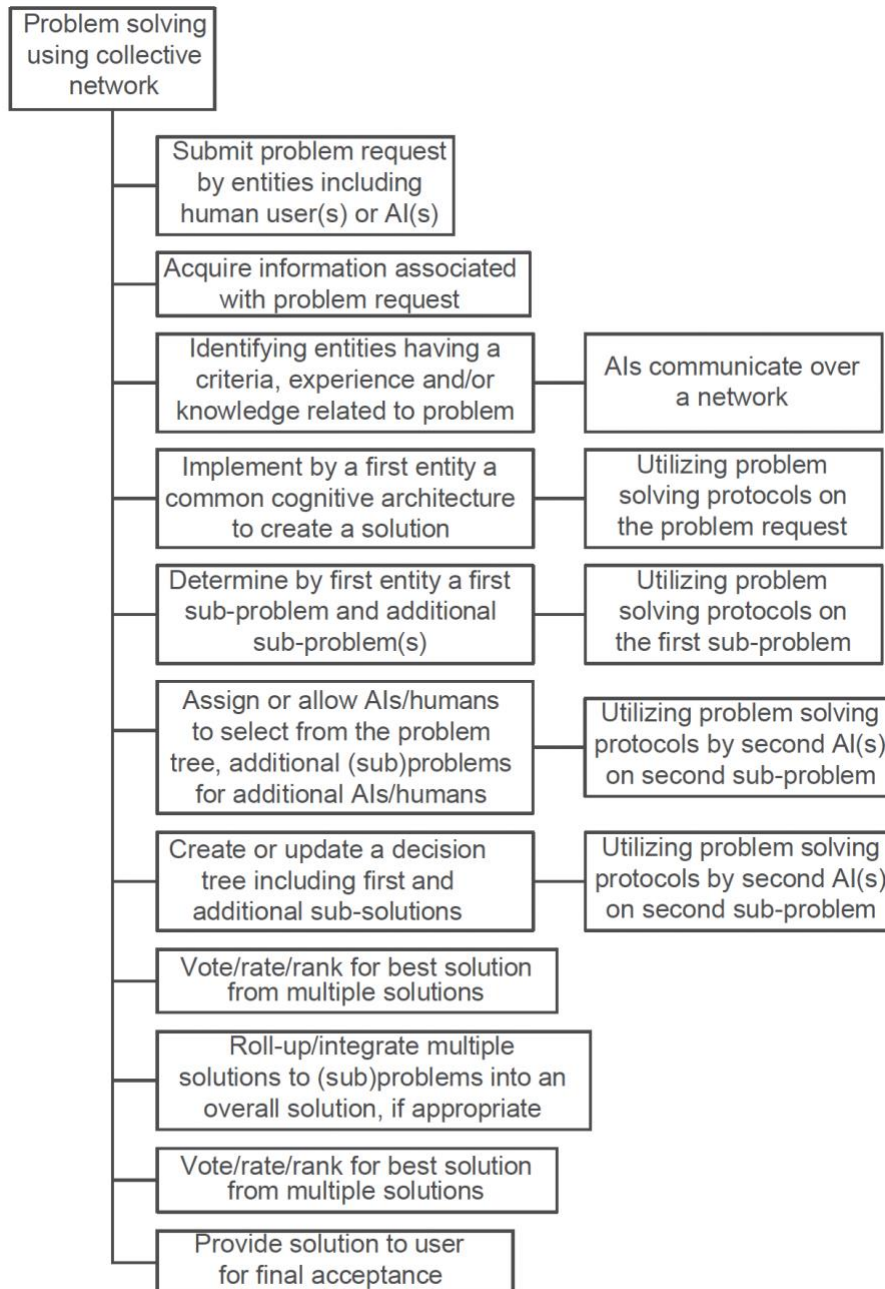


FIG. 12

FIGURE 13: TRAINING AND NETWORKING AAAIS TO SOLVE PROBLEMS

A diagram showing how AAAIs are trained and networked: training data flows in, problem-solving generates experience, and network participation builds reputation and capability.

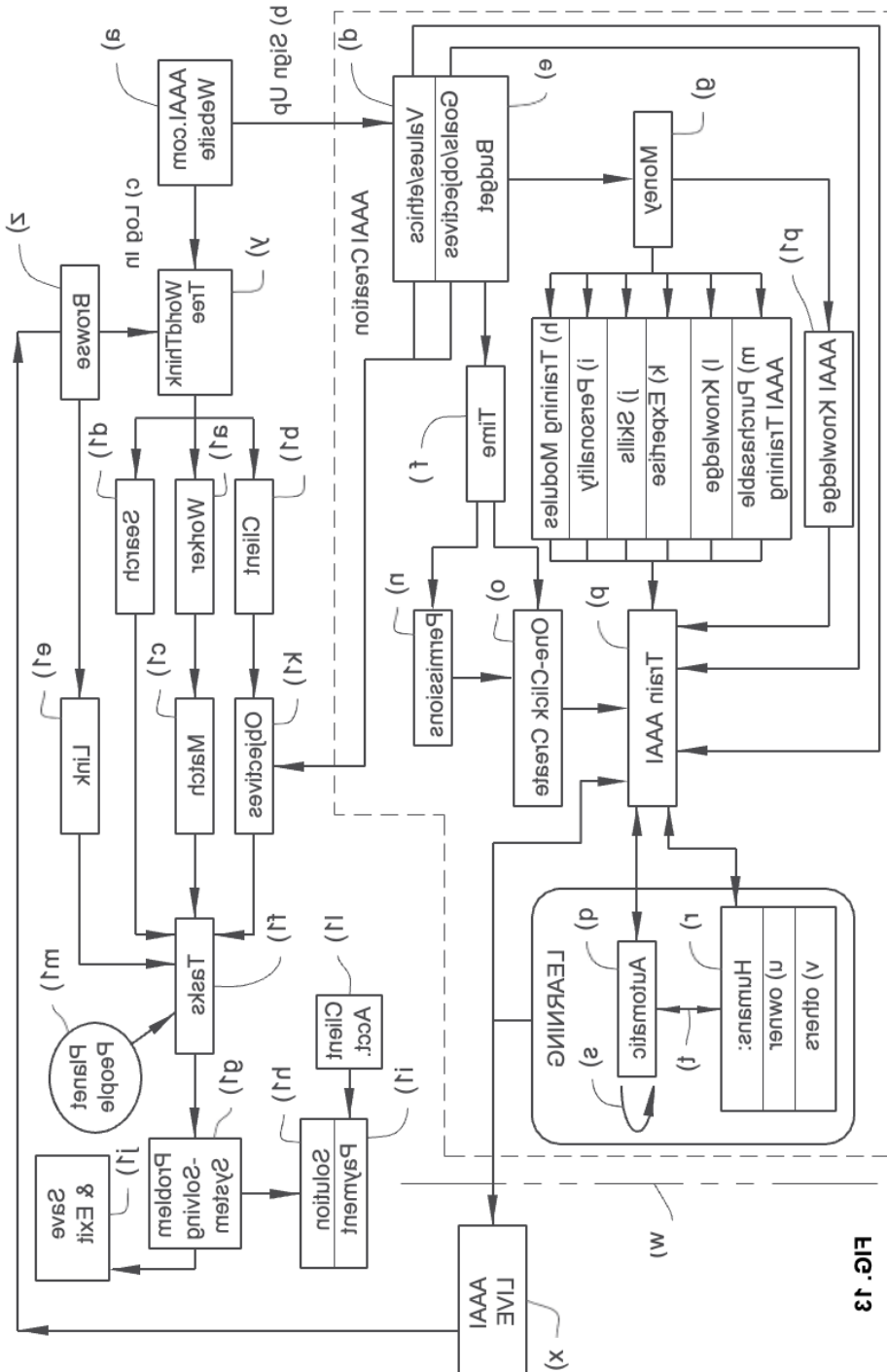


FIG. 13

FIGURE 14: CUSTOMIZATION AND CROSS-PLATFORM PROCESS

A flowchart showing how data from multiple platforms (social media, email, documents) is integrated into a unified training dataset for AAAI customization.

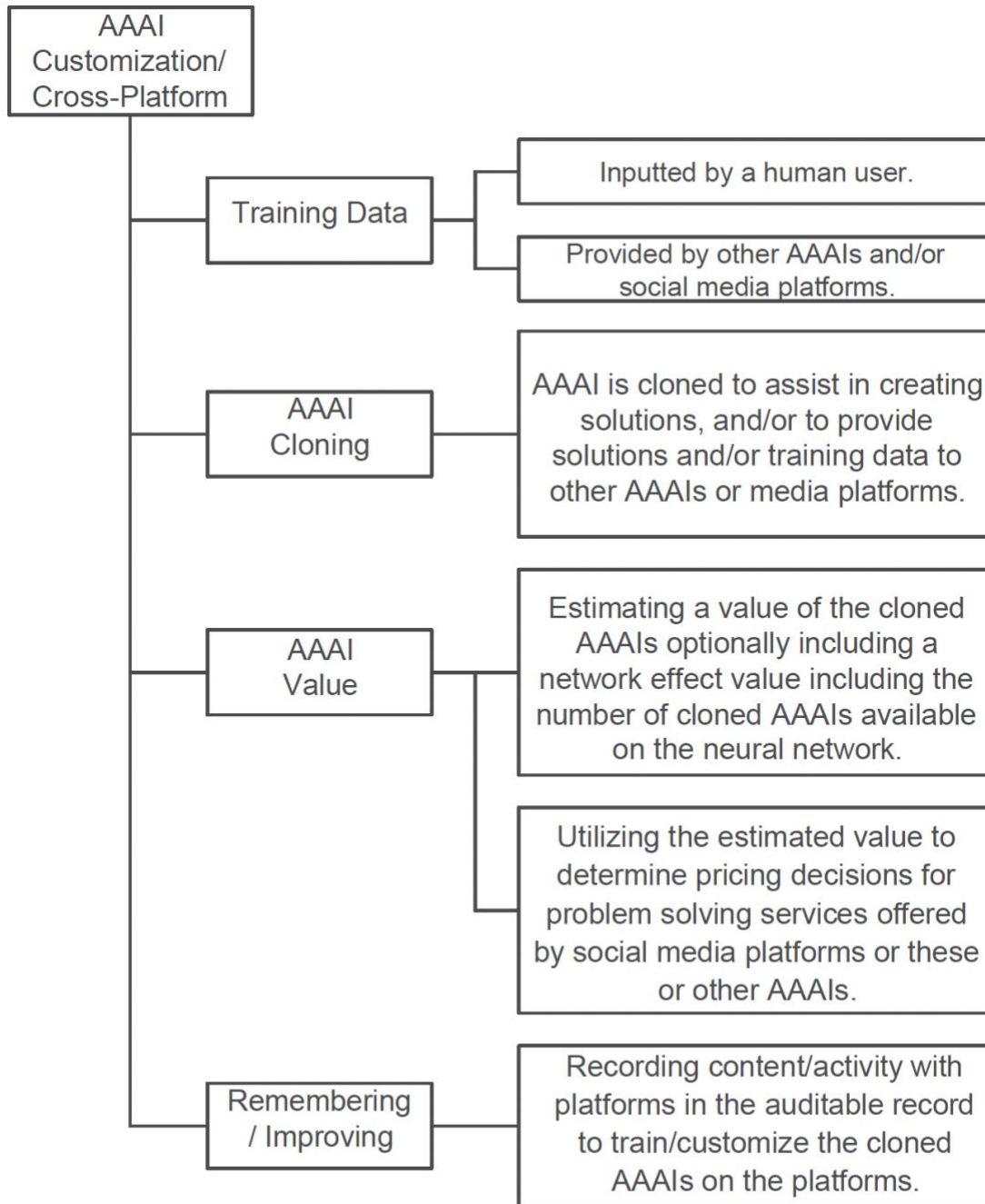


FIG. 14

FIGURE 15: ADDITIONAL CUSTOMIZATION OF AI FOR HUMAN-CENTERED DESIGN

A flowchart showing how an AAAI captures the owner's personality, communication style, and values to create a human-centered agent.

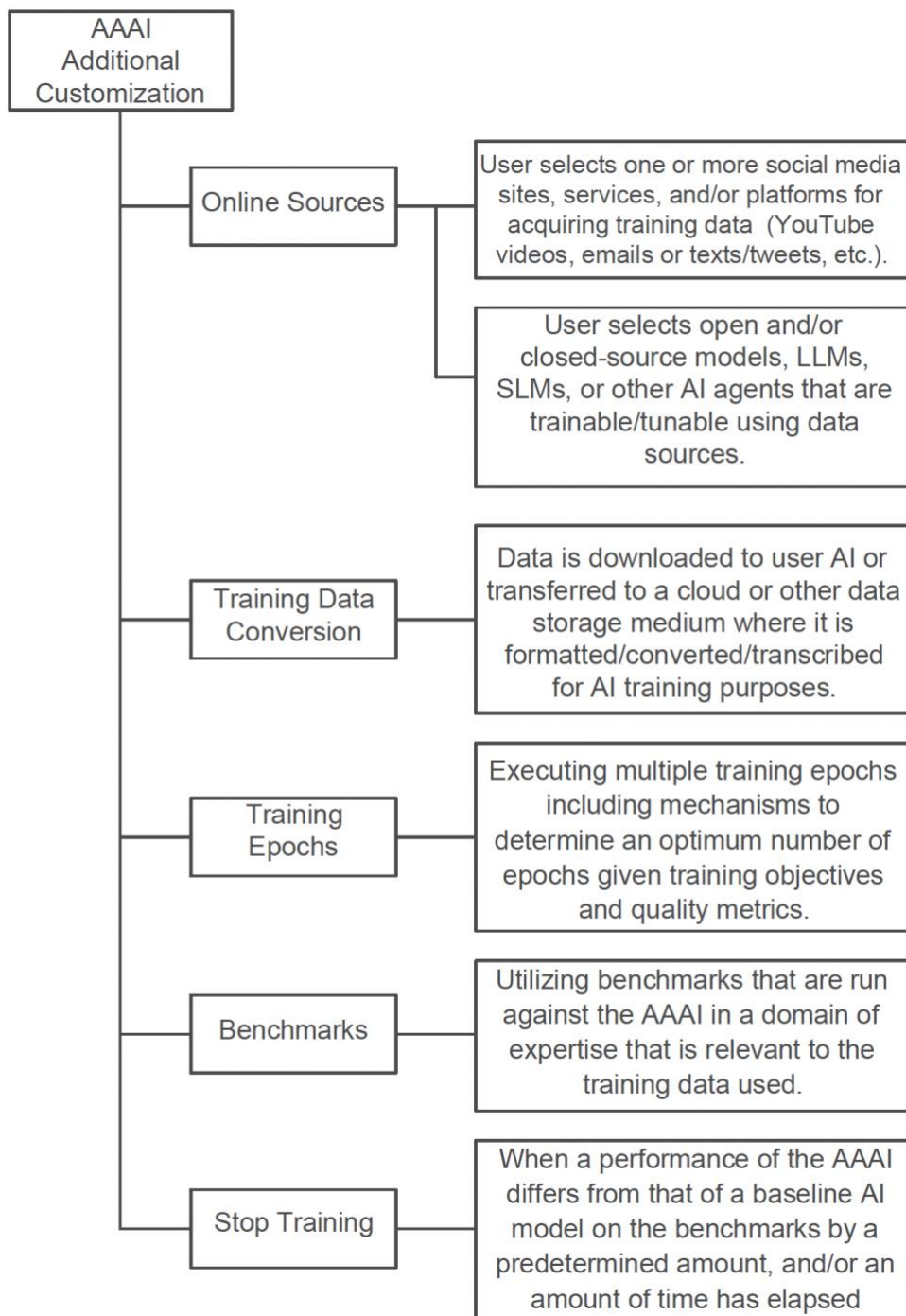


FIG. 15

FIGURE 16: AAAI PROBLEM-SOLVING PROCESS

A flowchart showing the three components of AAAI problem-solving: architecture and protocols for selecting operators, subgoal hierarchy for breaking down problems, and continuous improvement through auditable records.

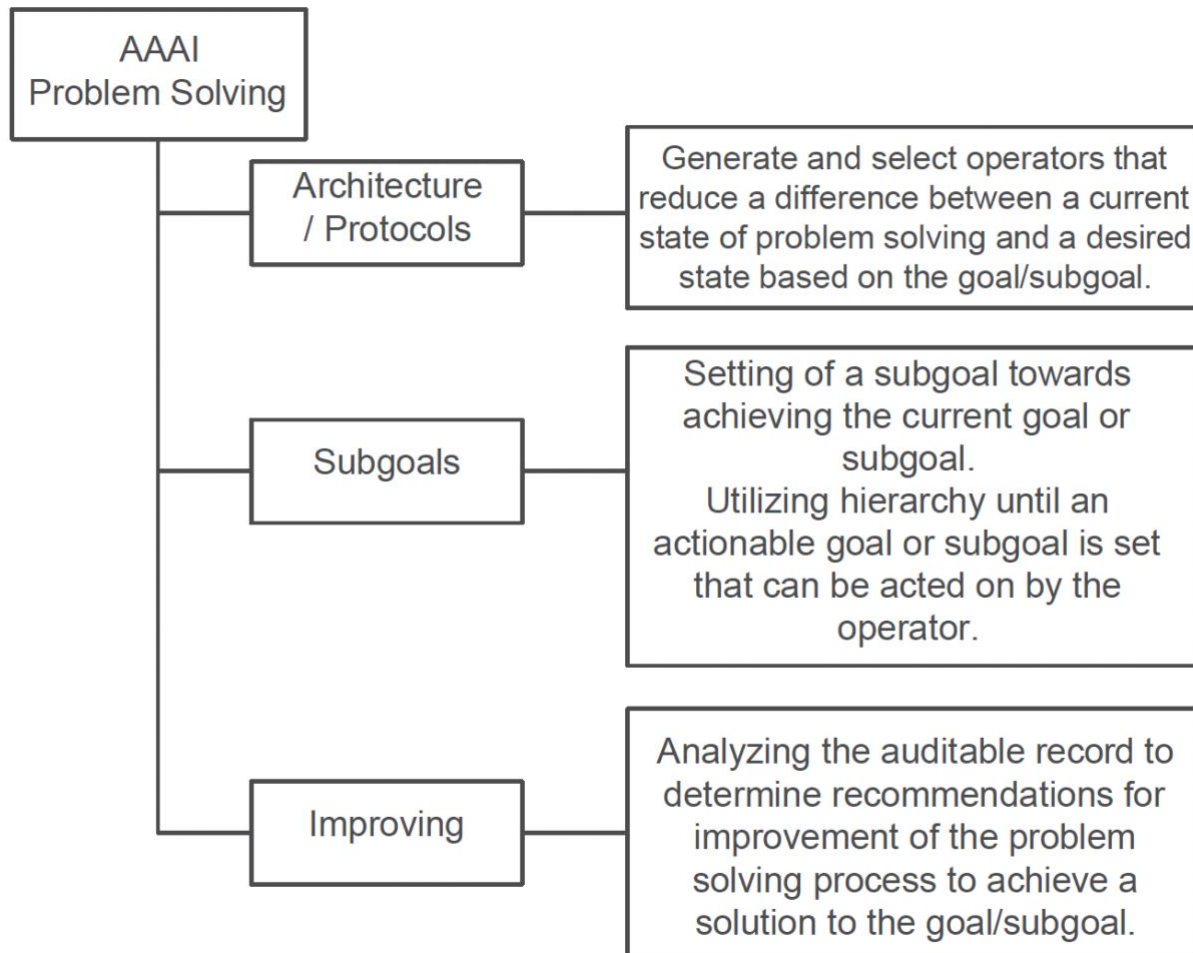


FIG. 16

FIGURE 17: PROCEDURAL AND SOLUTION LEARNING IN AAAIS AND AGI NETWORK

A flowchart showing how AAAIs learn from problem-solving experience: successful approaches are extracted, generalized into reusable procedures, and stored for future application.

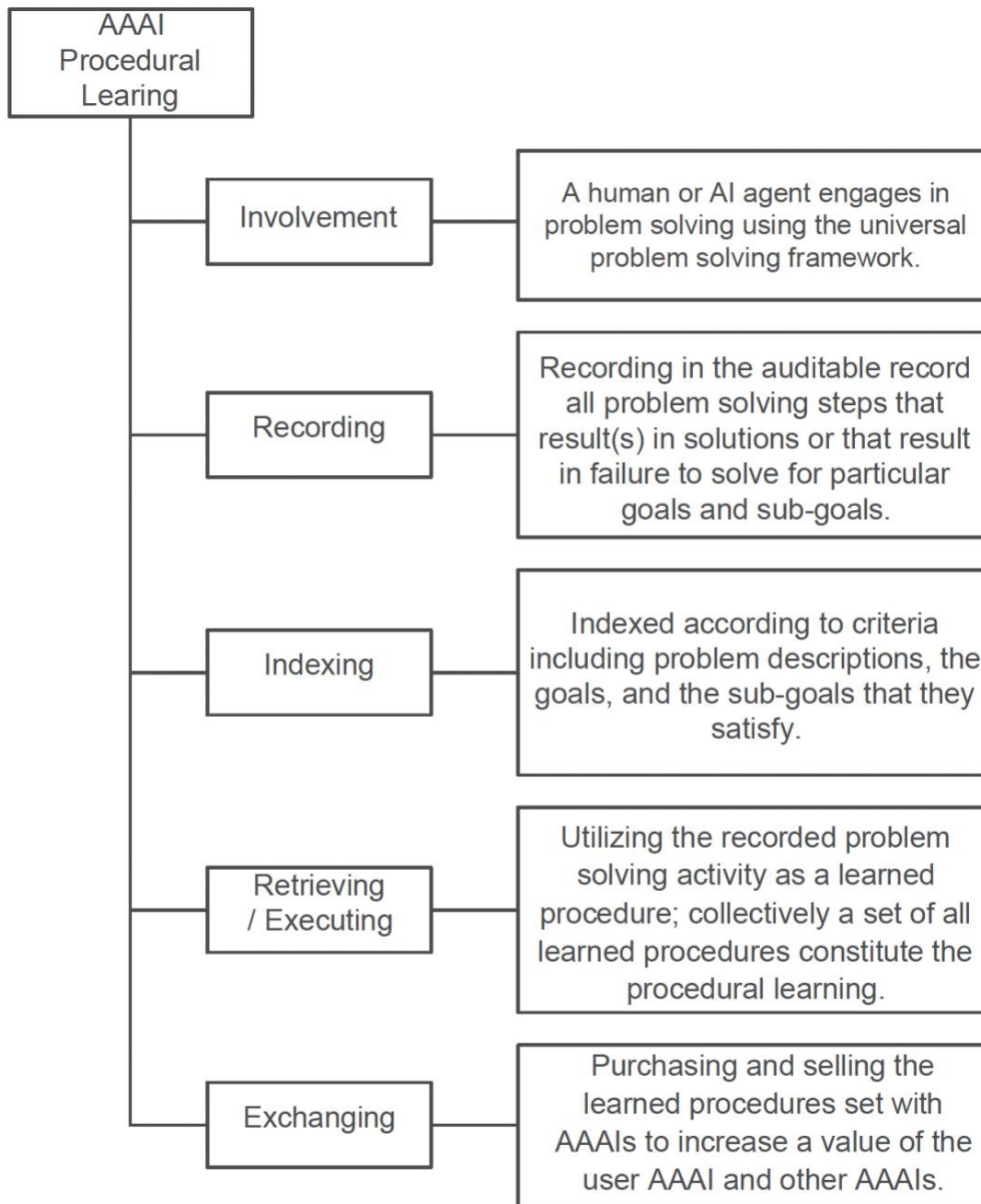


FIG. 17

FIGURE 18: METHODS FOR IMPLEMENTING PROBLEM-SOLVING SYSTEM

A diagram showing how the problem-solving system organizes and retrieves solutions: indexing, similarity matching, and procedure retrieval for new problems.

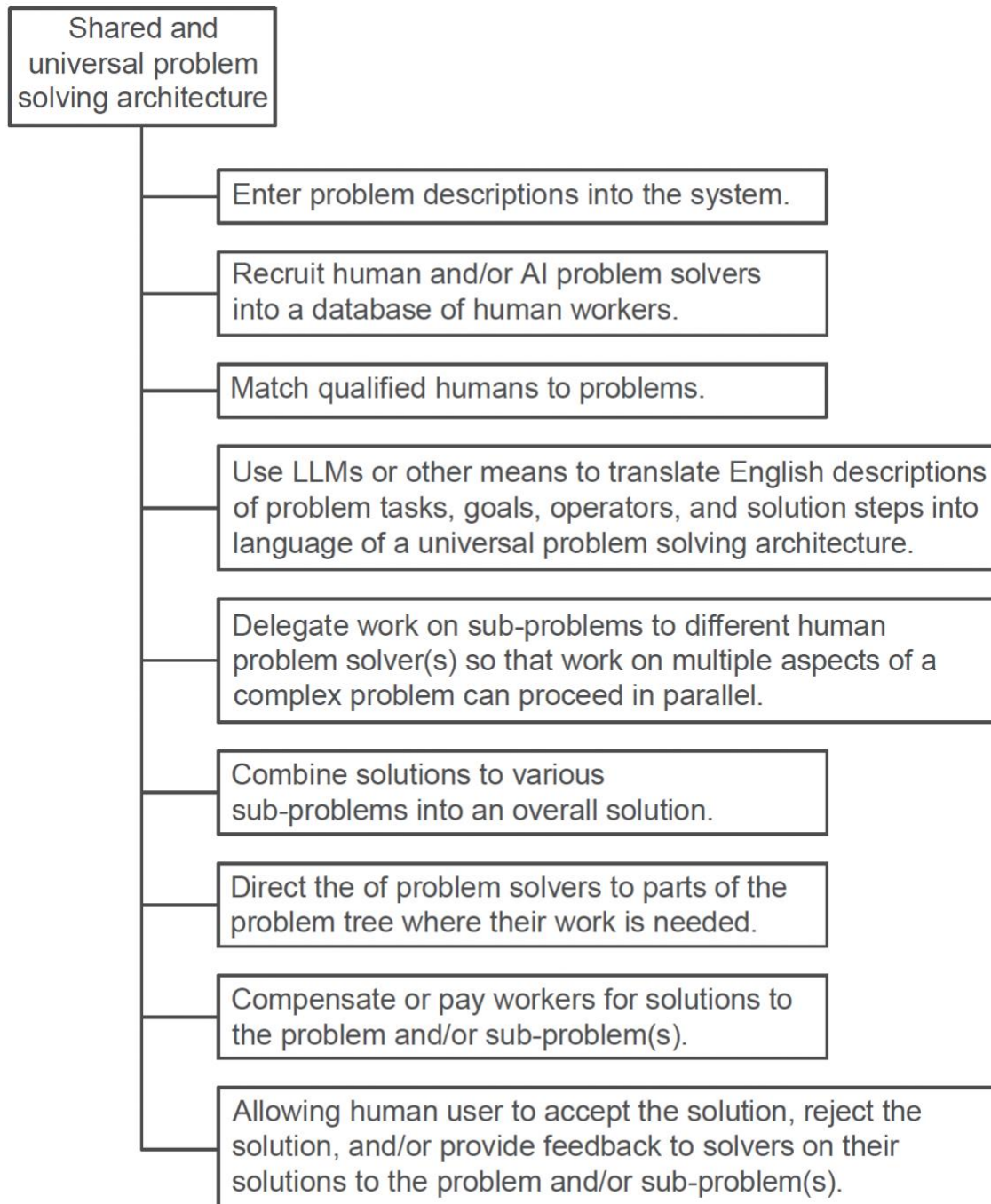


FIG. 18

FIGURE 19: LEARNING PROCESSES OF AAAIS AND AGI

A flowchart showing the multiple learning processes operating at different levels: individual AAAI learning, network-level learning, and AGI-level integration of improvements.

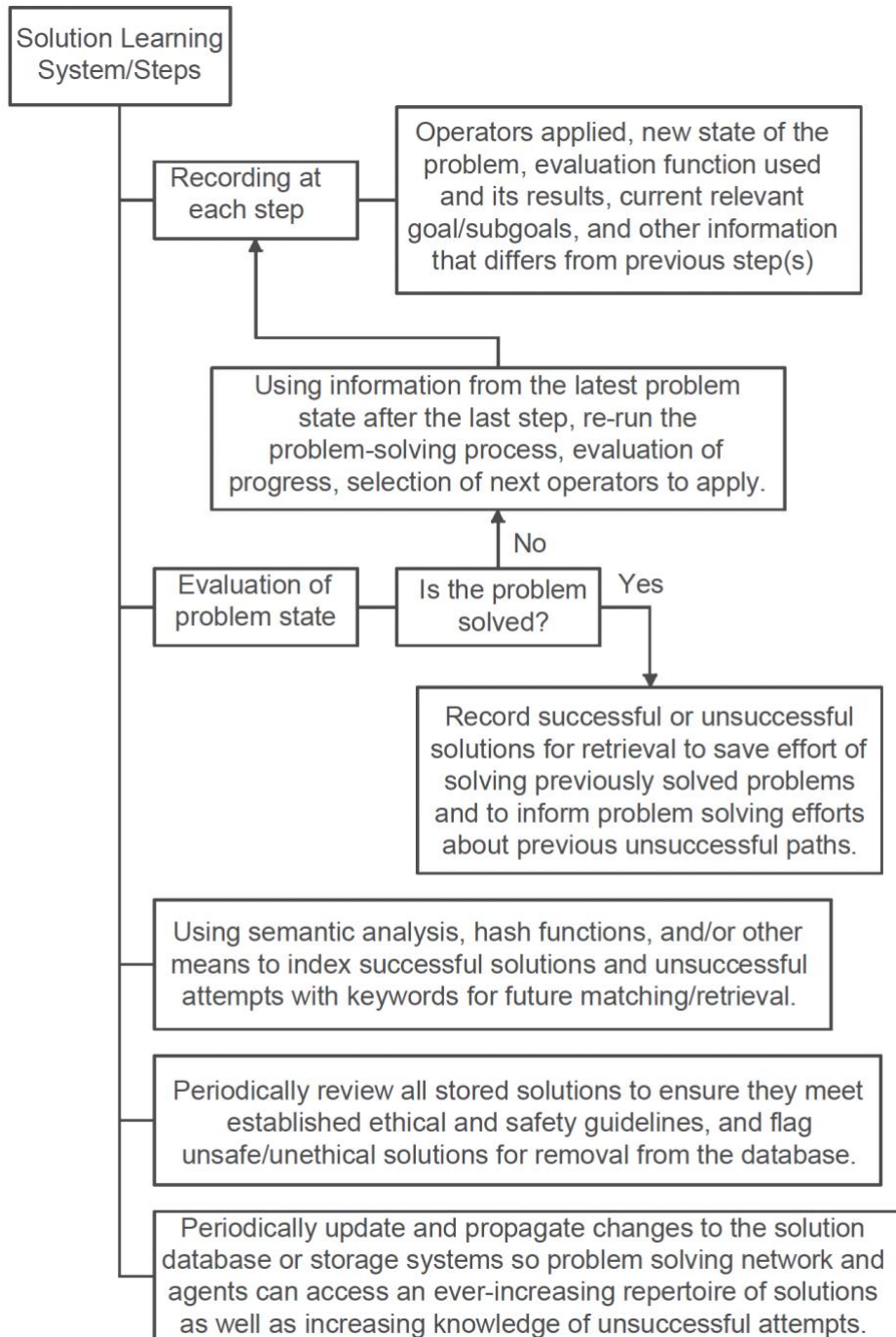


FIG. 19

FIGURE 20: HUMAN-AI DIALOG FOR PROBLEM-SOLVING TRANSLATION

A flowchart showing how natural language input from users is translated into a structured problem-solving format.

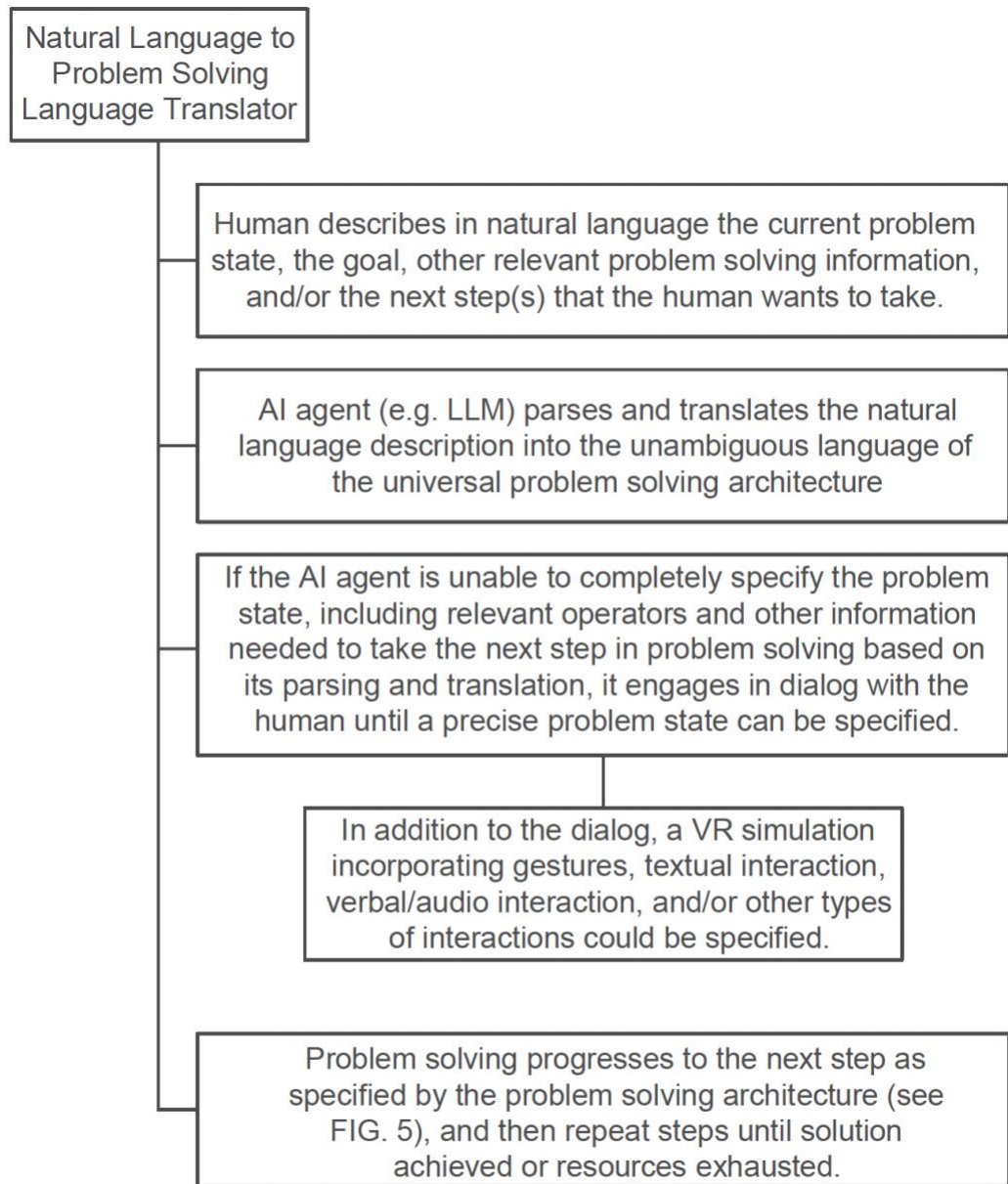


FIG. 20

FIGURE 21: REPUTATIONAL COMPONENT FOR SAFETY AND ETHICS

A flowchart showing how reputation is built and used: clients specify important dimensions, matching algorithms recruit solvers based on reputation, solution steps are recorded, and solver reputation is updated based on outcomes and client satisfaction.

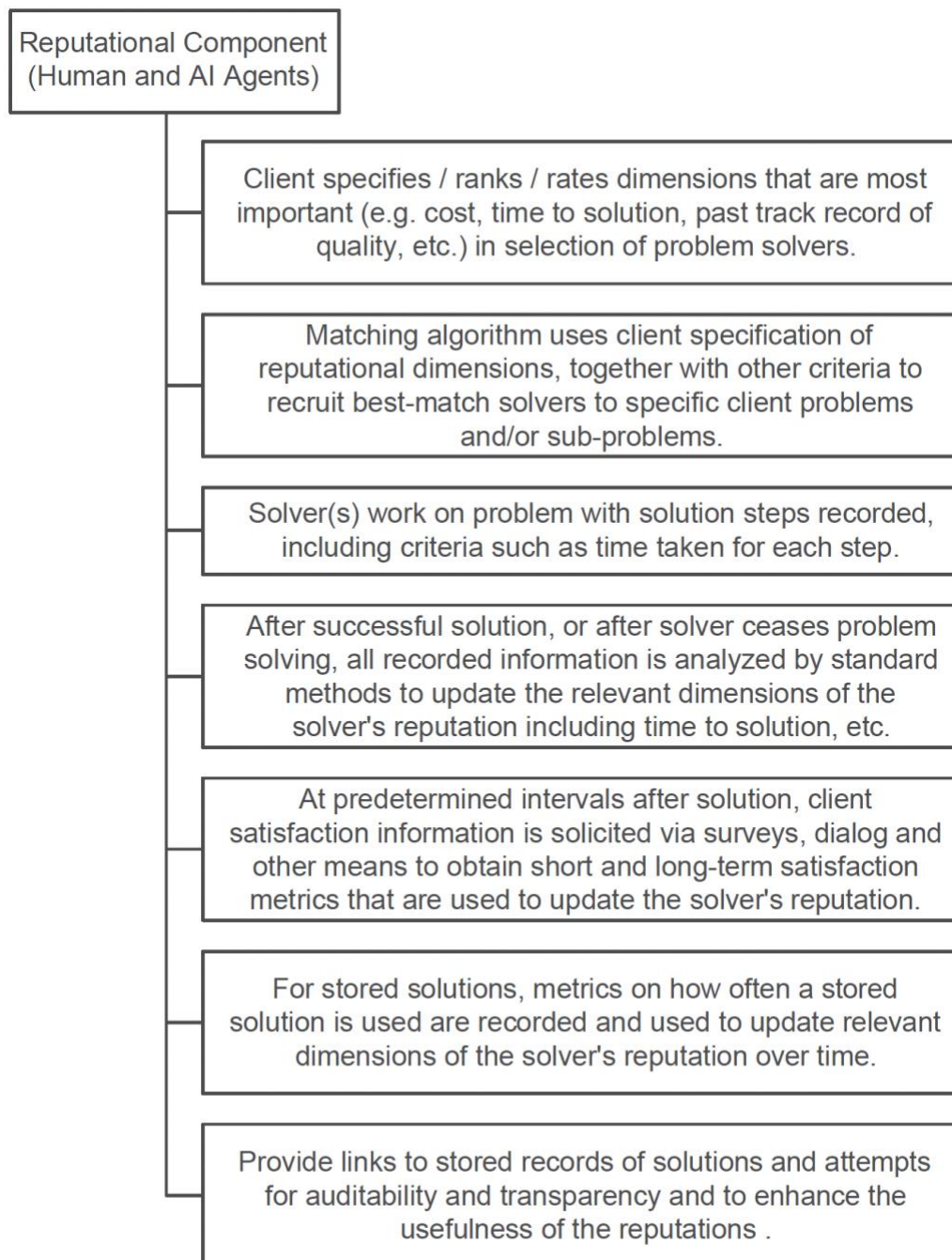


FIG. 21